PROJECT - IBM HR ANALYTICS EMPLOYEE ATTRITION & PERFORMANCE

Project Description:

This project aims to analyze the IBM HR Attrition dataset to understand the key factors driving employee attrition, to analyze factors influencing employee performance and to develop a predictive model capable of identifying employees at high risk of leaving the company, thus the company can develop targeted strategies for employee retention, improve workplace satisfaction, and optimize human resources planning.

Tech Stack Used:

Microsoft Excel: For dataset view

Python: For Data cleaning & preprocessing, EDA, machine learning for predictive modeling, data visualization

Power BI: For dashboard creation

Dataset Overview:

This is a fictional dataset created by IBM data scientists. This dataset is used to uncover the factors that lead to employee attrition.

Column Descriptions:

Col. No.	Column Name	Variable Type	Column Description
1	Age	Numerical	Employee's current age.
2	Attrition	Categorical	Target Variable: Whether the employee left the company (Yes/No).
3	BusinessTravel	Categorical	Frequency of business travel (e.g., Non-Travel, Travel_Rarely, Travel_Frequently).
4	DailyRate	Numerical	Daily rate of pay for the employee.

Col. No.	Column Name	Variable Type	Column Description
5	Department	Categorical	Department the employee works in (e.g., Sales, R&D).
6	DistanceFromHome	Numerical	Distance from home to work (in miles).
7	Education	Numerical (Ordinal)	Educational degree level (1='Below College' to 5='Doctor').
8	EducationField	Categorical	Field of study (e.g., Life Sciences, Marketing, Medical).
9	EmployeeCount	Numerical (Constant)	Employee Count (always 1).
10	EmployeeNumber	Numerical	Unique employee ID number.
11	EnvironmentSatisfaction	Numerical (Ordinal)	Work environment satisfaction rating (1='Low' to 4='Very High').
12	Gender	Categorical	Employee's gender (Male/Female).
13	HourlyRate	Numerical	Hourly rate of pay.
14	JobInvolvement	Numerical (Ordinal)	Job involvement rating (1='Low' to 4='Very High').
15	JobLevel	Numerical (Ordinal)	Level of the job (1='Entry' to 5='Executive').
16	JobRole	Categorical	Specific role performed by the employee (e.g., Sales Executive, Manager).
17	JobSatisfaction	Numerical (Ordinal)	Job satisfaction rating (1='Low' to 4='Very High').
18	MaritalStatus	Categorical	Marital status (Single, Married, Divorced).
19	MonthlyIncome	Numerical	Employee's monthly base salary.

Col. No.	Column Name	Variable Type	Column Description
20	MonthlyRate	Numerical	Monthly total compensation rate.
21	NumCompaniesWorked	Numerical	Number of companies worked at prior to current one.
22	Over18	Categorical (Constant)	Whether the employee is over 18 (always 'Y').
23	OverTime	Categorical	Whether the employee works overtime (Yes/No).
24	PercentSalaryHike	Numerical	Percentage salary increase received last year.
25	PerformanceRating	Numerical (Ordinal)	Employee's performance rating (3='Excellent', 4='Outstanding').
26	RelationshipSatisfaction	Numerical (Ordinal)	Relationship satisfaction rating (1='Low' to 4='Very High').
27	StandardHours	Numerical (Constant)	Standard working hours (always 80).
28	StockOptionLevel	Numerical (Ordinal)	Level of stock option allocation (0='None' to 3='High').
29	TotalWorkingYears	Numerical	Total years of professional work experience.
30	TrainingTimesLastYear	Numerical	Number of training sessions attended in the last year.
31	WorkLifeBalance	Numerical (Ordinal)	Work-Life Balance rating (1='Bad' to 4='Best').
32	YearsAtCompany	Numerical	Total number of years the employee has been with the current company.
33	YearsInCurrentRole	Numerical	Number of years spent in the employee's current job role.
34	YearsSinceLastPromotion	Numerical	Number of years since the employee's last promotion.

Col. No.	Column Name	Variable Type	Column Description
35	YearsWithCurrManager	Numerical	Number of years spent with the current manager.

Steps:

1. Importing Necessary Libraries
2. Data Collection & Data Understanding
3. Data Cleaning & Preprocessing
4. Exploratory Data Analysis (EDA)
5. Employee Attrition Prediction Model using Machine Learning

Below are the steps followed and python codes for doing the analysis.

1. Importing Necessary Libraries

```
#Importing necessary libraries

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')
pd.set_option('display.max_columns', 35)
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
```

2. Data Collection & Data Understanding

```
#Loading the dataset

df=pd.read_csv('WA_Fn-UseC_-HR-Employee-Attrition.csv')
df.head(5)
```

```
   Age Attrition     BusinessTravel  DailyRate           Department  \
0   41       Yes      Travel_Rarely       1102                Sales
1   49        No  Travel_Frequently        279  Research & Development
2   37       Yes      Travel_Rarely       1373  Research & Development
```

```
3   33       No  Travel_Frequently      1392  Research & Development
4   27       No     Travel_Rarely       591  Research & Development

   DistanceFromHome  Education EducationField  EmployeeCount  EmployeeNumber
\
0                 1          2  Life Sciences              1               1
1                 8          1  Life Sciences              1               2
2                 2          2          Other              1               4
3                 3          4  Life Sciences              1               5
4                 2          1        Medical              1               7

   EnvironmentSatisfaction  Gender  HourlyRate  JobInvolvement  JobLevel  \
0                        2  Female          94               3         2
1                        3    Male          61               2         2
2                        4    Male          92               2         1
3                        4  Female          56               3         1
4                        1    Male          40               3         1

              JobRole  JobSatisfaction MaritalStatus  MonthlyIncome  \
0       Sales Executive                4       Single           5993
1    Research Scientist                2      Married           5130
2  Laboratory Technician               3       Single           2090
3    Research Scientist                3      Married           2909
4  Laboratory Technician               2      Married           3468

   MonthlyRate  NumCompaniesWorked Over18 OverTime  PercentSalaryHike  \
0        19479                   8      Y      Yes                 11
1        24907                   1      Y       No                 23
2         2396                   6      Y      Yes                 15
3        23159                   1      Y      Yes                 11
4        16632                   9      Y       No                 12

   PerformanceRating  RelationshipSatisfaction  StandardHours  \
0                  3                         1             80
1                  4                         4             80
2                  3                         2             80
3                  3                         3             80
4                  3                         4             80

   StockOptionLevel  TotalWorkingYears  TrainingTimesLastYear  \
0                 0                  8                      0
1                 1                 10                      3
2                 0                  7                      3
3                 0                  8                      3
4                 1                  6                      3

   WorkLifeBalance  YearsAtCompany  YearsInCurrentRole  \
```

0	1	6	4
1	3	10	7
2	3	0	0
3	3	8	7
4	3	2	2

	YearsSinceLastPromotion	YearsWithCurrManager
0	0	5
1	1	7
2	0	0
3	3	0
4	2	2

```
# check data shape
df.shape
(1470, 35)

# check data types
df.dtypes
Age                        int64
Attrition                  object
BusinessTravel             object
DailyRate                  int64
Department                 object
DistanceFromHome           int64
Education                  int64
EducationField             object
EmployeeCount              int64
EmployeeNumber             int64
EnvironmentSatisfaction    int64
Gender                     object
HourlyRate                 int64
JobInvolvement             int64
JobLevel                   int64
JobRole                    object
JobSatisfaction            int64
MaritalStatus              object
MonthlyIncome              int64
MonthlyRate                int64
NumCompaniesWorked         int64
Over18                     object
OverTime                   object
PercentSalaryHike          int64
PerformanceRating          int64
RelationshipSatisfaction   int64
StandardHours              int64
StockOptionLevel           int64
```

```
TotalWorkingYears           int64
TrainingTimesLastYear       int64
WorkLifeBalance             int64
YearsAtCompany              int64
YearsInCurrentRole          int64
YearsSinceLastPromotion     int64
YearsWithCurrManager        int64
dtype: object
```

df.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1470 entries, 0 to 1469
Data columns (total 35 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   Age                       1470 non-null   int64
 1   Attrition                 1470 non-null   object
 2   BusinessTravel            1470 non-null   object
 3   DailyRate                 1470 non-null   int64
 4   Department                1470 non-null   object
 5   DistanceFromHome          1470 non-null   int64
 6   Education                 1470 non-null   int64
 7   EducationField            1470 non-null   object
 8   EmployeeCount             1470 non-null   int64
 9   EmployeeNumber            1470 non-null   int64
 10  EnvironmentSatisfaction   1470 non-null   int64
 11  Gender                    1470 non-null   object
 12  HourlyRate                1470 non-null   int64
 13  JobInvolvement            1470 non-null   int64
 14  JobLevel                  1470 non-null   int64
 15  JobRole                   1470 non-null   object
 16  JobSatisfaction           1470 non-null   int64
 17  MaritalStatus             1470 non-null   object
 18  MonthlyIncome             1470 non-null   int64
 19  MonthlyRate               1470 non-null   int64
 20  NumCompaniesWorked        1470 non-null   int64
 21  Over18                    1470 non-null   object
 22  OverTime                  1470 non-null   object
 23  PercentSalaryHike         1470 non-null   int64
 24  PerformanceRating         1470 non-null   int64
 25  RelationshipSatisfaction  1470 non-null   int64
 26  StandardHours             1470 non-null   int64
 27  StockOptionLevel          1470 non-null   int64
 28  TotalWorkingYears         1470 non-null   int64
 29  TrainingTimesLastYear     1470 non-null   int64
 30  WorkLifeBalance           1470 non-null   int64
```

```
 31  YearsAtCompany           1470 non-null   int64
 32  YearsInCurrentRole       1470 non-null   int64
 33  YearsSinceLastPromotion  1470 non-null   int64
 34  YearsWithCurrManager     1470 non-null   int64
dtypes: int64(26), object(9)
memory usage: 402.1+ KB
```

3. Data Cleaning & Preprocessing

```
# check number of duplicated data
print(f'Number of duplicated data: {df.duplicated().sum()}')
```

```
Number of duplicated data: 0
```

```
# check missing values
df.isnull().sum()
Age                         0
Attrition                   0
BusinessTravel              0
DailyRate                   0
Department                  0
DistanceFromHome            0
Education                   0
EducationField              0
EmployeeCount               0
EmployeeNumber              0
EnvironmentSatisfaction     0
Gender                      0
HourlyRate                  0
JobInvolvement              0
JobLevel                    0
JobRole                     0
JobSatisfaction             0
MaritalStatus               0
MonthlyIncome               0
MonthlyRate                 0
NumCompaniesWorked          0
Over18                      0
OverTime                    0
PercentSalaryHike           0
PerformanceRating           0
RelationshipSatisfaction    0
StandardHours               0
StockOptionLevel            0
TotalWorkingYears           0
TrainingTimesLastYear       0
WorkLifeBalance             0
```

```
YearsAtCompany            0
YearsInCurrentRole        0
YearsSinceLastPromotion   0
YearsWithCurrManager      0
dtype: int64
```

There are no null values and duplicated data in the dataset.

```
# check data description statistics
df.describe()
```

```
            Age   DailyRate  DistanceFromHome   Education  EmployeeCount  \
count  1470.000000  1470.000000      1470.000000  1470.000000        1470.0
mean     36.923810   802.485714         9.192517     2.912925           1.0
std       9.135373   403.509100         8.106864     1.024165           0.0
min      18.000000   102.000000         1.000000     1.000000           1.0
25%      30.000000   465.000000         2.000000     2.000000           1.0
50%      36.000000   802.000000         7.000000     3.000000           1.0
75%      43.000000  1157.000000        14.000000     4.000000           1.0
max      60.000000  1499.000000        29.000000     5.000000           1.0

       EmployeeNumber  EnvironmentSatisfaction   HourlyRate  JobInvolvement  \
count     1470.000000              1470.000000  1470.000000     1470.000000
mean      1024.865306                 2.721769    65.891156        2.729932
std        602.024335                 1.093082    20.329428        0.711561
min          1.000000                 1.000000    30.000000        1.000000
25%        491.250000                 2.000000    48.000000        2.000000
50%       1020.500000                 3.000000    66.000000        3.000000
75%       1555.750000                 4.000000    83.750000        3.000000
max       2068.000000                 4.000000   100.000000        4.000000

        JobLevel  JobSatisfaction  MonthlyIncome   MonthlyRate  \
count  1470.000000      1470.000000    1470.000000   1470.000000
mean      2.063946         2.728571    6502.931293  14313.103401
std       1.106940         1.102846    4707.956783   7117.786044
min       1.000000         1.000000    1009.000000   2094.000000
25%       1.000000         2.000000    2911.000000   8047.000000
50%       2.000000         3.000000    4919.000000  14235.500000
75%       3.000000         4.000000    8379.000000  20461.500000
max       5.000000         4.000000   19999.000000  26999.000000

       NumCompaniesWorked  PercentSalaryHike  PerformanceRating  \
count         1470.000000        1470.000000        1470.000000
mean             2.693197          15.209524           3.153741
std              2.498009           3.659938           0.360824
min              0.000000          11.000000           3.000000
25%              1.000000          12.000000           3.000000
50%              2.000000          14.000000           3.000000
```

```
75%          4.000000        18.000000        3.000000
max          9.000000        25.000000        4.000000

      RelationshipSatisfaction  StandardHours  StockOptionLevel  \
count            1470.000000        1470.0      1470.000000
mean               2.712245         80.0         0.793878
std                1.081209          0.0         0.852077
min                1.000000         80.0         0.000000
25%                2.000000         80.0         0.000000
50%                3.000000         80.0         1.000000
75%                4.000000         80.0         1.000000
max                4.000000         80.0         3.000000

      TotalWorkingYears  TrainingTimesLastYear  WorkLifeBalance  \
count       1470.000000           1470.000000      1470.000000
mean          11.279592              2.799320         2.761224
std           7.780782              1.289271         0.706476
min           0.000000              0.000000         1.000000
25%           6.000000              2.000000         2.000000
50%          10.000000              3.000000         3.000000
75%          15.000000              3.000000         3.000000
max          40.000000              6.000000         4.000000

      YearsAtCompany  YearsInCurrentRole  YearsSinceLastPromotion  \
count     1470.000000        1470.000000            1470.000000
mean        7.008163           4.229252               2.187755
std        6.126525           3.623137               3.222430
min        0.000000           0.000000               0.000000
25%        3.000000           2.000000               0.000000
50%        5.000000           3.000000               1.000000
75%        9.000000           7.000000               3.000000
max       40.000000          18.000000              15.000000

      YearsWithCurrManager
count          1470.000000
mean              4.123129
std               3.568136
min               0.000000
25%               2.000000
50%               3.000000
75%               7.000000
max              17.000000
```

#Checking the number of unique values in each columns to find out the irrelevant dat a for analysis

```
df.nunique()
```

```
Age                       43
Attrition                  2
BusinessTravel             3
DailyRate                886
Department                 3
DistanceFromHome          29
Education                  5
EducationField             6
EmployeeCount              1
EmployeeNumber          1470
EnvironmentSatisfaction    4
Gender                     2
HourlyRate                71
JobInvolvement             4
JobLevel                   5
JobRole                    9
JobSatisfaction            4
MaritalStatus              3
MonthlyIncome           1349
MonthlyRate             1427
NumCompaniesWorked        10
Over18                     1
OverTime                   2
PercentSalaryHike         15
PerformanceRating          2
RelationshipSatisfaction   4
StandardHours              1
StockOptionLevel           4
TotalWorkingYears         40
TrainingTimesLastYear      7
WorkLifeBalance            4
YearsAtCompany            37
YearsInCurrentRole        19
YearsSinceLastPromotion   16
YearsWithCurrManager      18
dtype: int64
```

```
#Dropping the irrelevant data
cols_to_drop = ['EmployeeCount', 'StandardHours', 'Over18', 'EmployeeNumber']
df = df.drop(columns=cols_to_drop)
```

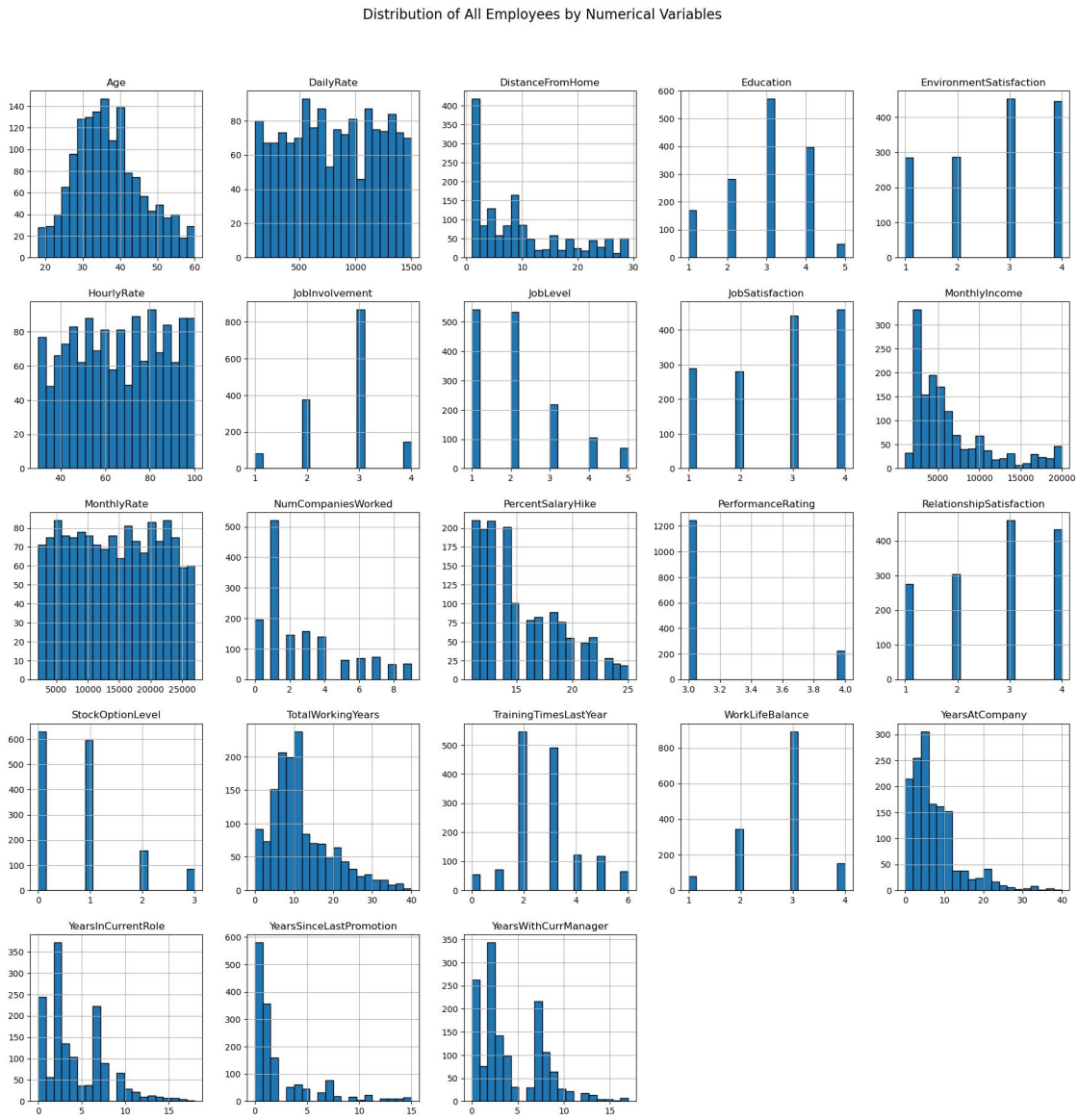
These columns are dropped because they are irrelevant for analysis. They have a single, constant value across all records and it is dropped as they provide no predictive power.

```
df.shape
(1470, 31)
```

4. Exploratory Data Analysis

#Distribution of numerical variables by count for overall employees

```
df.hist(figsize=(18, 18), bins=20, edgecolor='black')
plt.suptitle('Distribution of All Employees by Numerical Variables', y=1.02, fontsize=16)
plt.tight_layout(pad=2.0)
plt.show()
```



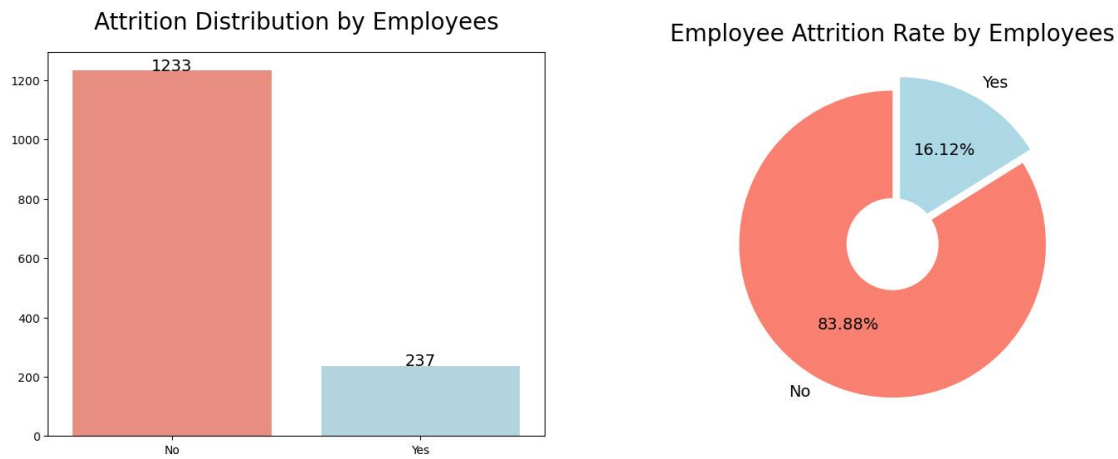
Attrition Distribution by Employees

```
attrition_counts = df['Attrition'].value_counts()

# Extract the counts for 'No' (Stayed) and 'Yes' (Left)
Employees_stayed_count = attrition_counts.get('No', 0)
Employees_left_count = attrition_counts.get('Yes', 0)
```

```
print(f"Number of Employees who stayed in the company (Attrition='No'): {Employees_stayed_count}")
print(f"Number of Employees who left the company (Attrition='Yes'): {Employees_left_count}")
```

```
Number of Employees who stayed in the company (Attrition='No'): 1233
Number of Employees who left the company (Attrition='Yes'): 237
```

```
#Distribution of Attrition

plt.figure(figsize=(17,6))
plt.subplot(1,2,1)
attrition_rate = df["Attrition"].value_counts()
sns.barplot(x=attrition_rate.index,y=attrition_rate.values,palette=['salmon', 'lightblue'])
plt.title("Attrition Distribution by Employees",fontweight="normal",size=20,pad=20)
for i, v in enumerate(attrition_rate.values):
    plt.text(i, v, v,ha="center", fontweight='normal', fontsize=14)


plt.subplot(1,2,2)
plt.pie(attrition_rate, labels=["No","Yes"], autopct="%.2f%%", textprops={"fontweight":"normal","size":14},
        colors = ['salmon', 'lightblue'],explode=[0,0.1],startangle=90)
center_circle = plt.Circle((0, 0), 0.3, fc='white')
fig = plt.gcf()
fig.gca().add_artist(center_circle)
plt.title("Employee Attrition Rate by Employees",fontweight="normal",size=20,pad=10)
plt.show()
```



Findings & Insights:

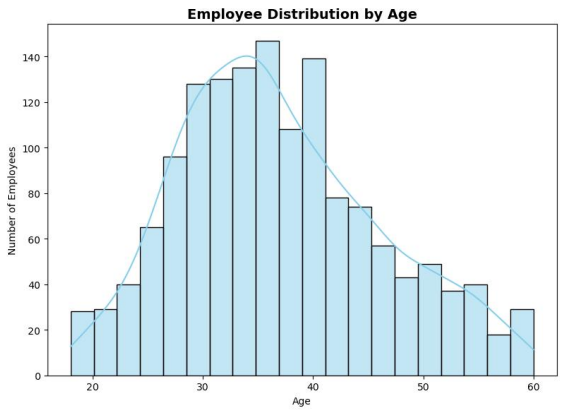
The dataset consists of 1,470 employees, out of which 237 employees left the company, resulting in an attrition rate of 16.12%. The remaining 1,233 employees stayed, representing a retention rate of 83.88%. This suggests that IBM experiences

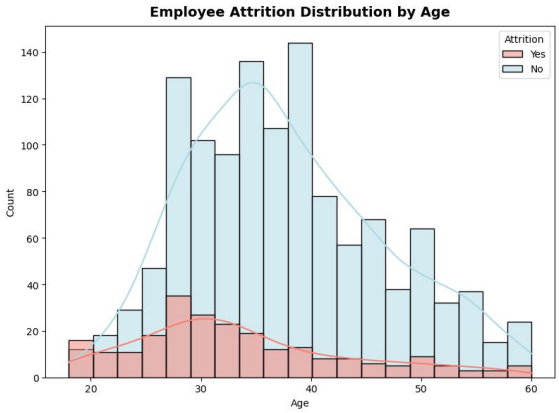
a noticeable moderate level of employee turnover, making it important to analyze the underlying factors influencing attrition such as overtime, job satisfaction, salary hikes, and work-life balance.

```
# Defining a function to calculate and return attrition rate (%) for each category in a given column

def calculate_attrition_rate(df, column):
    # Group by the column and Attrition status
    attrition_count = df.groupby([column, 'Attrition']).size().unstack(fill_value=0)

    # Calculate attrition rate (%)
    attrition_rate = attrition_count['Yes'] / attrition_count.sum(axis=1) * 100

    # Create a clean DataFrame
    attrition_rate_df = attrition_rate.reset_index()
    attrition_rate_df.columns = [column, 'AttritionRate']

    return attrition_rate_df
```

1) Attrition vs Age

```
#Attrition vs Age
plt.figure(figsize=(16, 6))


plt.subplot(1, 2, 1)
sns.histplot(df['Age'], bins=20, kde=True, color='skyblue')
plt.title('Employee Distribution by Age', fontsize=14, fontweight='bold')
plt.xlabel('Age')
plt.ylabel('Number of Employees')



plt.subplot(1,2,2)
sns.histplot(x="Age",hue="Attrition",data=df,kde=True,palette=['salmon', 'lightblue'])
plt.title("Employee Attrition Distribution by Age",fontweight="black",size=14,pad=10)

plt.tight_layout()
plt.show()
```



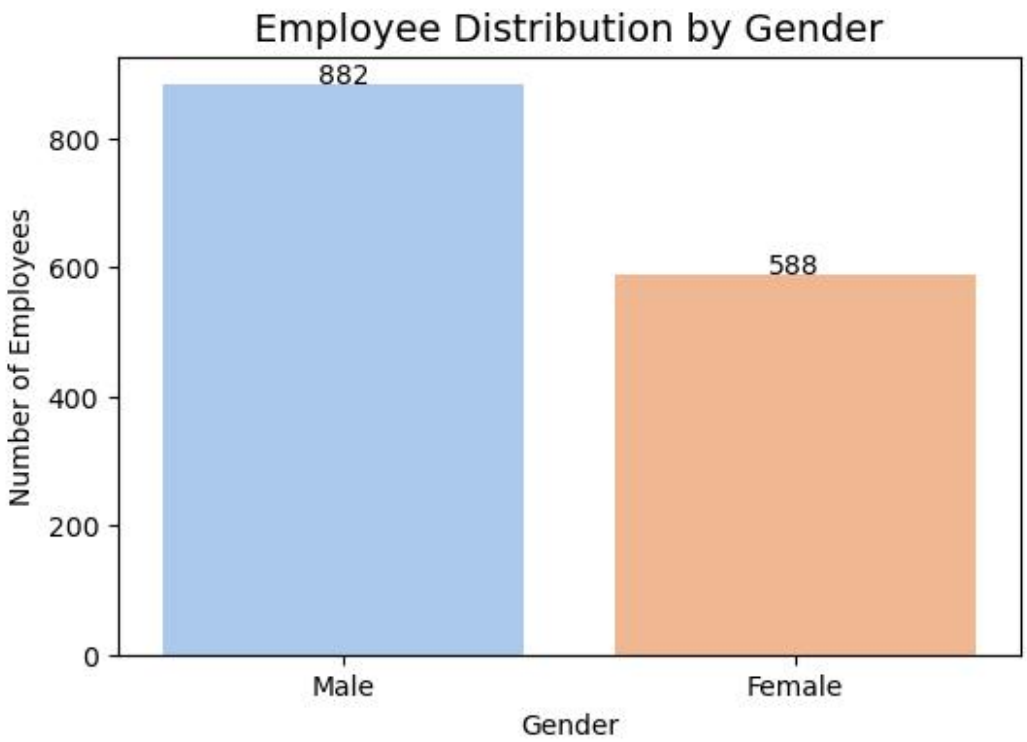


Findings & Insights:

Most of the employees are between age late 20s to 40. We can see from the graph that as the age is increasing the attrition is decreasing. Employees in the late 20s and early 30s tends to leave the company. This indicates that the employees with young age leaves the company more compared to elder employees.

2) Attrition vs Gender

```
# Attrition vs Gender

df['Gender'].value_counts()
Male      882
Female    588
Name: Gender, dtype: int64
gender_counts = df['Gender'].value_counts()

plt.figure(figsize=(6,4))
sns.barplot(x=gender_counts.index, y=gender_counts.values, palette='pastel')


for i, value in enumerate(gender_counts.values):
    plt.text(i, value + 2, str(value), ha='center', fontweight='normal')

plt.title('Employee Distribution by Gender', fontsize=14, fontweight='normal')
plt.xlabel('Gender')
plt.ylabel('Number of Employees')
plt.show()
```

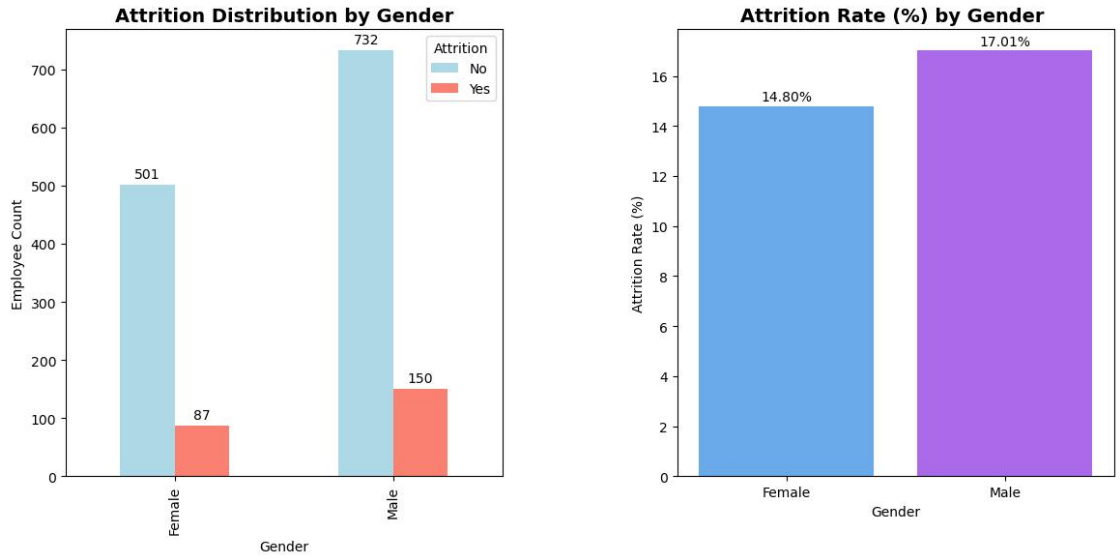


Findings:

Out of the total employees, 882 employees are males and 588 employees are females.

```
df[df['Attrition'] == 'Yes']['Gender'].value_counts()
Male      150
Female     87
Name: Gender, dtype: int64
calculate_attrition_rate(df, 'Gender').round(2)
  Gender  AttritionRate
0  Female         14.80
1    Male         17.01

fig, axes = plt.subplots(1, 2, figsize=(14, 6))
plt.subplots_adjust(wspace=0.4)

# Attrition Distribution (Yes/No) by Gender
attrition_gender = df.groupby(['Gender', 'Attrition']).size().unstack(fill_value=0)
attrition_gender.plot(kind='bar', ax=axes[0], color=['lightblue', 'salmon'])
axes[0].set_title('Attrition Distribution by Gender', fontsize=14, fontweight='bold')
axes[0].set_xlabel('Gender')
axes[0].set_ylabel('Employee Count')
axes[0].legend(title='Attrition')
for container in axes[0].containers:
    axes[0].bar_label(container, fmt='%d', label_type='edge', fontsize=10, padding=3)

# Attrition Rate (%) by Gender
```

```
attrition_rate_gender = calculate_attrition_rate(df, 'Gender')
sns.barplot(x='Gender', y='AttritionRate', data=attrition_rate_gender, palette='cool',
ax=axes[1])
axes[1].set_title('Attrition Rate (%) by Gender ', fontsize=14, fontweight='bold')
axes[1].set_xlabel('Gender')
axes[1].set_ylabel('Attrition Rate (%)')
for i, v in enumerate(attrition_rate_gender['AttritionRate']):
    axes[1].text(i, v + 0.2 , f'{v:.2f}%', ha='center', fontweight='normal')

plt.show()
```



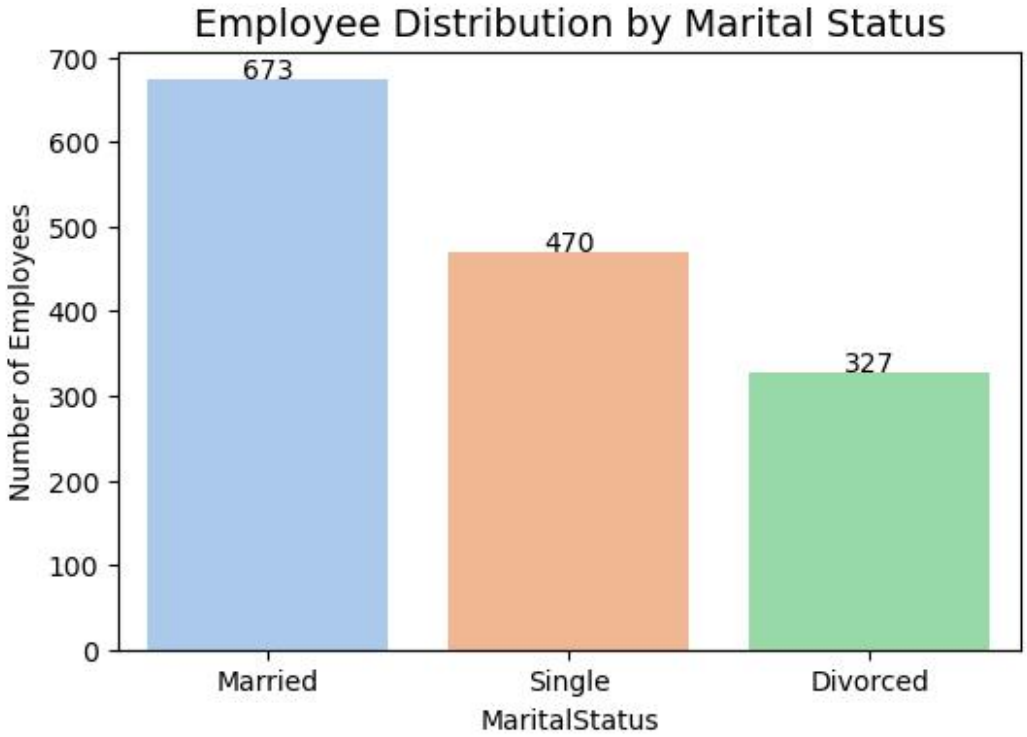
Findings & Insights:

The number of male employees in the company accounts for a higher proportion than female employees. Male employees are leaving the company more compared to female employees.

3) Attrition vs Marital Status

```
df['MaritalStatus'].value_counts()
Married     673
Single      470
Divorced    327
Name: MaritalStatus, dtype: int64
maritalstatus_counts = df['MaritalStatus'].value_counts()

plt.figure(figsize=(6,4))
sns.barplot(x=maritalstatus_counts.index, y=maritalstatus_counts.values, palette='p
astel')


for i, value in enumerate(maritalstatus_counts.values):
    plt.text(i, value + 2, str(value), ha='center', fontweight='normal')
```

```
plt.title('Employee Distribution by Marital Status', fontsize=14, fontweight='normal')
plt.xlabel('MaritalStatus')
plt.ylabel('Number of Employees')
plt.show()
```

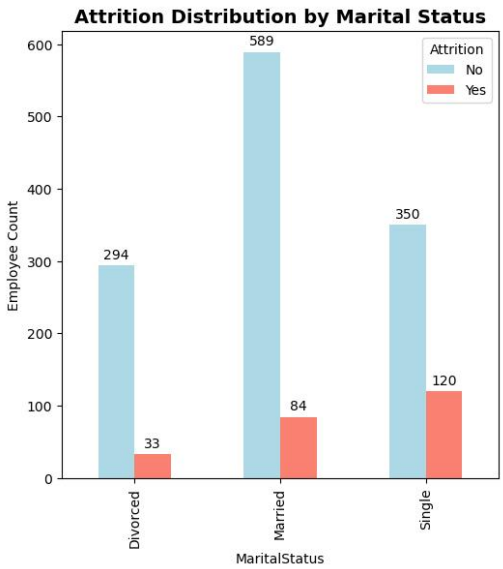


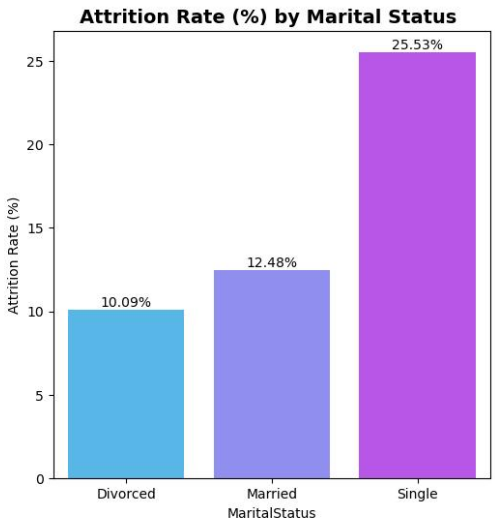
Findings:

Most of the employees in the company are married.

```
df[df['Attrition'] == 'Yes']['MaritalStatus'].value_counts()
Single      120
Married      84
Divorced     33
Name: MaritalStatus, dtype: int64
calculate_attrition_rate(df, 'MaritalStatus').round(2)
  MaritalStatus  AttritionRate
0     Divorced          10.09
1      Married          12.48
2       Single          25.53

fig, axes = plt.subplots(1, 2, figsize=(14, 6))
plt.subplots_adjust(wspace=0.4)

#Attrition Distribution (Yes/No) by MaritalStatus
attrition_maritalstatus = df.groupby(['MaritalStatus', 'Attrition']).size().unstack(fill_va
lue=0)
attrition_maritalstatus.plot(kind='bar', ax=axes[0], color=['lightblue', 'salmon'])
axes[0].set_title('Attrition Distribution by Marital Status', fontsize=14, fontweight='b
```

```
old')
axes[0].set_xlabel('MaritalStatus')
axes[0].set_ylabel('Employee Count')
axes[0].legend(title='Attrition')
for container in axes[0].containers:
    axes[0].bar_label(container, fmt='%d', label_type='edge', fontsize=10, padding=3)

#Attrition Rate (%) by MaritalStatus
attrition_rate_maritalstatus = calculate_attrition_rate(df, 'MaritalStatus')
sns.barplot(x='MaritalStatus', y='AttritionRate', data=attrition_rate_maritalstatus, pa
lette='cool', ax=axes[1])
axes[1].set_title('Attrition Rate (%) by Marital Status ', fontsize=14, fontweight='bold')
axes[1].set_xlabel('MaritalStatus')
axes[1].set_ylabel('Attrition Rate (%)')
for i, v in enumerate(attrition_rate_maritalstatus['AttritionRate']):
    axes[1].text(i, v + 0.2 , f'{v:.2f}%', ha='center', fontweight='normal')

plt.show()
```





Findings & Insights:

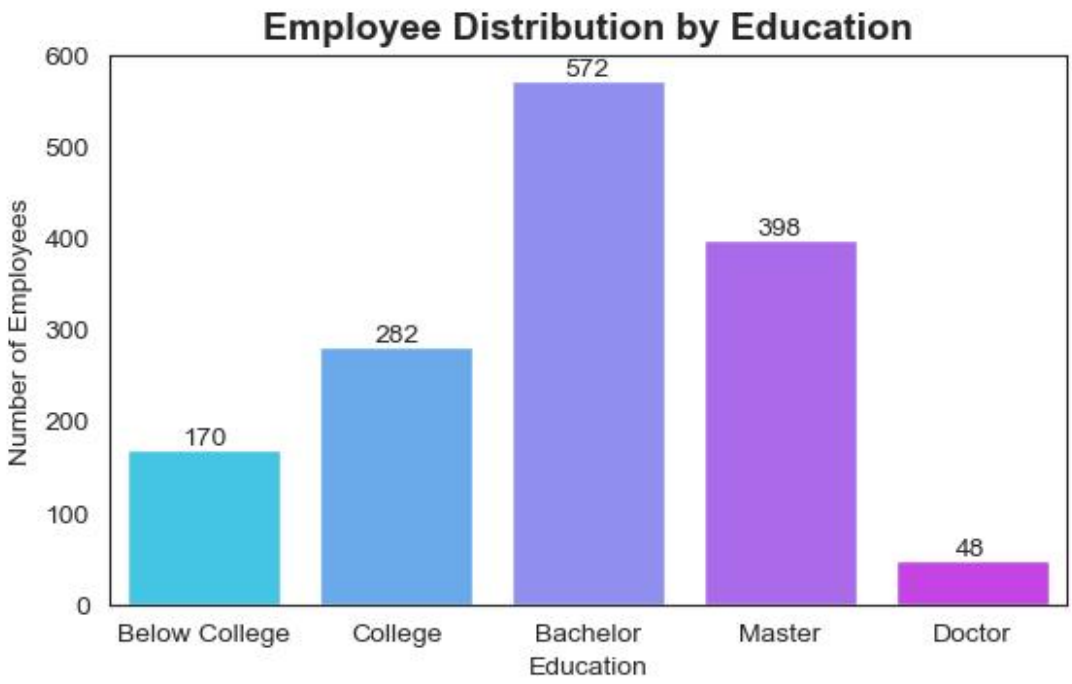
The attrition rate is high for employees who are single and low for employees who are divorced. Single employees have the highest attrition because they are usually younger, early in their careers, and have fewer family responsibilities. This makes them more flexible and more willing to change jobs or companies for better pay, growth, or location, while married and divorced employees prefer job stability.

4) Attrition vs Education Level

```
df['Education'].value_counts()
3    572
4    398
2    282
```

```
1    170
5     48
Name: Education, dtype: int64
```

```
education_labels = {1: "Below College",2: "College",3: "Bachelor",4: "Master",5: "Doc
tor"}


edu_counts = df['Education'].value_counts().sort_index()

sns.set_style("white")

plt.figure(figsize=(6, 4))

sns.barplot(x=[education_labels[i] for i in edu_counts.index],
        y=edu_counts.values,
        palette='cool')

plt.title('Employee Distribution by Education', fontsize=14, fontweight='bold')
plt.xlabel('Education')
plt.ylabel('Number of Employees')

plt.grid(False)

for i, v in enumerate(edu_counts.values):
   plt.text(i, v + 5, str(v), ha='center', fontweight='normal')


plt.tight_layout(pad=2.0)
plt.show()
```

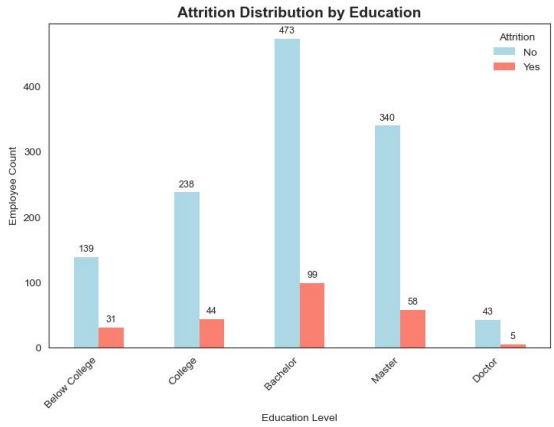


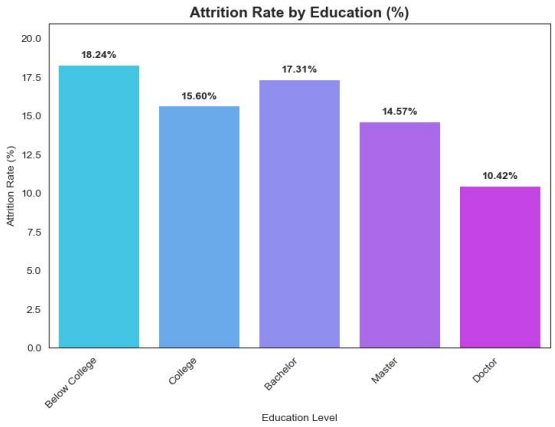
Findings:

Most of the employees have a Bachelor's degree as their education qualification, followed by a Masters degree.

```
df[df['Attrition'] == 'Yes']['Education'].value_counts()
3    99
4    58
2    44
1    31
5     5
Name: Education, dtype: int64
calculate_attrition_rate(df, 'Education').round(2)
   Education  AttritionRate
0          1          18.24
1          2          15.60
2          3          17.31
3          4          14.57
4          5          10.42

education_labels = {
    1: "Below College",
    2: "College",
    3: "Bachelor",
    4: "Master",
    5: "Doctor"
}
```

```
sns.set_style("white")


fig, axes = plt.subplots(1, 2, figsize=(15, 6))
plt.subplots_adjust(wspace=0.4)

# Attrition Distribution (Yes/No) by Education
attrition_edu = df.groupby(['Education', 'Attrition']).size().unstack(fill_value=0)
attrition_edu.plot(kind='bar', ax=axes[0], color=['lightblue', 'salmon'], edgecolor='no
ne')
axes[0].set_title('Attrition Distribution by Education', fontsize=14, fontweight='bold')
axes[0].set_xlabel('Education Level')
axes[0].set_ylabel('Employee Count')
axes[0].set_xticklabels([education_labels[i] for i in attrition_edu.index], rotation=45,
ha='right')
axes[0].legend(title='Attrition', frameon=False)
axes[0].grid(False)
for container in axes[0].containers:
    axes[0].bar_label(container, fmt='%d', label_type='edge', fontsize=9, padding=3)

# Attrition Rate (%) by Education
attrition_rate_edu = calculate_attrition_rate(df, 'Education')
sns.barplot(x='Education', y='AttritionRate', data=attrition_rate_edu, palette='cool',
ax=axes[1], edgecolor='none')
axes[1].set_title('Attrition Rate by Education (%)', fontsize=14, fontweight='bold')
axes[1].set_xlabel('Education Level')
axes[1].set_ylabel('Attrition Rate (%)')
axes[1].set_xticklabels([education_labels[i] for i in attrition_rate_edu['Education']], r
otation=45, ha='right')
axes[1].set_ylim(0, attrition_rate_edu['AttritionRate'].max() * 1.15)
axes[1].grid(False)
for i, v in enumerate(attrition_rate_edu['AttritionRate']):
    axes[1].text(i, v + 0.5, f'{v:.2f}%', ha='center', fontweight='bold')

plt.tight_layout(pad=2.0)
plt.show()
```



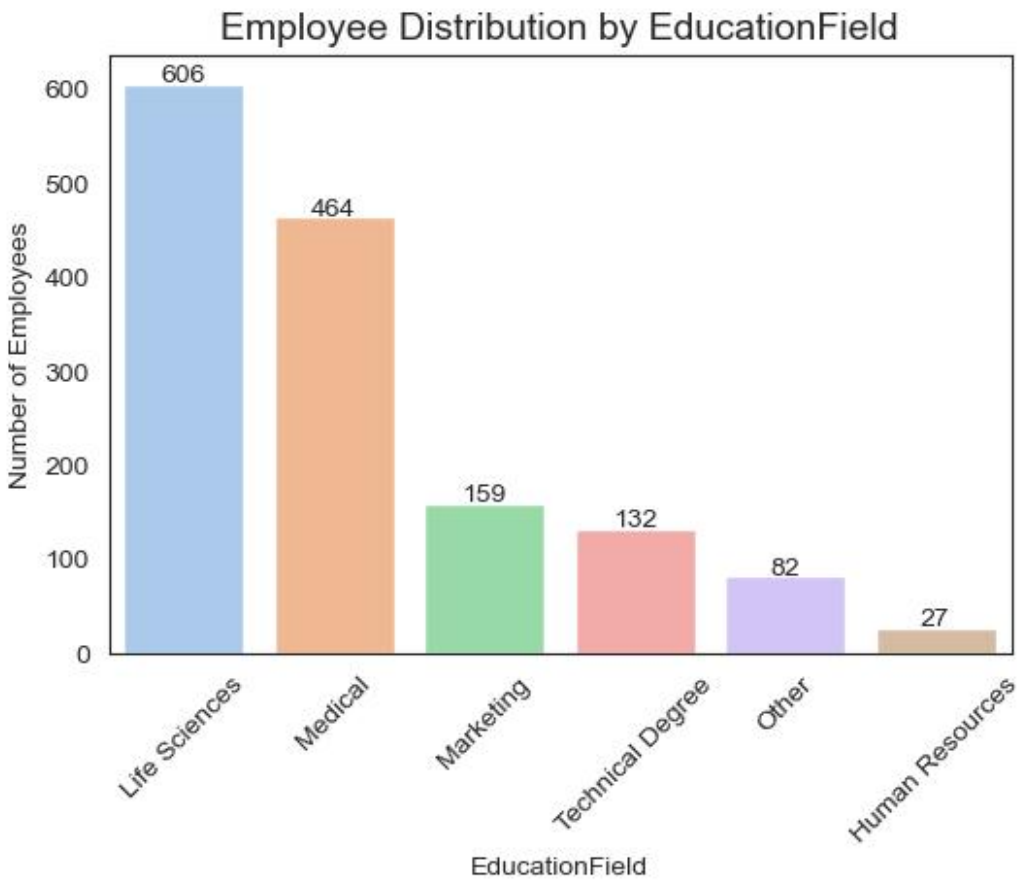


Findings & Insights:

From the graph we can observe a trend of decreasing in attrition rate as the education qualification increases. The employees with lower education levels are more likely to leave the company, while employees with higher education tend to stay longer.

5) Attrition vs Education Field

```
df['EducationField'].value_counts()
Life Sciences      606
Medical            464
Marketing          159
Technical Degree   132
Other              82
Human Resources    27
Name: EducationField, dtype: int64
edufield_counts = df['EducationField'].value_counts()

plt.figure(figsize=(6,4))
sns.barplot(x=edufield_counts.index, y=edufield_counts, palette='pastel')


for i, value in enumerate(edufield_counts.values):
    plt.text(i, value + 2, str(value), ha='center', fontweight='normal')

plt.title('Employee Distribution by EducationField', fontsize=14, fontweight='normal')
plt.xlabel('EducationField')
plt.ylabel('Number of Employees')
plt.xticks(rotation=45)
plt.show()
```

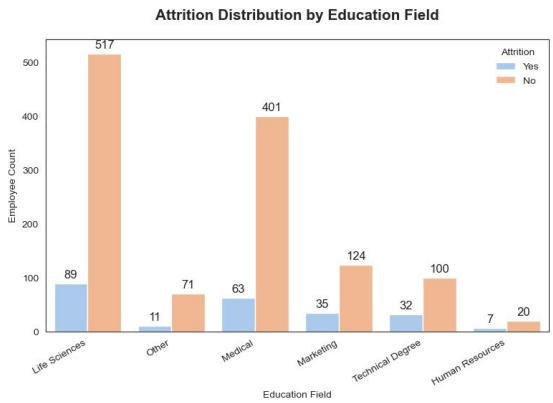


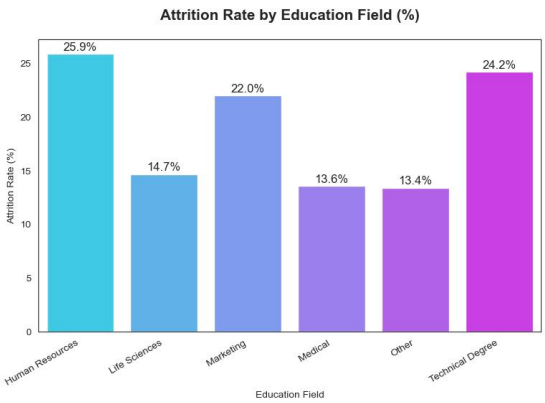
Findings:

Most of the employees are either from Life Science or Medical Education Field and very few employees are from Human Resources Education Field.

```
df[df['Attrition'] == 'Yes']['EducationField'].value_counts()
Life Sciences      89
Medical            63
Marketing          35
Technical Degree   32
Other              11
Human Resources     7
Name: EducationField, dtype: int64
calculate_attrition_rate(df, 'EducationField').round(2)
    EducationField  AttritionRate
0   Human Resources         25.93
1    Life Sciences         14.69
2        Marketing         22.01
3          Medical         13.58
4            Other         13.41
5  Technical Degree         24.24

sns.set_style("white")
```

```
fig, axes = plt.subplots(1, 2, figsize=(16, 6))
plt.subplots_adjust(wspace=0.4)

#  Attrition Distribution (Yes/No) by Education Field ---
sns.countplot(data=df, x='EducationField', hue='Attrition', palette='pastel', ax=axes
[0])
axes[0].set_title('Attrition Distribution by Education Field', fontsize=15, fontweight='
bold', pad=20)
axes[0].set_xlabel('Education Field')
axes[0].set_ylabel('Employee Count')
axes[0].set_xticklabels(axes[0].get_xticklabels(), rotation=30, ha='right')
axes[0].grid(False)
axes[0].legend(title='Attrition', frameon=False)


for container in axes[0].containers:
    axes[0].bar_label(container, fmt='%d', label_type='edge', fontsize=12, padding=3)

#  Attrition Rate (%) by Education Field ---
attrition_rate_edufield = calculate_attrition_rate(df, 'EducationField')

sns.barplot(x='EducationField', y='AttritionRate', data=attrition_rate_edufield,
            palette='cool', ax=axes[1])
axes[1].set_title('Attrition Rate by Education Field (%)', fontsize=15, fontweight='bol
d', pad=20)
axes[1].set_xlabel('Education Field')
axes[1].set_ylabel('Attrition Rate (%)')
axes[1].set_xticklabels(attrition_rate_edufield['EducationField'], rotation=30, ha='rig
ht')
axes[1].grid(False)


for i, v in enumerate(attrition_rate_edufield['AttritionRate']):
    axes[1].text(i, v + 0.3, f'{v:.1f}%', ha='center', fontsize=12)

plt.tight_layout(pad=2.0)
plt.show()
```



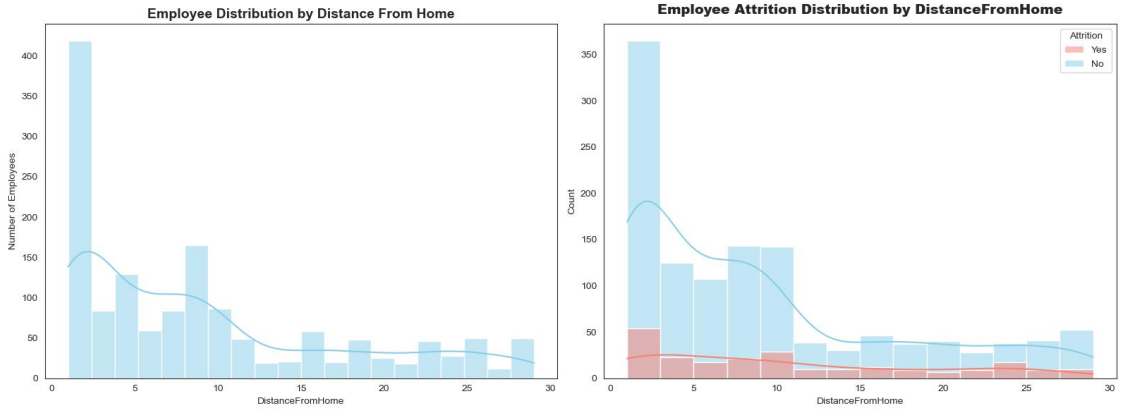


Findings & Insights:

Education Fields like Human Resources, Technical Degree and Marketing is having very high attrition rate. This may be because of work load as there are very few employees in these education fields compared to education field with low attrition rate.

6) Attrition vs Distance from Home

```
#Attition vs DistanceFromHome
plt.figure(figsize=(16, 6))

#  Employee distribution by DistanceFromHome
plt.subplot(1, 2, 1)
sns.histplot(df['DistanceFromHome'], bins=20, kde=True, color='skyblue')
plt.title('Employee Distribution by Distance From Home', fontsize=14, fontweight='bold')
plt.xlabel('DistanceFromHome')
plt.ylabel('Number of Employees')

#  Attrition distribution by DistanceFromHome

plt.subplot(1,2,2)
sns.histplot(x="DistanceFromHome",hue="Attrition",data=df,kde=True,palette=['salmon', 'skyblue'])
plt.title("Employee Attrition Distribution by DistanceFromHome",fontweight="black",size=14,pad=10)

plt.tight_layout()
plt.show()
```

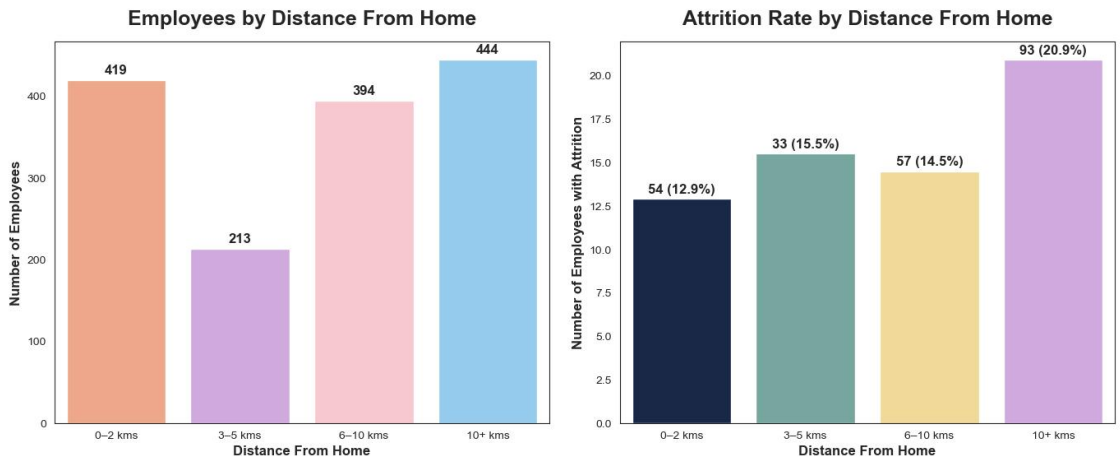


Findings:

Most of the employees are having a distance upto 10 km from home to the office.

```
df["DistanceFromHome"].describe().to_frame().T
                count     mean      std  min  25%  50%   75%   max
DistanceFromHome  1470.0  9.192517  8.106864  1.0  2.0  7.0  14.0  29.0

bin_edges = [0, 2, 5, 10, 30]
bin_labels = ['0–2 kms', '3–5 kms', '6–10 kms', '10+ kms']


distance_groups = pd.cut(df['DistanceFromHome'], bins=bin_edges, labels=bin_label
s)

# Employee Distribution by Distance ---
plt.figure(figsize=(14,6))
plt.subplot(1,2,1)
value_1 = distance_groups.value_counts().sort_index()
sns.barplot(x=value_1.index, y=value_1.values, palette=["#FFA07A", "#D4A1E7", "#F
FC0CB", "#87CEFA"])
plt.title("Employees by Distance From Home", fontweight="bold", pad=15, size=18)
plt.xlabel("Distance From Home", fontsize=12, fontweight="bold")
plt.ylabel("Number of Employees", fontsize=12, fontweight="bold")
for index, value in enumerate(value_1.values):
   plt.text(index, value + 5, value, ha="center", va="bottom", fontweight="bold", size
=12)

#  Attrition Rate by Distance ---
plt.subplot(1,2,2)
distance_groups_attrition = pd.cut(df[df["Attrition"] == "Yes"]["DistanceFromHome
"], bins=bin_edges, labels=bin_labels)
value_2 = distance_groups_attrition.value_counts().sort_index()
attrition_rate = np.round((value_2 / value_1) * 100, 1).values

sns.barplot(x=value_2.index, y=attrition_rate, palette=["#11264e","#6faea4","#FEE0
8B","#D4A1E7"])
```

```
plt.title("Attrition Rate by Distance From Home", fontweight="bold", pad=15, size=1
8)
plt.xlabel("Distance From Home", fontsize=12, fontweight="bold")
plt.ylabel("Number of Employees with Attrition", fontsize=12, fontweight="bold")
for i, rate in enumerate(attrition_rate):
    plt.text(i, rate + 0.2, f"{value_2.values[i]} ({rate}%)",
             ha="center", va="bottom", fontweight="bold", size=12)

plt.tight_layout(pad=2.0)
plt.show()
```



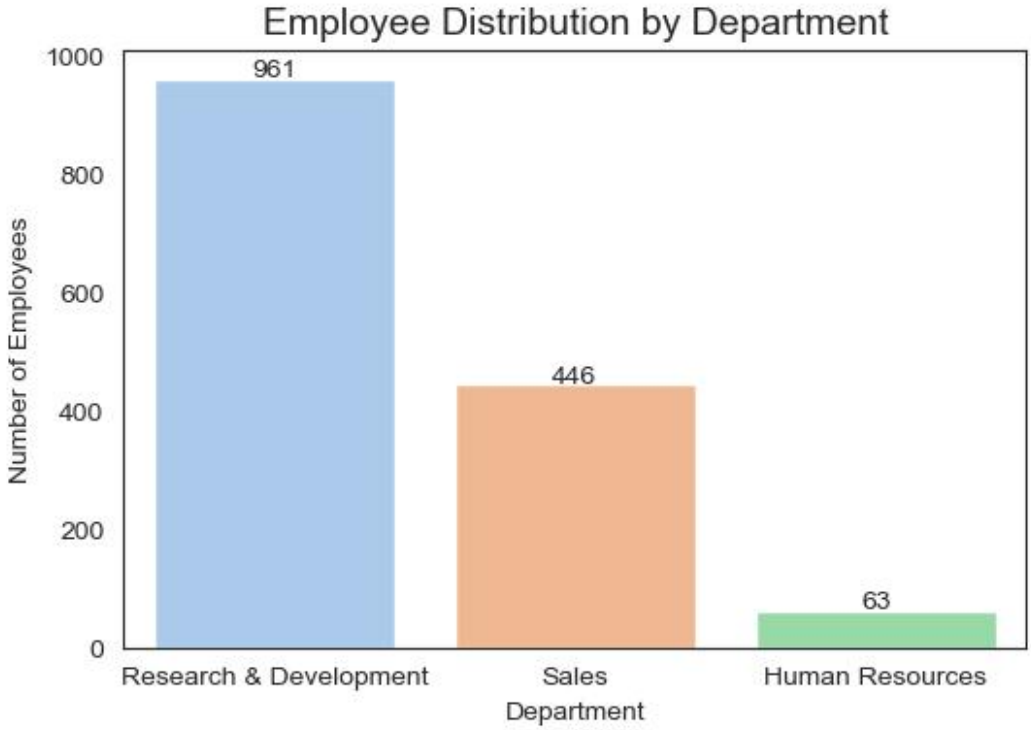
Findings & Insights:

From the bar chart we can understand that the employees who stay far away from the office are having attrition rate and they are more likely to leave the company. They have long travel time that causes fatigue, stress, and poor work-life balance. Overall, the longer or more stressful the daily commute, the higher the chance that employees will leave the company.

7) Attrition vs Department

```
df['Department'].value_counts()
Research & Development    961
Sales                     446
Human Resources           63
Name: Department, dtype: int64
Department_counts = df['Department'].value_counts()

plt.figure(figsize=(6,4))
sns.barplot(x=Department_counts.index, y=Department_counts.values, palette='pas
tel')


for i, value in enumerate(Department_counts.values):
    plt.text(i, value + 2, str(value), ha='center', fontweight='normal')
```

```
plt.title('Employee Distribution by Department', fontsize=14, fontweight='normal')
plt.xlabel('Department')
plt.ylabel('Number of Employees')
plt.show()
```

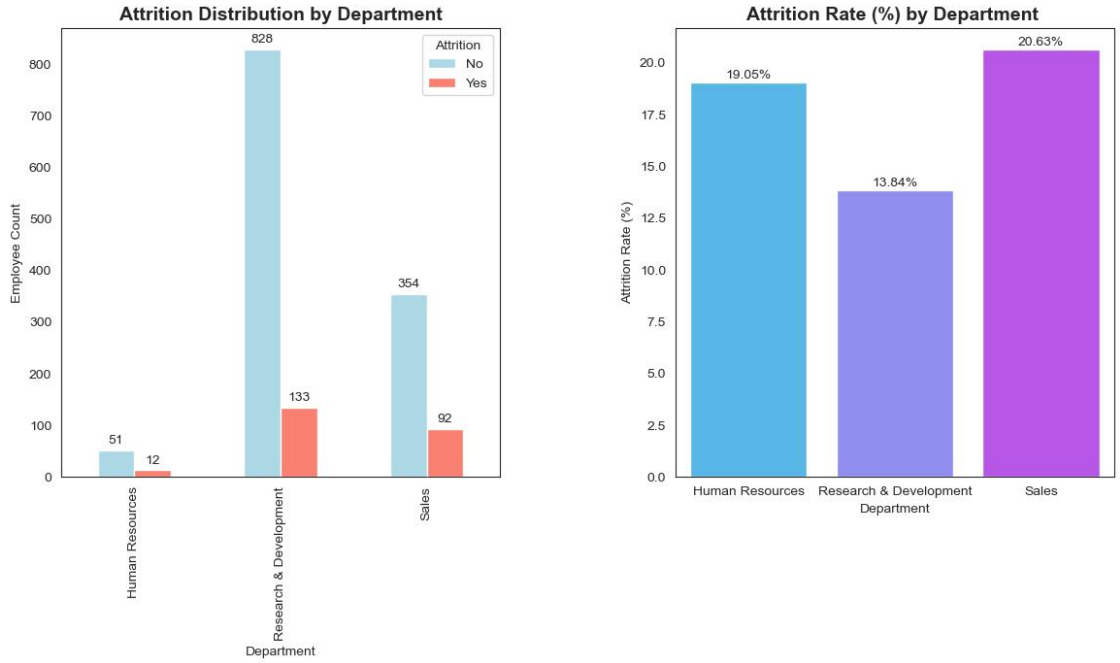


Findings:

Most of the employees are in the Research & Development department and least in the HR department.

```
df[df['Attrition'] == 'Yes']['Department'].value_counts()
Research & Development    133
Sales                      92
Human Resources            12
Name: Department, dtype: int64
calculate_attrition_rate(df, 'Department').round(2)
            Department  AttritionRate
0        Human Resources         19.05
1  Research & Development        13.84
2                Sales         20.63

fig, axes = plt.subplots(1, 2, figsize=(14, 6))
plt.subplots_adjust(wspace=0.4)

# Attrition Distribution (Yes/No) by Department
attrition_dept = df.groupby(['Department', 'Attrition']).size().unstack(fill_value=0)
attrition_dept.plot(kind='bar', ax=axes[0], color=['lightblue', 'salmon'])
axes[0].set_title('Attrition Distribution by Department', fontsize=14, fontweight='bol
```

```
d')
axes[0].set_xlabel('Department')
axes[0].set_ylabel('Employee Count')
axes[0].legend(title='Attrition')

for container in axes[0].containers:
    axes[0].bar_label(container, fmt='%d', label_type='edge', fontsize=10, padding=3)

# Attrition Rate (%) by Department
attrition_rate_dept = calculate_attrition_rate(df, 'Department')
sns.barplot(x='Department', y='AttritionRate', data=attrition_rate_dept, palette='coo
l', ax=axes[1])
axes[1].set_title('Attrition Rate (%) by Department ', fontsize=14, fontweight='bold')
axes[1].set_xlabel('Department')
axes[1].set_ylabel('Attrition Rate (%)')
for i, v in enumerate(attrition_rate_dept['AttritionRate']):
    axes[1].text(i, v + 0.2 , f'{v:.2f}%', ha='center', fontweight='normal')

plt.show()
```



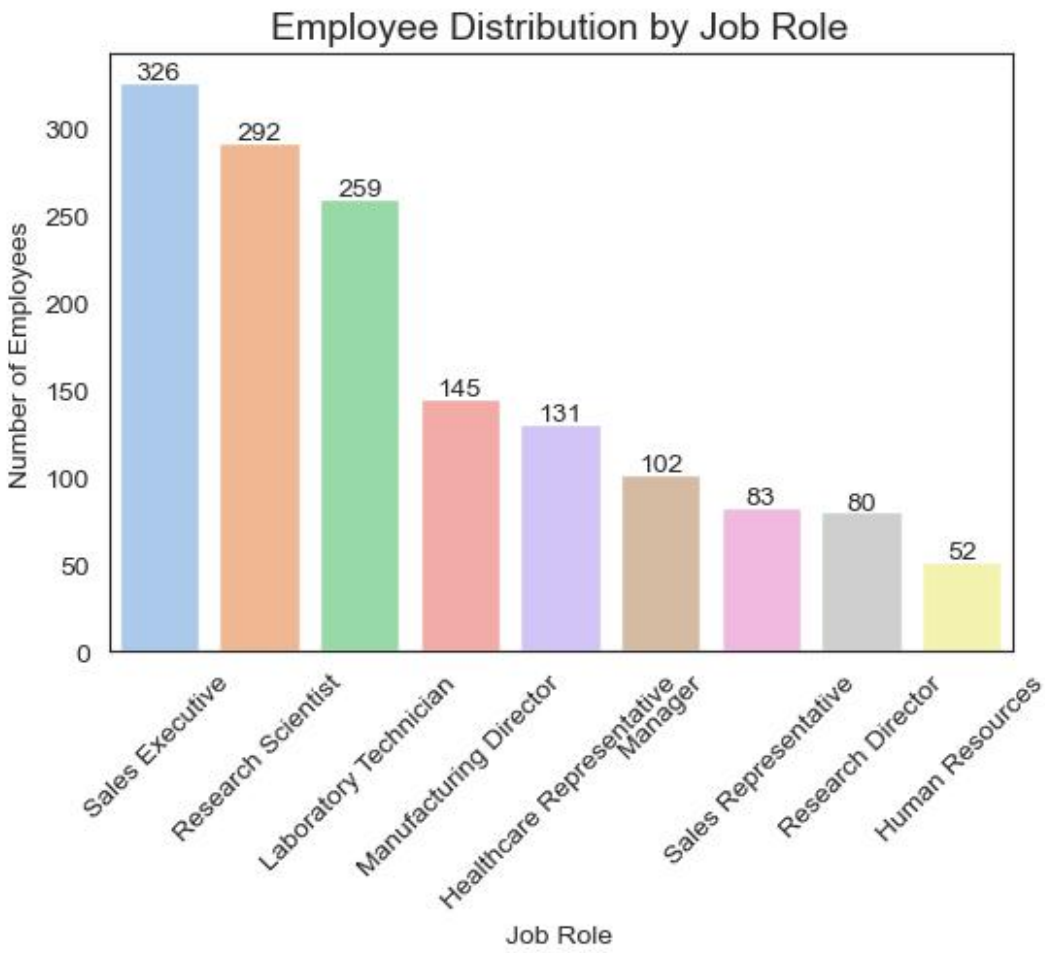
Findings & Insights:

Employees in the Sales department have high attrition rate and employees in the Research & Development department have low attrition rate. Even though most employees work in R&D, attrition rate is highest in Sales, followed by HR, and lowest in R&D. This shows that Sales roles are more stressful and unstable, while R&D jobs offer better stability and career security.

8) Attrition vs Job Role

```
df['JobRole'].value_counts()
```

```
Sales Executive            326
Research Scientist          292
Laboratory Technician        259
Manufacturing Director       145
Healthcare Representative    131
Manager                    102
Sales Representative         83
Research Director           80
Human Resources             52
Name: JobRole, dtype: int64
jobrole_counts = df['JobRole'].value_counts()

plt.figure(figsize=(6,4))
sns.barplot(x=jobrole_counts.index, y=jobrole_counts.values, palette='pastel')


for i, value in enumerate(jobrole_counts.values):
    plt.text(i, value + 2, str(value), ha='center', fontweight='normal')

plt.title('Employee Distribution by Job Role', fontsize=14, fontweight='normal')
plt.xlabel('Job Role')
plt.ylabel('Number of Employees')
plt.xticks(rotation=45)
plt.show()
```



Findings:

Most of the employees in the company are having job roles such as Sales Executive, Research Scientist, and Laboratory Technician.

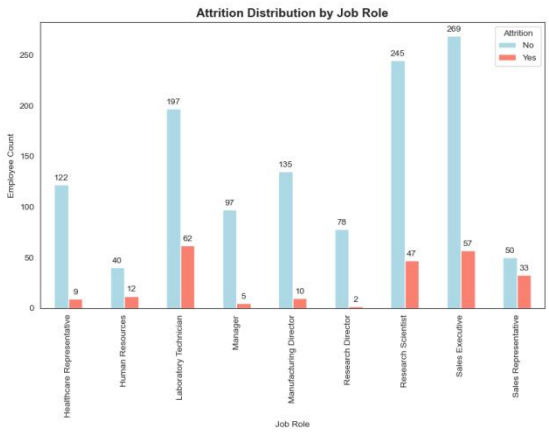
```
df[df['Attrition'] == 'Yes']['JobRole'].value_counts()
Laboratory Technician       62
Sales Executive             57
Research Scientist          47
Sales Representative        33
Human Resources             12
Manufacturing Director      10
Healthcare Representative    9
Manager                      5
Research Director            2
Name: JobRole, dtype: int64
```

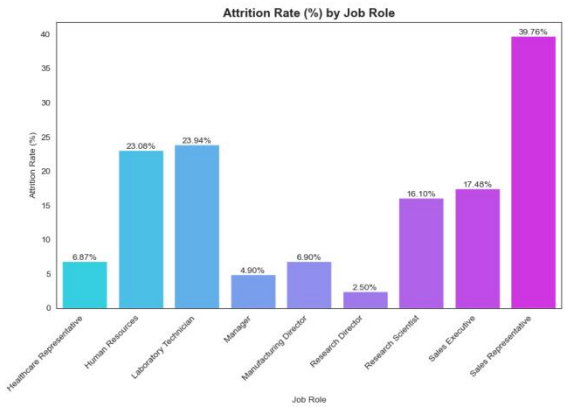
```
calculate_attrition_rate(df, 'JobRole').round(2)
```

```
                JobRole  AttritionRate
0  Healthcare Representative       6.87
1          Human Resources      23.08
```

```
2     Laboratory Technician        23.94
3                Manager        4.90
4    Manufacturing Director        6.90
5        Research Director        2.50
6       Research Scientist        16.10
7         Sales Executive        17.48
8      Sales Representative        39.76
```

```
fig, axes = plt.subplots(1, 2, figsize=(18, 7))
plt.subplots_adjust(wspace=0.4)

# Attrition Distribution (Yes/No) by Job Role
attrition_jobrole = df.groupby(['JobRole', 'Attrition']).size().unstack(fill_value=0)
attrition_jobrole.plot(kind='bar', ax=axes[0], color=['lightblue', 'salmon'])
axes[0].set_title('Attrition Distribution by Job Role', fontsize=14, fontweight='bold')
axes[0].set_xlabel('Job Role')
axes[0].set_ylabel('Employee Count')
axes[0].legend(title='Attrition')

for container in axes[0].containers:
    axes[0].bar_label(container, fmt='%d', label_type='edge', fontsize=10, padding=3)

# Attrition Rate (%) by Job Role
attrition_rate_jobrole = calculate_attrition_rate(df, 'JobRole')
sns.barplot(
    x='JobRole',
    y='AttritionRate',
    data=attrition_rate_jobrole,
    palette='cool',
    ax=axes[1]
)

axes[1].set_title('Attrition Rate (%) by Job Role', fontsize=14, fontweight='bold')
axes[1].set_xlabel('Job Role')
axes[1].set_ylabel('Attrition Rate (%)')


for i, v in enumerate(attrition_rate_jobrole['AttritionRate']):
    axes[1].text(i, v + 0.2, f'{v:.2f}%', ha='center', fontsize=10)


axes[1].set_xticklabels(axes[1].get_xticklabels(), rotation=45, ha='right')

plt.tight_layout()
plt.show()
```





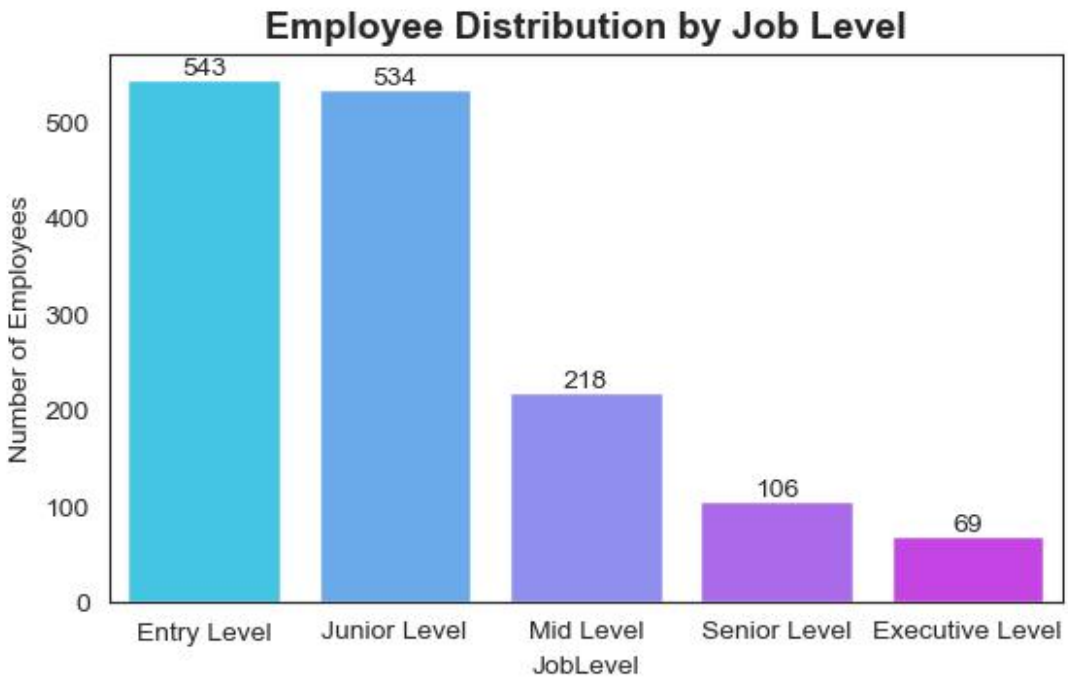
Findings & Insights:

Sales Representatives have the highest attrition because their jobs are target-driven, stressful, and high-pressure. Roles like Sales Representatives, Lab Technicians, and Human Resources have the highest attrition because they are more stressful, lower paid, and offer fewer growth opportunities. Senior roles such as Managers and Research Directors have the lowest attrition due to better pay, stability, and career security.

9)Attrition vs Job Level

```
df['JobLevel'].value_counts()
1    543
2    534
3    218
4    106
5     69
Name: JobLevel, dtype: int64

joblevel_labels = {1:"Entry Level",2:"Junior Level",3:"Mid Level",4:"Senior Level",5:"Executive Level"}


jl_counts = df['JobLevel'].value_counts().sort_index()

sns.set_style("white")

plt.figure(figsize=(6, 4))

sns.barplot(x=[joblevel_labels[i] for i in jl_counts.index],
        y=jl_counts.values,
        palette='cool')

plt.title('Employee Distribution by Job Level', fontsize=14, fontweight='bold')
plt.xlabel('JobLevel')
plt.ylabel('Number of Employees')
```

```
plt.grid(False)

for i, v in enumerate(jl_counts.values):
    plt.text(i, v + 5, str(v), ha='center', fontweight='normal')


plt.tight_layout(pad=2.0)
plt.show()
```

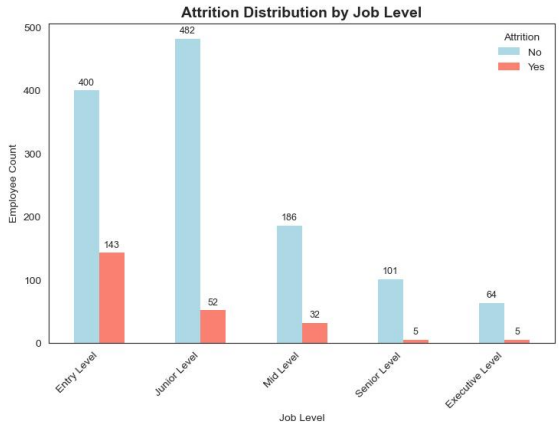


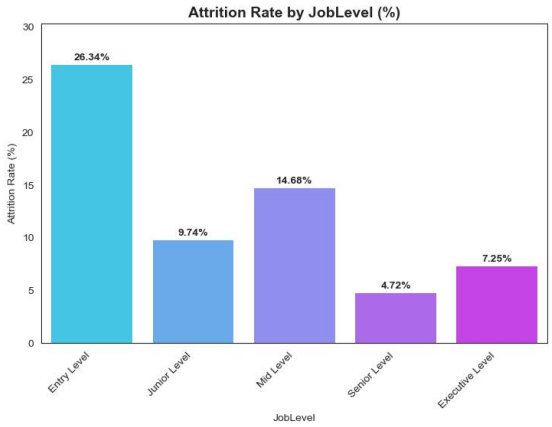
Findings:

Most of the employees in the company are in entry level and junior level positions.

```
df[df['Attrition'] == 'Yes']['JobLevel'].value_counts()
1    143
2     52
3     32
5      5
4      5
Name: JobLevel, dtype: int64

calculate_attrition_rate(df, 'JobLevel').round(2)

   JobLevel  AttritionRate
0         1          26.34
1         2           9.74
2         3          14.68
3         4           4.72
4         5           7.25
```

```
joblevel_labels = {1:"Entry Level",2:"Junior Level",3:"Mid Level",4:"Senior Level",5:"E
xecutive Level"}



sns.set_style("white")


fig, axes = plt.subplots(1, 2, figsize=(15, 6))
plt.subplots_adjust(wspace=0.4)

#  Attrition Distribution (Yes/No) by Job Level
attrition_jl = df.groupby(['JobLevel', 'Attrition']).size().unstack(fill_value=0)
attrition_jl.plot(kind='bar', ax=axes[0], color=['lightblue', 'salmon'], edgecolor='none')
axes[0].set_title('Attrition Distribution by Job Level', fontsize=14, fontweight='bold')
axes[0].set_xlabel('Job Level')
axes[0].set_ylabel('Employee Count')
axes[0].set_xticklabels([joblevel_labels[i] for i in attrition_jl.index], rotation=45, ha='
right')
axes[0].legend(title='Attrition', frameon=False)
axes[0].grid(False)
for container in axes[0].containers:
    axes[0].bar_label(container, fmt='%d', label_type='edge', fontsize=9, padding=3)

#  Attrition Rate (%) by Job Level
attrition_rate_jl = calculate_attrition_rate(df, 'JobLevel')
sns.barplot(x='JobLevel', y='AttritionRate', data=attrition_rate_jl, palette='cool', ax=a
xes[1], edgecolor='none')
axes[1].set_title('Attrition Rate by JobLevel (%)', fontsize=14, fontweight='bold')
axes[1].set_xlabel('JobLevel')
axes[1].set_ylabel('Attrition Rate (%)')
axes[1].set_xticklabels([joblevel_labels[i] for i in attrition_rate_jl['JobLevel']], rotatio
n=45, ha='right')
axes[1].set_ylim(0, attrition_rate_jl['AttritionRate'].max() * 1.15)
axes[1].grid(False)
for i, v in enumerate(attrition_rate_jl['AttritionRate']):
    axes[1].text(i, v + 0.5, f'{v:.2f}%', ha='center', fontweight='bold')

plt.tight_layout(pad=2.0)
plt.show()
```



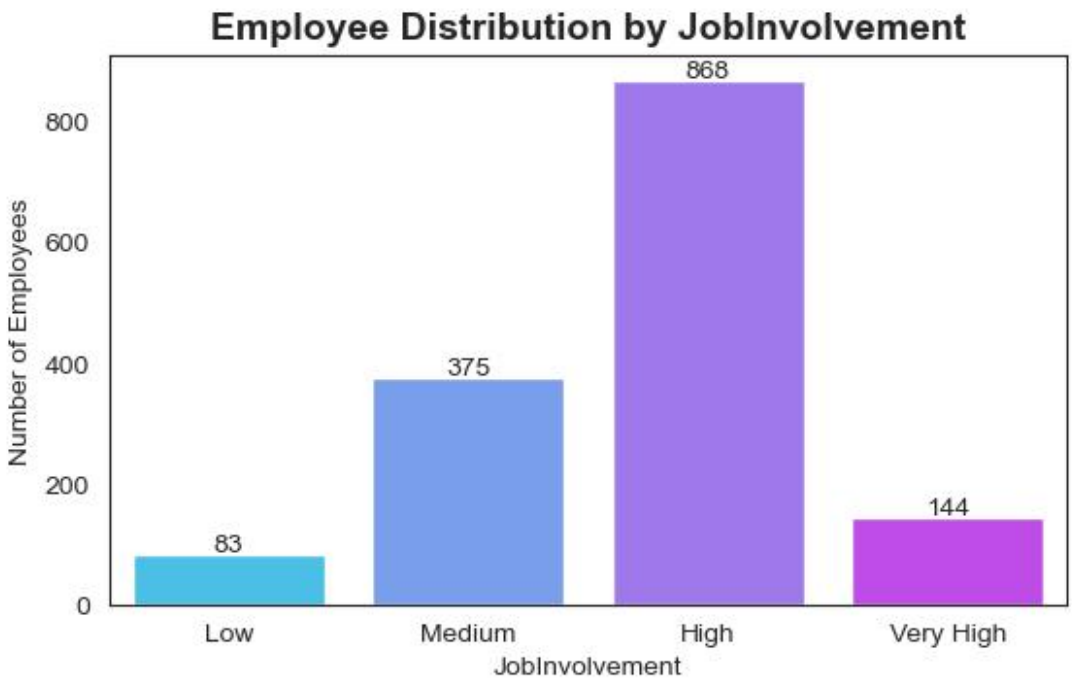


Findings & Insights:

Employees in the entry level positions are having high attrition rate. Entry-level employees leave the most because they are still exploring careers, receive lower pay, and look for better opportunities.

10) Attrition vs Job Involvement

```
df['JobInvolvement'].value_counts()
3    868
2    375
4    144
1     83
Name: JobInvolvement, dtype: int64

jobinvo_labels = {1:"Low",2:"Medium",3:"High",4:"Very High"}

jin_counts = df['JobInvolvement'].value_counts().sort_index()
sns.set_style("white")
plt.figure(figsize=(6, 4))
sns.barplot(x=[jobinvo_labels[i] for i in jin_counts.index],
        y=jin_counts.values,
        palette='cool')

plt.title('Employee Distribution by JobInvolvement', fontsize=14, fontweight='bold')
plt.xlabel('JobInvolvement')
plt.ylabel('Number of Employees')

plt.grid(False)

for i, v in enumerate(jin_counts.values):
   plt.text(i, v + 5, str(v), ha='center', fontweight='normal')


plt.tight_layout(pad=2.0)
plt.show()
```

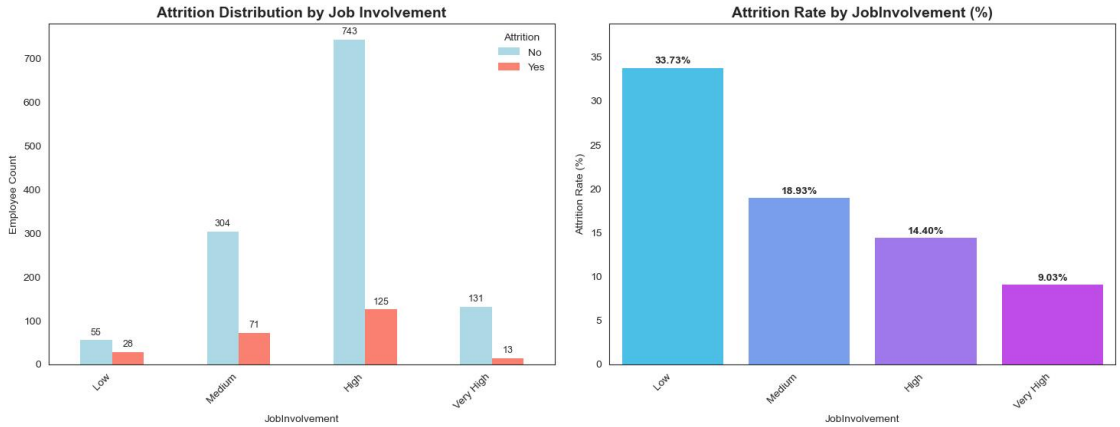


Findings & Insights:

Job involvement is the degree to which an employee identifies with their work and actively participates in it. For most employees the level is High.

```
df[df['Attrition'] == 'Yes']['JobInvolvement'].value_counts()
3    125
2     71
1     28
4     13
Name: JobInvolvement, dtype: int64
calculate_attrition_rate(df, 'JobInvolvement').round(2)
  JobInvolvement  AttritionRate
0              1          33.73
1              2          18.93
2              3          14.40
3              4           9.03

jobinvo_labels = {1:"Low",2:"Medium",3:"High",4:"Very High"}



sns.set_style("white")


fig, axes = plt.subplots(1, 2, figsize=(15, 6))
plt.subplots_adjust(wspace=0.4)

#  Attrition Distribution (Yes/No) by JobInvolvement
```

```
attrition_jinv = df.groupby(['JobInvolvement', 'Attrition']).size().unstack(fill_value=0)
attrition_jinv.plot(kind='bar', ax=axes[0], color=['lightblue', 'salmon'], edgecolor='no
ne')
axes[0].set_title('Attrition Distribution by Job Involvement', fontsize=14, fontweight=
'bold')
axes[0].set_xlabel('JobInvolvement')
axes[0].set_ylabel('Employee Count')
axes[0].set_xticklabels([jobinvo_labels[i] for i in attrition_jinv.index], rotation=45, ha
='right')
axes[0].legend(title='Attrition', frameon=False)
axes[0].grid(False)
for container in axes[0].containers:
    axes[0].bar_label(container, fmt='%d', label_type='edge', fontsize=9, padding=3)

# Attrition Rate (%) by JobInvolvement
attrition_rate_jinv = calculate_attrition_rate(df, 'JobInvolvement')
sns.barplot(x='JobInvolvement', y='AttritionRate', data=attrition_rate_jinv, palette='c
ool', ax=axes[1], edgecolor='none')
axes[1].set_title('Attrition Rate by JobInvolvement (%)', fontsize=14, fontweight='bol
d')
axes[1].set_xlabel('JobInvolvement')
axes[1].set_ylabel('Attrition Rate (%)')
axes[1].set_xticklabels([jobinvo_labels[i] for i in attrition_rate_jinv['JobInvolvement
']], rotation=45, ha='right')
axes[1].set_ylim(0, attrition_rate_jinv['AttritionRate'].max() * 1.15)
axes[1].grid(False)
for i, v in enumerate(attrition_rate_jinv['AttritionRate']):
    axes[1].text(i, v + 0.5, f'{v:.2f}%', ha='center', fontweight='bold')

plt.tight_layout(pad=2.0)
plt.show()
```



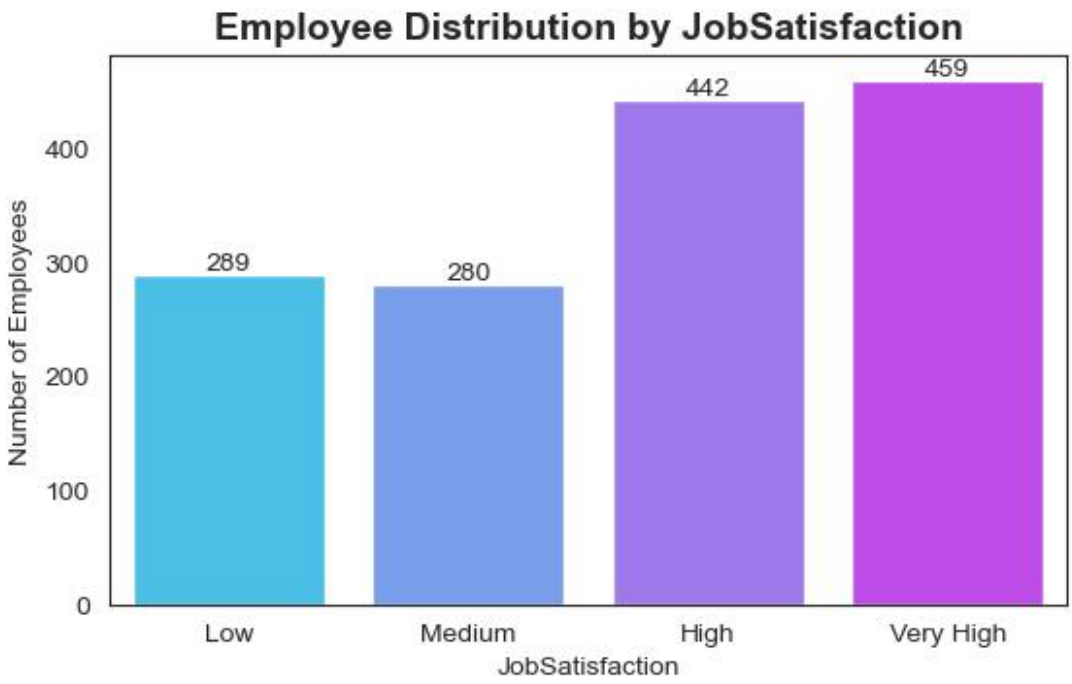
Findings & Insights:

Employees with a low level of Job Involvement are having high attrition rate. We can observe from the chart that as the level increases attrition rate decreases. It means

that the more engaged and involved employees are in their work, the less likely they are to resign.

11) Attrition vs Job Satisfaction

```
df['JobSatisfaction'].value_counts()
4    459
3    442
1    289
2    280
Name: JobSatisfaction, dtype: int64

jobsat_labels = {1:"Low",2:"Medium",3:"High",4:"Very High"}


jsat_counts = df['JobSatisfaction'].value_counts().sort_index()

sns.set_style("white")

plt.figure(figsize=(6, 4))

sns.barplot(x=[jobsat_labels[i] for i in jsat_counts.index],
            y=jsat_counts.values,
            palette='cool')

plt.title('Employee Distribution by JobSatisfaction', fontsize=14, fontweight='bold')
plt.xlabel('JobSatisfaction')
plt.ylabel('Number of Employees')

plt.grid(False)

for i, v in enumerate(jsat_counts.values):
    plt.text(i, v + 5, str(v), ha='center', fontweight='normal')


plt.tight_layout(pad=2.0)
plt.show()
```



Findings:

Most of the employees in the company are having high levels of job satisfaction.

```
df[df['Attrition'] == 'Yes']['JobSatisfaction'].value_counts()
3    73
1    66
4    52
2    46
Name: JobSatisfaction, dtype: int64
```

```
calculate_attrition_rate(df, 'JobSatisfaction').round(2)
```

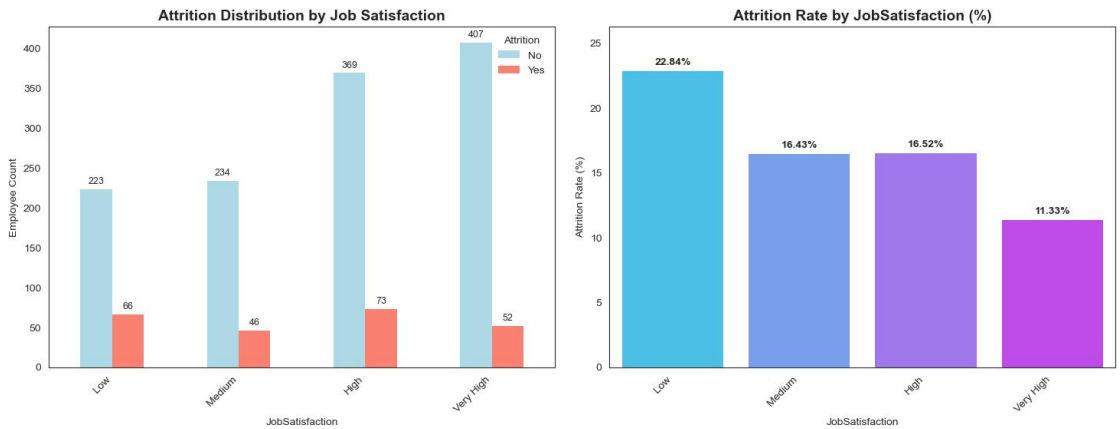
	JobSatisfaction	AttritionRate
0	1	22.84
1	2	16.43
2	3	16.52
3	4	11.33

```
jobsat_labels = {1:"Low",2:"Medium",3:"High",4:"Very High"}
```

```
sns.set_style("white")
```

```
fig, axes = plt.subplots(1, 2, figsize=(15, 6))
plt.subplots_adjust(wspace=0.4)
```

```
# Attrition Distribution (Yes/No) by JobSatisfaction
attrition_jsat = df.groupby(['JobSatisfaction', 'Attrition']).size().unstack(fill_value=0)
attrition_jsat.plot(kind='bar', ax=axes[0], color=['lightblue', 'salmon'], edgecolor='no
ne')
axes[0].set_title('Attrition Distribution by Job Satisfaction', fontsize=14, fontweight='
bold')
axes[0].set_xlabel('JobSatisfaction')
axes[0].set_ylabel('Employee Count')
axes[0].set_xticklabels([jobsat_labels[i] for i in attrition_jsat.index], rotation=45, ha='
right')
axes[0].legend(title='Attrition', frameon=False)
axes[0].grid(False)
for container in axes[0].containers:
    axes[0].bar_label(container, fmt='%d', label_type='edge', fontsize=9, padding=3)

# Attrition Rate (%) by JobSatisfaction
attrition_rate_jsat = calculate_attrition_rate(df, 'JobSatisfaction')
sns.barplot(x='JobSatisfaction', y='AttritionRate', data=attrition_rate_jsat, palette='c
ool', ax=axes[1], edgecolor='none')
axes[1].set_title('Attrition Rate by JobSatisfaction (%)', fontsize=14, fontweight='bold
')
axes[1].set_xlabel('JobSatisfaction')
axes[1].set_ylabel('Attrition Rate (%)')
axes[1].set_xticklabels([jobsat_labels[i] for i in attrition_rate_jsat['JobSatisfaction']],
rotation=45, ha='right')
axes[1].set_ylim(0, attrition_rate_jsat['AttritionRate'].max() * 1.15)
axes[1].grid(False)
for i, v in enumerate(attrition_rate_jsat['AttritionRate']):
    axes[1].text(i, v + 0.5, f'{v:.2f}%', ha='center', fontweight='bold')

plt.tight_layout(pad=2.0)
plt.show()
```

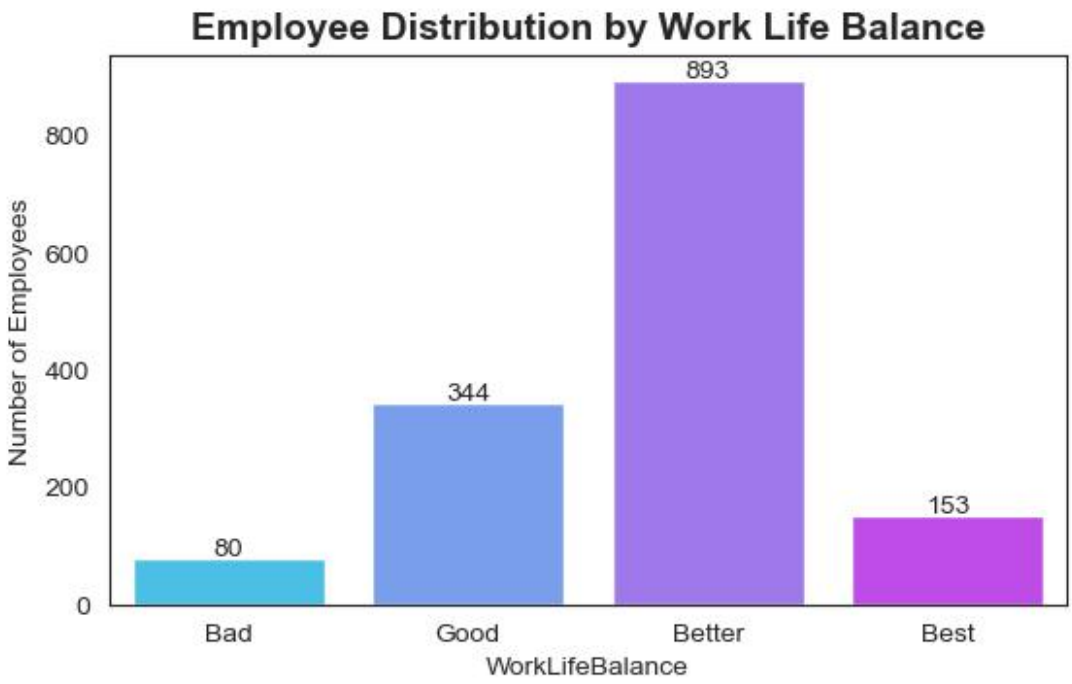


Findings & Insights:

Employees with low levels of job satisfaction have high attrition rate. As the level increases attrition rate decreases. Employees who are not satisfied with their jobs tend to leave the company more.

12) Attrition vs Work-Life Balance

```
df['WorkLifeBalance'].value_counts()
3    893
2    344
4    153
1     80
Name: WorkLifeBalance, dtype: int64

wlb_labels = {1:"Bad",2:"Good",3:"Better",4:"Best"}


wlb_counts = df['WorkLifeBalance'].value_counts().sort_index()

sns.set_style("white")

plt.figure(figsize=(6, 4))

sns.barplot(x=[wlb_labels[i] for i in wlb_counts.index],
        y=wlb_counts.values,
        palette='cool')

plt.title('Employee Distribution by Work Life Balance', fontsize=14, fontweight='bold')
plt.xlabel('WorkLifeBalance')
plt.ylabel('Number of Employees')

plt.grid(False)

for i, v in enumerate(wlb_counts.values):
   plt.text(i, v + 5, str(v), ha='center', fontweight='normal')


plt.tight_layout(pad=2.0)
plt.show()
```



Findings:

Most of the employees in the company are having a better work-life balance.

```
df[df['Attrition'] == 'Yes']['WorkLifeBalance'].value_counts()
3    127
2     58
4     27
1     25
Name: WorkLifeBalance, dtype: int64
```

```
calculate_attrition_rate(df, 'WorkLifeBalance').round(2)
```

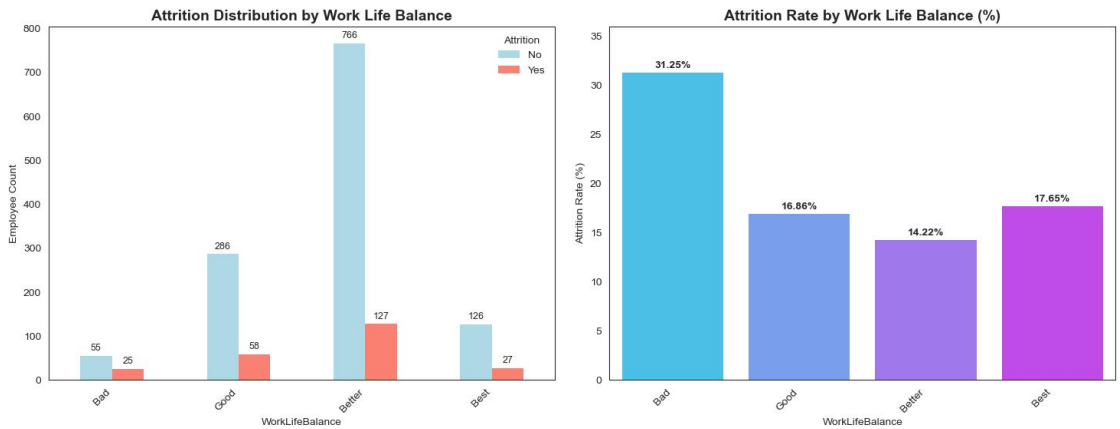
	WorkLifeBalance	AttritionRate
0	1	31.25
1	2	16.86
2	3	14.22
3	4	17.65

```
wlb_labels = {1:"Bad",2:"Good",3:"Better",4:"Best"}
```

```
sns.set_style("white")
```

```
fig, axes = plt.subplots(1, 2, figsize=(15, 6))
plt.subplots_adjust(wspace=0.4)
```

```
# Attrition Distribution (Yes/No) by WorkLifeBalance
attrition_wlb = df.groupby(['WorkLifeBalance', 'Attrition']).size().unstack(fill_value=0)
attrition_wlb.plot(kind='bar', ax=axes[0], color=['lightblue', 'salmon'], edgecolor='no
ne')
axes[0].set_title('Attrition Distribution by Work Life Balance', fontsize=14, fontweigh
t='bold')
axes[0].set_xlabel('WorkLifeBalance')
axes[0].set_ylabel('Employee Count')
axes[0].set_xticklabels([wlb_labels[i] for i in attrition_wlb.index], rotation=45, ha='ri
ght')
axes[0].legend(title='Attrition', frameon=False)
axes[0].grid(False)
for container in axes[0].containers:
    axes[0].bar_label(container, fmt='%d', label_type='edge', fontsize=9, padding=3)

# Attrition Rate (%) by WorkLifeBalance
attrition_rate_wlb = calculate_attrition_rate(df, 'WorkLifeBalance')
sns.barplot(x='WorkLifeBalance', y='AttritionRate', data=attrition_rate_wlb, palette='
cool', ax=axes[1], edgecolor='none')
axes[1].set_title('Attrition Rate by Work Life Balance (%)', fontsize=14, fontweight='b
old')
axes[1].set_xlabel('WorkLifeBalance')
axes[1].set_ylabel('Attrition Rate (%)')
axes[1].set_xticklabels([wlb_labels[i] for i in attrition_rate_wlb['WorkLifeBalance']], r
otation=45, ha='right')
axes[1].set_ylim(0, attrition_rate_wlb['AttritionRate'].max() * 1.15)
axes[1].grid(False)
for i, v in enumerate(attrition_rate_wlb['AttritionRate']):
    axes[1].text(i, v + 0.5, f'{v:.2f}%', ha='center', fontweight='bold')

plt.tight_layout(pad=2.0)
plt.show()
```

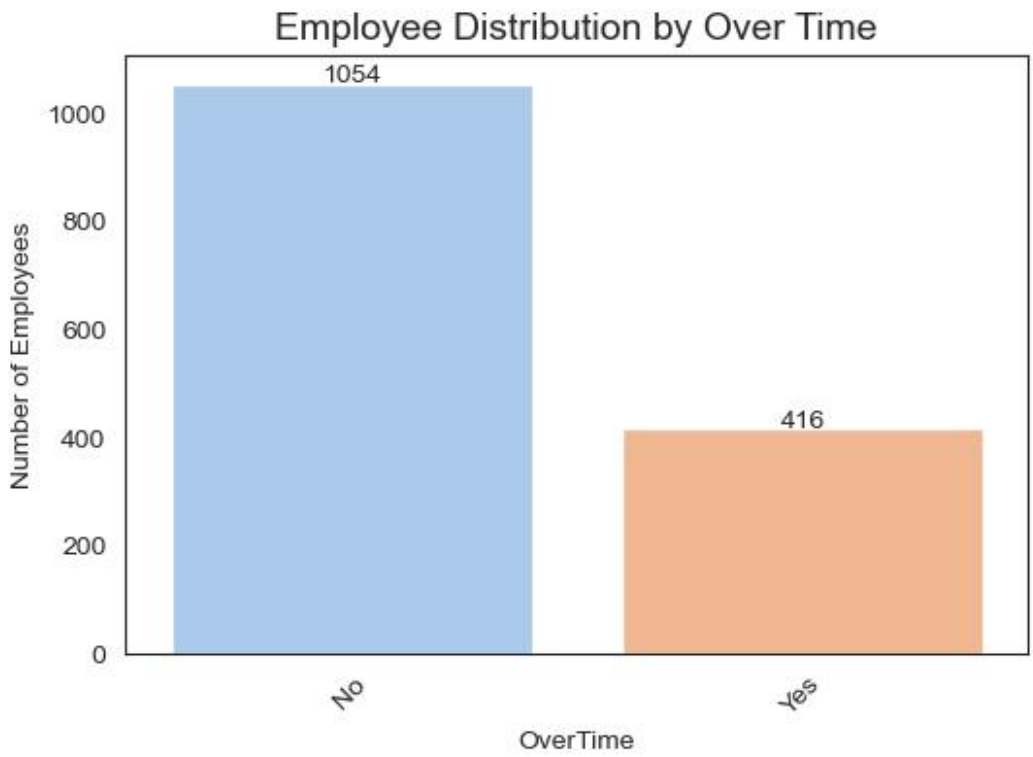


Findings:

Employees with a bad work-life balance have higher attrition rate. They tends to leave the company more.

13) Attrition vs OverTime

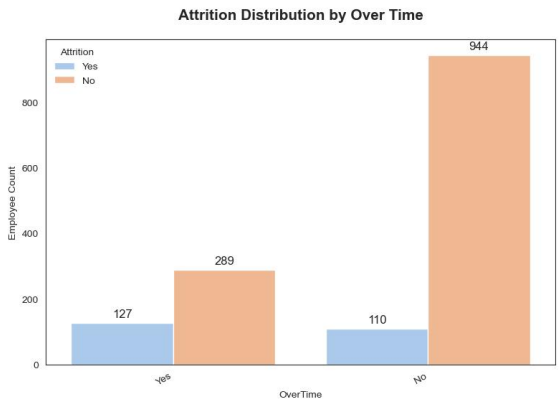
```
df['OverTime'].value_counts()
No     1054
Yes     416
Name: OverTime, dtype: int64
ot_counts = df['OverTime'].value_counts()

plt.figure(figsize=(6,4))
sns.barplot(x=ot_counts.index, y=ot_counts.values, palette='pastel')


for i, value in enumerate(ot_counts.values):
   plt.text(i, value + 2, str(value), ha='center', fontweight='normal')

plt.title('Employee Distribution by Over Time', fontsize=14, fontweight='normal')
plt.xlabel('OverTime')
plt.ylabel('Number of Employees')
plt.xticks(rotation=45)
plt.show()
```

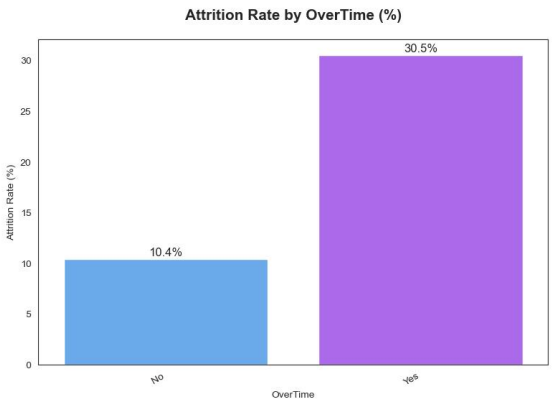


Findings:

Most of the employees doesn't work overtime.

```
df[df['Attrition'] == 'Yes']['OverTime'].value_counts()
Yes    127
No     110
Name: OverTime, dtype: int64
calculate_attrition_rate(df, 'OverTime').round(2)
  OverTime  AttritionRate
0      No          10.44
1     Yes          30.53

sns.set_style("white")


fig, axes = plt.subplots(1, 2, figsize=(16, 6))
plt.subplots_adjust(wspace=0.4)

#  Attrition Distribution (Yes/No) by OverTime ---
sns.countplot(data=df, x='OverTime', hue='Attrition', palette='pastel', ax=axes[0])
axes[0].set_title('Attrition Distribution by Over Time', fontsize=15, fontweight='bold',
 pad=20)
axes[0].set_xlabel('OverTime')
axes[0].set_ylabel('Employee Count')
axes[0].set_xticklabels(axes[0].get_xticklabels(), rotation=30, ha='right')
axes[0].grid(False)
axes[0].legend(title='Attrition', frameon=False)


for container in axes[0].containers:
    axes[0].bar_label(container, fmt='%d', label_type='edge', fontsize=12, padding=3)

#  Attrition Rate (%) by OverTime ---
attrition_rate_ot = calculate_attrition_rate(df, 'OverTime')

sns.barplot(x='OverTime', y='AttritionRate', data=attrition_rate_ot,
            palette='cool', ax=axes[1])
axes[1].set_title('Attrition Rate by OverTime (%)', fontsize=15, fontweight='bold', pa
d=20)
axes[1].set_xlabel('OverTime')
axes[1].set_ylabel('Attrition Rate (%)')
axes[1].set_xticklabels(attrition_rate_ot['OverTime'], rotation=30, ha='right')
axes[1].grid(False)


for i, v in enumerate(attrition_rate_ot['AttritionRate']):
    axes[1].text(i, v + 0.3, f'{v:.1f}%', ha='center', fontsize=12)

plt.tight_layout(pad=2.0)
plt.show()
```



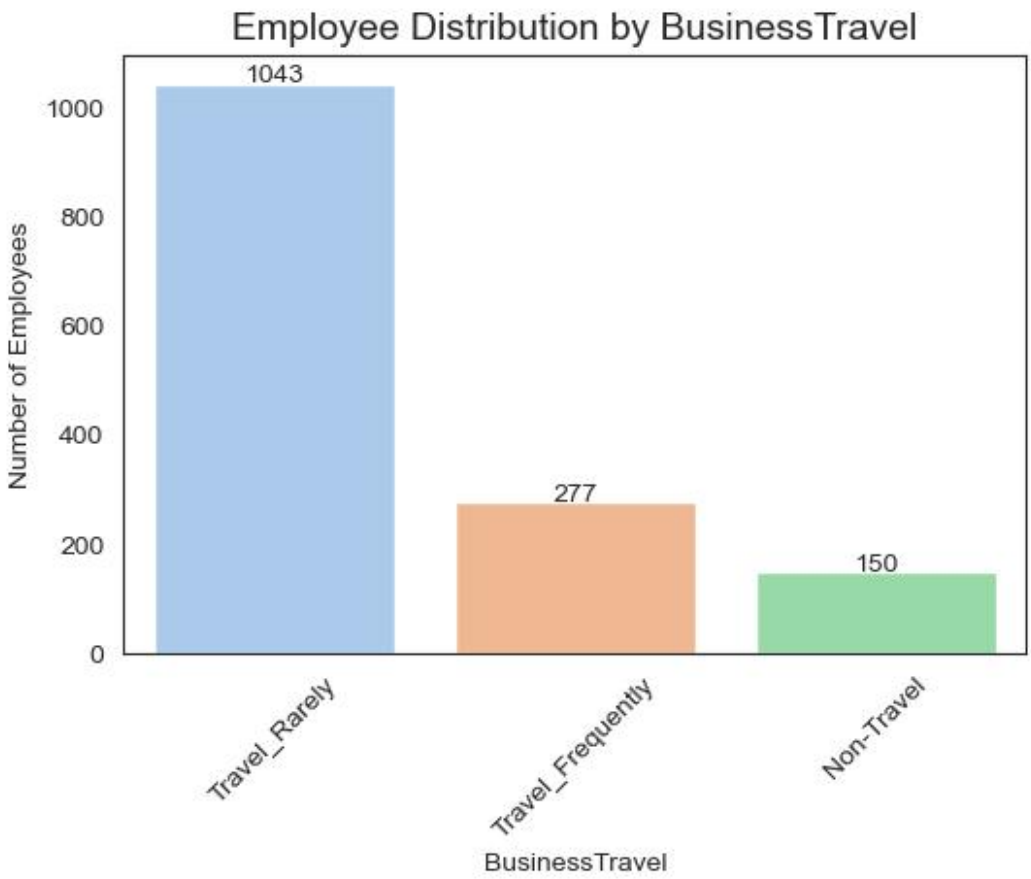


Findings:

Employees who are working overtime or have to work frequent overtimes are having higher attrition rates. They tend to leave the company.

14) Attrition vs Business Travel

```
df['BusinessTravel'].value_counts()
Travel_Rarely       1043
Travel_Frequently    277
Non-Travel           150
Name: BusinessTravel, dtype: int64
bt_counts = df['BusinessTravel'].value_counts()

plt.figure(figsize=(6,4))
sns.barplot(x=bt_counts.index, y=bt_counts.values, palette='pastel')


for i, value in enumerate(bt_counts.values):
    plt.text(i, value + 2, str(value), ha='center', fontweight='normal')

plt.title('Employee Distribution by BusinessTravel', fontsize=14, fontweight='normal')
plt.xlabel('BusinessTravel')
plt.ylabel('Number of Employees')
plt.xticks(rotation=45)
plt.show()
```



Findings:

Most of the employees in the company Travel Rarely.

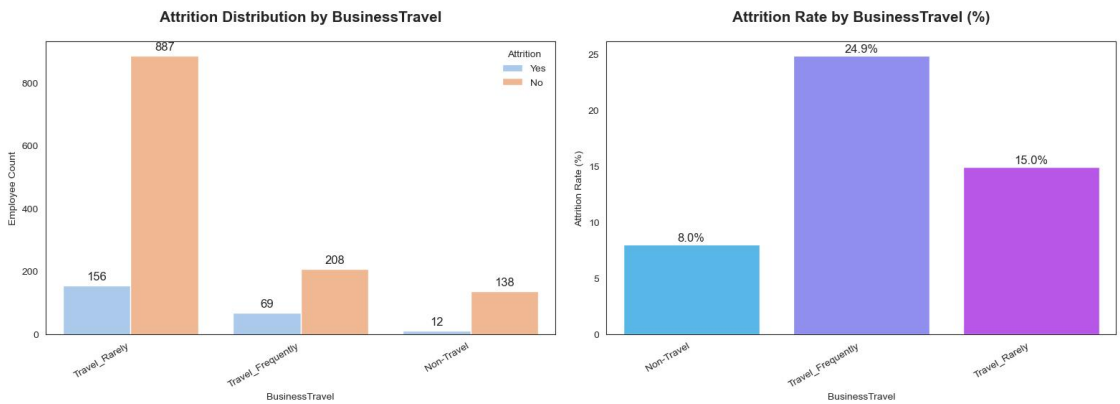
```
df[df['Attrition'] == 'Yes']['BusinessTravel'].value_counts()
Travel_Rarely       156
Travel_Frequently    69
Non-Travel           12
Name: BusinessTravel, dtype: int64
```

```
calculate_attrition_rate(df, 'BusinessTravel').round(2)
```

```
     BusinessTravel  AttritionRate
0        Non-Travel           8.00
1  Travel_Frequently         24.91
2      Travel_Rarely         14.96
```

```
sns.set_style("white")


fig, axes = plt.subplots(1, 2, figsize=(16, 6))
plt.subplots_adjust(wspace=0.4)

# Attrition Distribution (Yes/No) by BusinessTravel ---
```

```
sns.countplot(data=df, x='BusinessTravel', hue='Attrition', palette='pastel', ax=axes
[0])
axes[0].set_title('Attrition Distribution by BusinessTravel', fontsize=15, fontweight='b
old', pad=20)
axes[0].set_xlabel('BusinessTravel')
axes[0].set_ylabel('Employee Count')
axes[0].set_xticklabels(axes[0].get_xticklabels(), rotation=30, ha='right')
axes[0].grid(False)
axes[0].legend(title='Attrition', frameon=False)


for container in axes[0].containers:
    axes[0].bar_label(container, fmt='%d', label_type='edge', fontsize=12, padding=3)

# Attrition Rate (%) by BusinessTravel ---
attrition_rate_bt = calculate_attrition_rate(df, 'BusinessTravel')

sns.barplot(x='BusinessTravel', y='AttritionRate', data=attrition_rate_bt,
            palette='cool', ax=axes[1])
axes[1].set_title('Attrition Rate by BusinessTravel (%)', fontsize=15, fontweight='bold
', pad=20)
axes[1].set_xlabel('BusinessTravel')
axes[1].set_ylabel('Attrition Rate (%)')
axes[1].set_xticklabels(attrition_rate_bt['BusinessTravel'], rotation=30, ha='right')
axes[1].grid(False)


for i, v in enumerate(attrition_rate_bt['AttritionRate']):
    axes[1].text(i, v + 0.3, f'{v:.1f}%', ha='center', fontsize=12)

plt.tight_layout(pad=2.0)
plt.show()
```



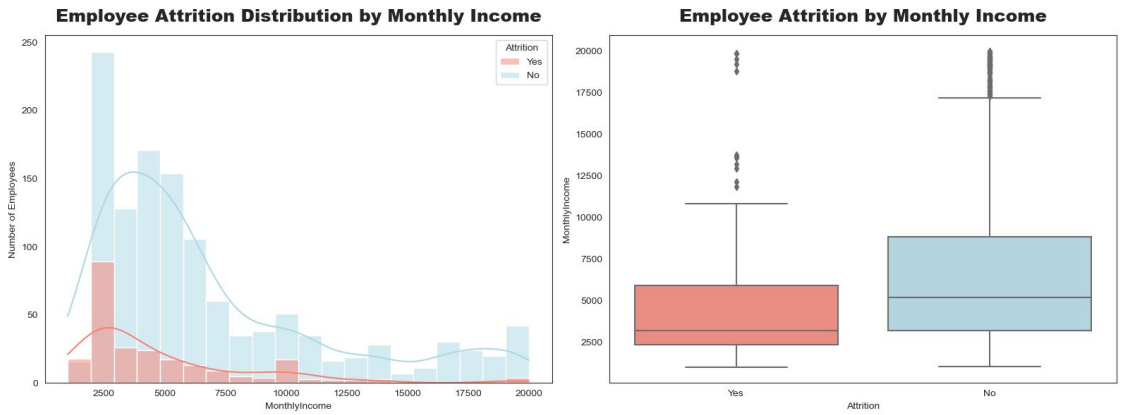
Findings & Insights:

High attrition rates can be observed by those employees who Travels Frequently. Low attrition rates can be observed by those employees who are Non-Travel.

Frequent business travel increases stress and work-life imbalance, leading to higher resignations.

15) Attrition vs Monthly Income

```
plt.figure(figsize=(16, 6))

# Employee distribution by Monthly Income
plt.subplot(1, 2, 1)
sns.histplot(x="MonthlyIncome",hue="Attrition",data=df,kde=True,palette=['salmon',
 'lightblue'])
plt.title("Employee Attrition Distribution by Monthly Income",fontweight="black",siz
e=18,pad=14)
plt.xlabel('MonthlyIncome')
plt.ylabel('Number of Employees')

# Attrition distribution by Monthly Income

plt.subplot(1,2,2)
sns.boxplot(x="Attrition",y="MonthlyIncome",data=df,palette=["salmon","lightblue
"])
plt.title("Employee Attrition by Monthly Income",fontweight="black",size=18,pad=14)

plt.tight_layout()
plt.show()
```



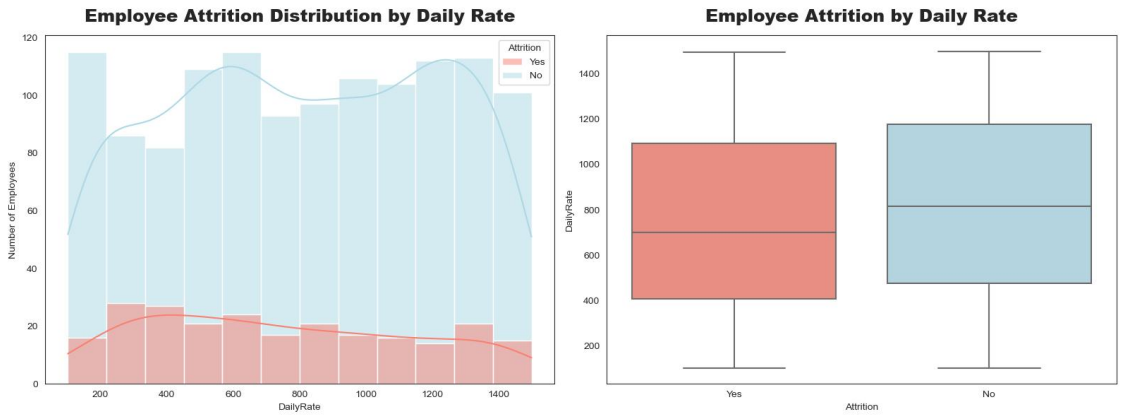
Findings:

Employees with lower monthly income are having high attrition and they are more likely to leave the company.

16) Attrition vs Daily Rate

```
plt.figure(figsize=(16, 6))

# Employee distribution by DailyRate
plt.subplot(1, 2, 1)
sns.histplot(x="DailyRate",hue="Attrition",data=df,kde=True,palette=['salmon', 'light
```

```
blue'])
plt.title("Employee Attrition Distribution by Daily Rate",fontweight="black",size=18,pad=14)
plt.xlabel('DailyRate')
plt.ylabel('Number of Employees')

# Attrition distribution by DailyRate

plt.subplot(1,2,2)
sns.boxplot(x="Attrition",y="DailyRate",data=df,palette=["salmon","lightblue"])
plt.title("Employee Attrition by Daily Rate",fontweight="black",size=18,pad=14)

plt.tight_layout()
plt.show()
```



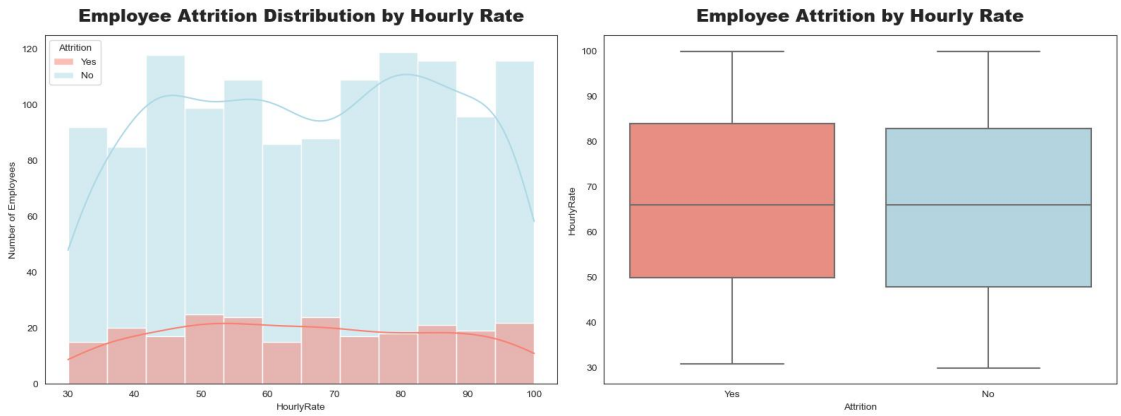
Findings:

Employees with low daily rate tend to have high attrition and they are more likely to leave the company.

17) Attrition vs Hourly Rate

```
plt.figure(figsize=(16, 6))

# Employee distribution by HourlyRate
plt.subplot(1, 2, 1)
sns.histplot(x="HourlyRate",hue="Attrition",data=df,kde=True,palette=['salmon', 'lightblue'])
plt.title("Employee Attrition Distribution by Hourly Rate",fontweight="black",size=18,pad=14)
plt.xlabel('HourlyRate')
plt.ylabel('Number of Employees')

#  Attrition distribution by HourlyRate

plt.subplot(1,2,2)
sns.boxplot(x="Attrition",y="HourlyRate",data=df,palette=["salmon","lightblue"])
```

```
plt.title("Employee Attrition by Hourly Rate",fontweight="black",size=18,pad=14)

plt.tight_layout()
plt.show()
```

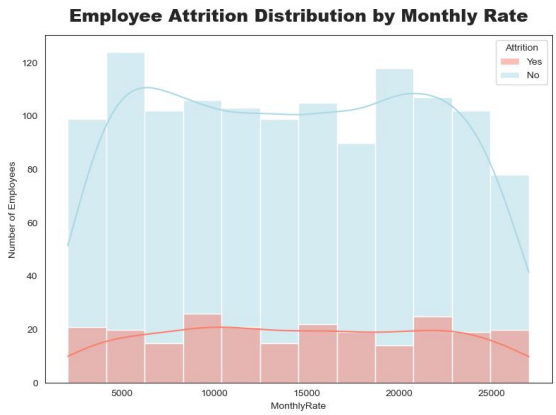


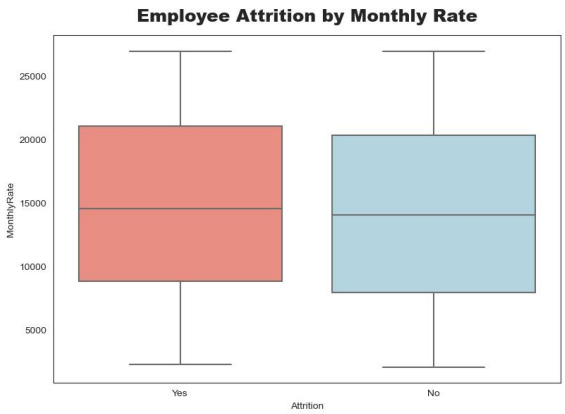
Findings:

The criteria hourly rate doesn't seem to affect the employee attrition.

18) Attrition vs Monthly Rate

```
plt.figure(figsize=(16, 6))

# Employee distribution by Monthly Rate
plt.subplot(1, 2, 1)
sns.histplot(x="MonthlyRate",hue="Attrition",data=df,kde=True,palette=['salmon', 'li
ghtblue'])
plt.title("Employee Attrition Distribution by Monthly Rate",fontweight="black",size=
18,pad=14)
plt.xlabel('MonthlyRate')
plt.ylabel('Number of Employees')

# Attrition distribution by MonthlyRate

plt.subplot(1,2,2)
sns.boxplot(x="Attrition",y="MonthlyRate",data=df,palette=["salmon","lightblue"])
plt.title("Employee Attrition by Monthly Rate",fontweight="black",size=18,pad=14)

plt.tight_layout()
plt.show()
```



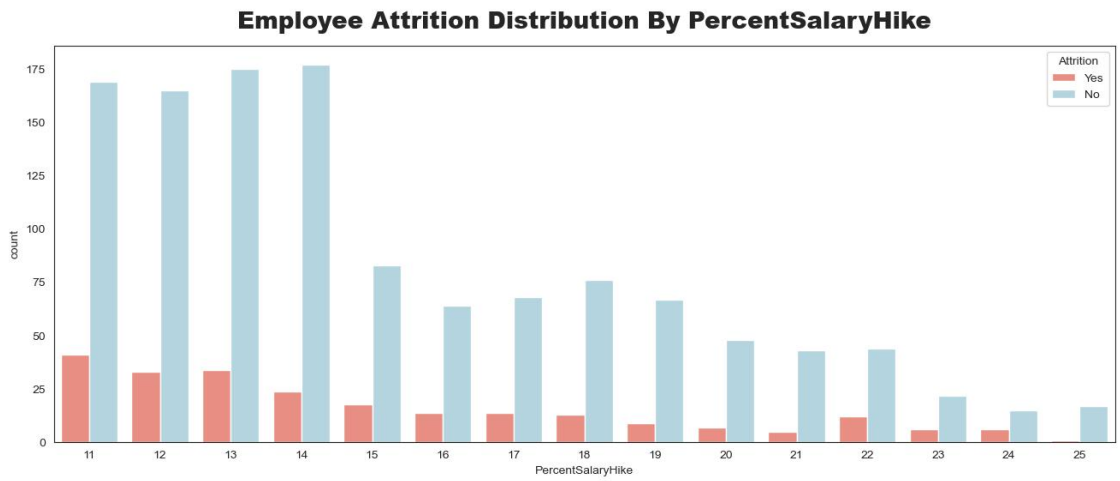


Findings:

The criteria monthly rate doesn't seem to affect the employee attrition.

19) Attrition vs Percent Salary Hike

```
#Employee Attrition Distribution By PercentSalaryHike
plt.figure(figsize=(16,6))
sns.countplot(x="PercentSalaryHike", hue="Attrition", data=df, palette=["salmon","lightblue"])
plt.title("Employee Attrition Distribution By PercentSalaryHike",fontweight="black",size=20,pad=15)
plt.show()
```



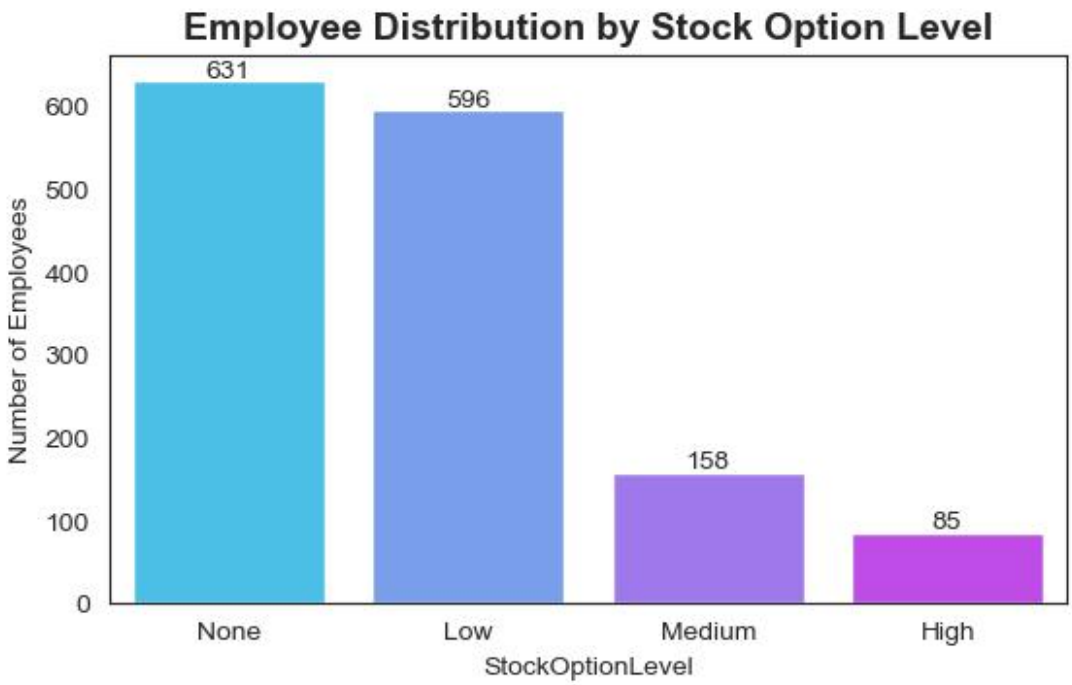
Findings & Insights:

Most of the employees are getting a salary hike in the range 11% to 14%. Employees with low salary hike tend to leave the company. And employees with high salary hike are likely to stay.

20) Attrition vs Stock Option Level

```
df['StockOptionLevel'].value_counts()
0    631
1    596
2    158
```

```
3    85
Name: StockOptionLevel, dtype: int64

sol_labels = {0:"None",1:"Low",2:"Medium",3:"High"}


sol_counts = df['StockOptionLevel'].value_counts().sort_index()

sns.set_style("white")

plt.figure(figsize=(6, 4))

sns.barplot(x=[sol_labels[i] for i in sol_counts.index],
            y=sol_counts.values,
            palette='cool')

plt.title('Employee Distribution by Stock Option Level', fontsize=14, fontweight='bold
')
plt.xlabel('StockOptionLevel')
plt.ylabel('Number of Employees')

plt.grid(False)

for i, v in enumerate(sol_counts.values):
   plt.text(i, v + 5, str(v), ha='center', fontweight='normal')


plt.tight_layout(pad=2.0)
plt.show()
```

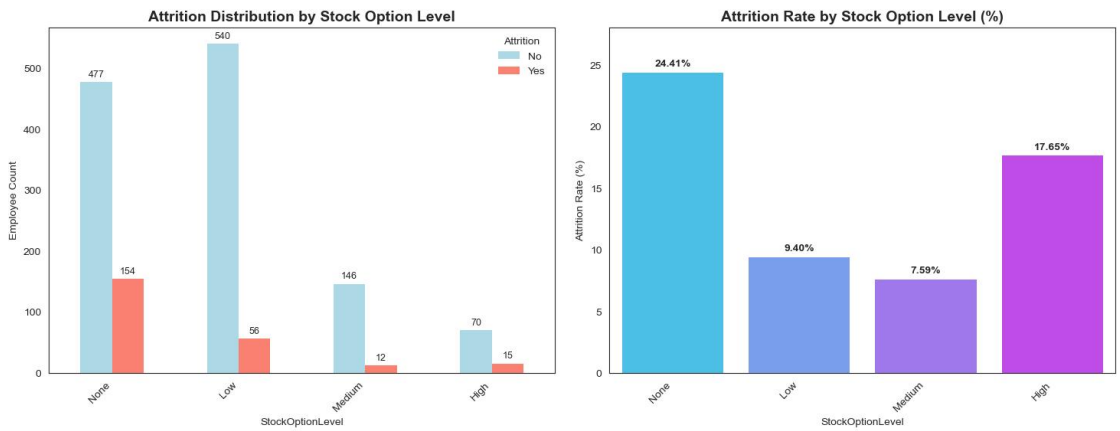


Findings:

Most of the employees in the company don't get any stock options.

```
df[df['Attrition'] == 'Yes']['StockOptionLevel'].value_counts()
0    154
1     56
3     15
2     12
Name: StockOptionLevel, dtype: int64
calculate_attrition_rate(df, 'StockOptionLevel').round(2)
   StockOptionLevel  AttritionRate
0                 0          24.41
1                 1           9.40
2                 2           7.59
3                 3          17.65

sol_labels = {0:"None",1:"Low",2:"Medium",3:"High"}



sns.set_style("white")


fig, axes = plt.subplots(1, 2, figsize=(15, 6))
plt.subplots_adjust(wspace=0.4)

#  Attrition Distribution (Yes/No) by StockOptionLevel
attrition_sol = df.groupby(['StockOptionLevel', 'Attrition']).size().unstack(fill_value=0)
```

```
attrition_sol.plot(kind='bar', ax=axes[0], color=['lightblue', 'salmon'], edgecolor='non
e')
axes[0].set_title('Attrition Distribution by Stock Option Level', fontsize=14, fontweigh
t='bold')
axes[0].set_xlabel('StockOptionLevel')
axes[0].set_ylabel('Employee Count')
axes[0].set_xticklabels([sol_labels[i] for i in attrition_sol.index], rotation=45, ha='righ
t')
axes[0].legend(title='Attrition', frameon=False)
axes[0].grid(False)
for container in axes[0].containers:
    axes[0].bar_label(container, fmt='%d', label_type='edge', fontsize=9, padding=3)

# Attrition Rate (%) by StockOptionLevel
attrition_rate_sol = calculate_attrition_rate(df, 'StockOptionLevel')
sns.barplot(x='StockOptionLevel', y='AttritionRate', data=attrition_rate_sol, palette='
cool', ax=axes[1], edgecolor='none')
axes[1].set_title('Attrition Rate by Stock Option Level (%)', fontsize=14, fontweight='
bold')
axes[1].set_xlabel('StockOptionLevel')
axes[1].set_ylabel('Attrition Rate (%)')
axes[1].set_xticklabels([sol_labels[i] for i in attrition_rate_sol['StockOptionLevel']], r
otation=45, ha='right')
axes[1].set_ylim(0, attrition_rate_sol['AttritionRate'].max() * 1.15)
axes[1].grid(False)
for i, v in enumerate(attrition_rate_sol['AttritionRate']):
    axes[1].text(i, v + 0.5, f'{v:.2f}%', ha='center', fontweight='bold')

plt.tight_layout(pad=2.0)
plt.show()
```

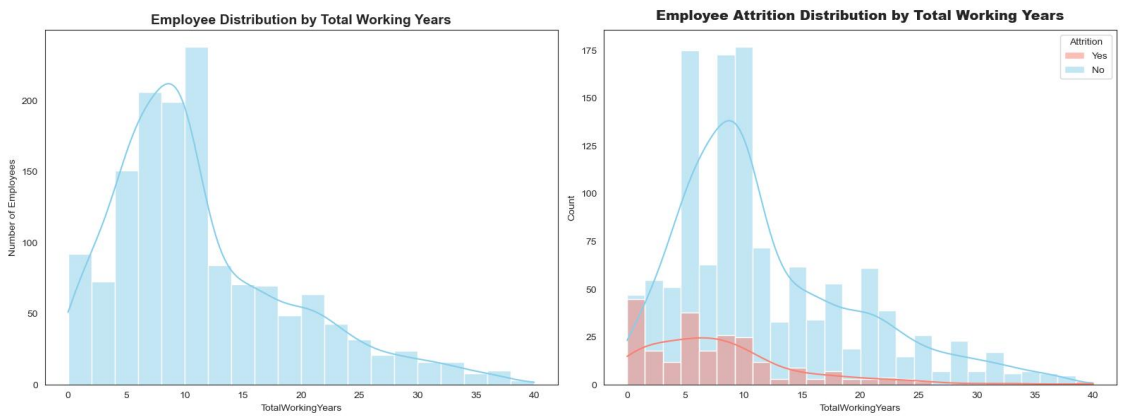


Findings & Insights:

The stock option levels doesn't seem to affect the employee attrition. Employees with stock benefits are more likely to stay in the company.

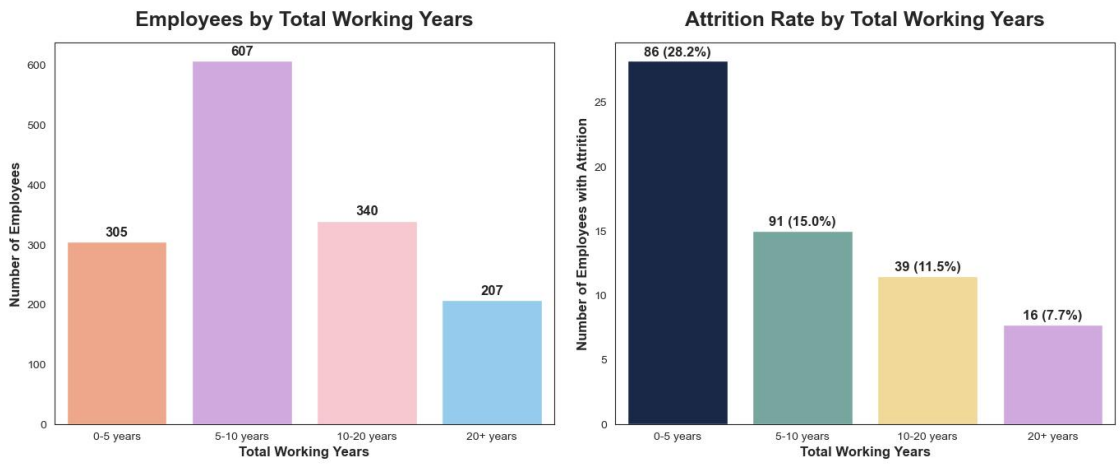
21) Attrition vs Total Working Years

```
plt.figure(figsize=(16, 6))

# Employee distribution by TotalWorkingYears
plt.subplot(1, 2, 1)
sns.histplot(df['TotalWorkingYears'], bins=20, kde=True, color='skyblue')
plt.title('Employee Distribution by Total Working Years', fontsize=14, fontweight='bol
d')
plt.xlabel('TotalWorkingYears')
plt.ylabel('Number of Employees')

#  Attrition distribution by TotalWorkingYears

plt.subplot(1,2,2)
sns.histplot(x="TotalWorkingYears",hue="Attrition",data=df,kde=True,palette=['salm
on', 'skyblue'])
plt.title("Employee Attrition Distribution by Total Working Years",fontweight="black",
size=14,pad=10)

plt.tight_layout()
plt.show()
```




```
df["TotalWorkingYears"].describe().to_frame().T
                count      mean      std  min  25%   50%   75%   max
TotalWorkingYears  1470.0  11.279592  7.780782  0.0  6.0  10.0  15.0  40.0

bin_edges = [0, 5, 10, 20, 50]
bin_labels = ['0-5 years', '5-10 years', '10-20 years', "20+ years"]


twy_groups = pd.cut(df['TotalWorkingYears'], bins=bin_edges, labels=bin_labels)

#  Employee Distribution by TotalWorkingYears ---
plt.figure(figsize=(14,6))
plt.subplot(1,2,1)
value_1 = twy_groups.value_counts().sort_index()
sns.barplot(x=value_1.index, y=value_1.values, palette=["#FFA07A", "#D4A1E7", "#F
```

```
FC0CB", "#87CEFA"])
plt.title("Employees by Total Working Years", fontweight="bold", pad=15, size=18)
plt.xlabel("Total Working Years", fontsize=12, fontweight="bold")
plt.ylabel("Number of Employees", fontsize=12, fontweight="bold")
for index, value in enumerate(value_1.values):
    plt.text(index, value + 5, value, ha="center", va="bottom", fontweight="bold", size
=12)

# Attrition Rate by TotalWorkingYears ---
plt.subplot(1,2,2)
twy_groups_attrition = pd.cut(df[df["Attrition"] == "Yes"]["TotalWorkingYears"], bins
=bin_edges, labels=bin_labels)
value_2 = twy_groups_attrition.value_counts().sort_index()
attrition_rate = np.round((value_2 / value_1) * 100,1).values

sns.barplot(x=value_2.index,y=attrition_rate, palette=["#11264e","#6faea4","#FEE08
B","#D4A1E7"])
plt.title("Attrition Rate by Total Working Years", fontweight="bold", pad=15, size=18)
plt.xlabel("Total Working Years", fontsize=12, fontweight="bold")
plt.ylabel("Number of Employees with Attrition", fontsize=12, fontweight="bold")
for i, rate in enumerate(attrition_rate):
    plt.text(i, rate + 0.2, f"{value_2.values[i]} ({rate}%)",
            ha="center", va="bottom", fontweight="bold", size=12)

plt.tight_layout(pad=2.0)
plt.show()
```



Findings & Insights:

Most of the employees in the company are having a total of 5 to 10 years of working experience. Employees with 0-5 years of total work experience are having high attrition rate, followed by employees with 5-10 years of total work experience. Employees with 20+ years of work experience are having less attrition rates.

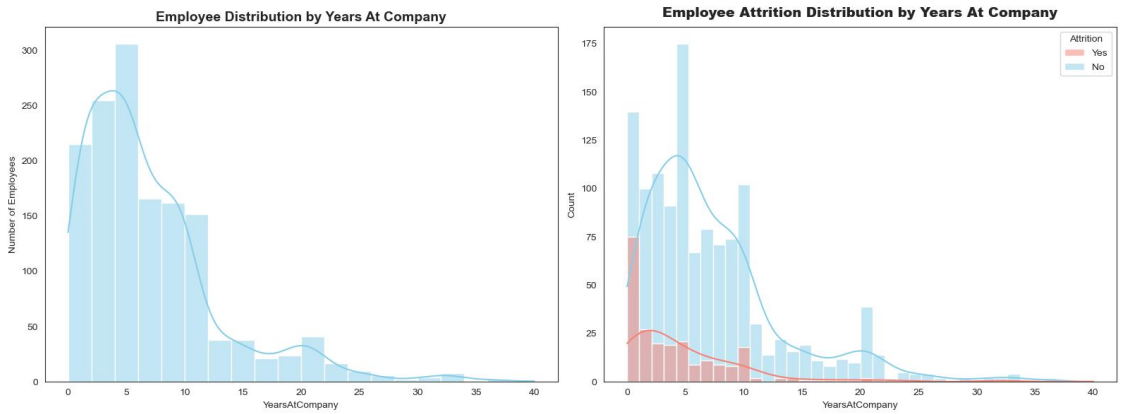
Employees who are in the early stages of their career will try to find better career opportunities.

```
bin_edges = [0, 5, 10, 20, 50]
bin_labels = ['0-5 years', '5-10 years', '10-20 years', '20+ years']

twy_groups = pd.cut(df['TotalWorkingYears'], bins=bin_edges, labels=bin_labels)
twy_groups_attrition = pd.cut(df[df["Attrition"] == "Yes"]["TotalWorkingYears"],
                        bins=bin_edges, labels=bin_labels)
total_counts = twy_groups.value_counts().sort_index()
total_counts
0-5 years      305
5-10 years     607
10-20 years    340
20+ years      207
Name: TotalWorkingYears, dtype: int64
attrition_counts = twy_groups_attrition.value_counts().sort_index()
attrition_counts
0-5 years      86
5-10 years     91
10-20 years    39
20+ years      16
Name: TotalWorkingYears, dtype: int64
attrition_ra = (attrition_counts / total_counts * 100).round(2)
attrition_ra
0-5 years      28.20
5-10 years     14.99
10-20 years    11.47
20+ years      7.73
Name: TotalWorkingYears, dtype: float64
```

22) Attrition vs Years at Company

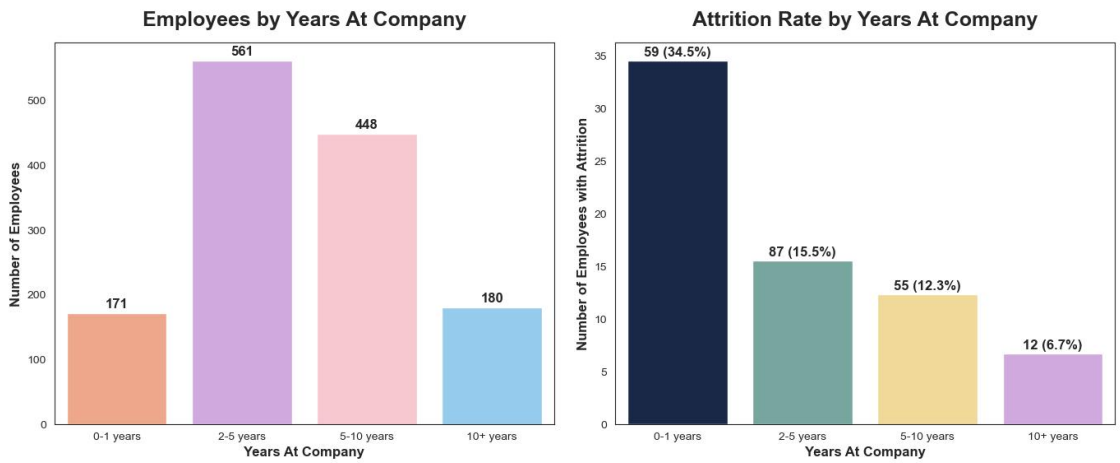
```
plt.figure(figsize=(16, 6))

# Employee distribution by YearsAtCompany
plt.subplot(1, 2, 1)
sns.histplot(df['YearsAtCompany'], bins=20, kde=True, color='skyblue')
plt.title('Employee Distribution by Years At Company', fontsize=14, fontweight='bold')
plt.xlabel('YearsAtCompany')
plt.ylabel('Number of Employees')

# Attrition distribution by YearsAtCompany

plt.subplot(1,2,2)
sns.histplot(x="YearsAtCompany",hue="Attrition",data=df,kde=True,palette=['salmo
n', 'skyblue'])
plt.title("Employee Attrition Distribution by Years At Company",fontweight="black",s
```

```
ize=14,pad=10)

plt.tight_layout()
plt.show()
```




```
bin_edges = [0, 1, 5, 10, 20]
bin_labels = ['0-1 years', '2-5 years', '5-10 years', "10+ years"]
yac_groups = pd.cut(df['YearsAtCompany'], bins=bin_edges, labels=bin_labels)
yac_groups_attrition = pd.cut(df[df["Attrition"] == "Yes"]["YearsAtCompany"],
                        bins=bin_edges, labels=bin_labels)


# Employee Distribution by YearsAtCompany ---
plt.figure(figsize=(14,6))
plt.subplot(1,2,1)
value_1 = yac_groups.value_counts().sort_index()
sns.barplot(x=value_1.index, y=value_1.values, palette=["#FFA07A", "#D4A1E7", "#F
FC0CB", "#87CEFA"])
plt.title("Employees by Years At Company", fontweight="bold", pad=15, size=18)
plt.xlabel("Years At Company", fontsize=12, fontweight="bold")
plt.ylabel("Number of Employees", fontsize=12, fontweight="bold")
for index, value in enumerate(value_1.values):
   plt.text(index, value + 5, value, ha="center", va="bottom", fontweight="bold", size
=12)

# Attrition Rate by YearsAtCompany ---
plt.subplot(1,2,2)
yac_groups_attrition = pd.cut(df[df["Attrition"] == "Yes"]["YearsAtCompany"], bins=
bin_edges, labels=bin_labels)
value_2 = yac_groups_attrition.value_counts().sort_index()
attrition_rate = np.round((value_2 / value_1) * 100,1).values

sns.barplot(x=value_2.index,y=attrition_rate, palette=["#11264e","#6faea4","#FEE08
B","#D4A1E7"])
plt.title("Attrition Rate by Years At Company", fontweight="bold", pad=15, size=18)
```

```
plt.xlabel("Years At Company", fontsize=12, fontweight="bold")
plt.ylabel("Number of Employees with Attrition", fontsize=12, fontweight="bold")
for i, rate in enumerate(attrition_rate):
   plt.text(i, rate + 0.2, f"{value_2.values[i]} ({rate}%)",
         ha="center", va="bottom", fontweight="bold", size=12)

plt.tight_layout(pad=2.0)
plt.show()
```



Findings & Insights:

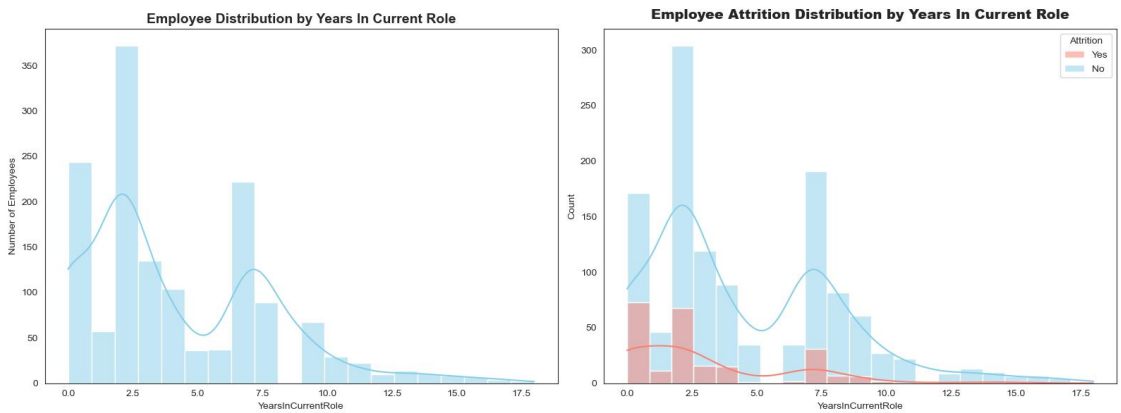
Most employees have worked for 2 to 5 years in the same company. Very few employees has working for 1 or less than 1 year or more than 10 years. Employees who have worked for 0-1 years are having very high attrition rate. Employees who have worked for 10+ years are having low attrition rate.

```
total_yaccounts = yac_groups.value_counts().sort_index()
total_yaccounts
0-1 years     171
2-5 years     561
5-10 years    448
10+ years     180
Name: YearsAtCompany, dtype: int64
attrition_yaccounts = yac_groups_attrition.value_counts().sort_index()
attrition_yaccounts
0-1 years     59
2-5 years     87
5-10 years    55
10+ years     12
Name: YearsAtCompany, dtype: int64
attrition_yacra = (attrition_yaccounts / total_yaccounts * 100).round(2)
attrition_yacra
0-1 years     34.50
2-5 years     15.51
5-10 years    12.28
```

10+ years 6.67
Name: YearsAtCompany, dtype: float64

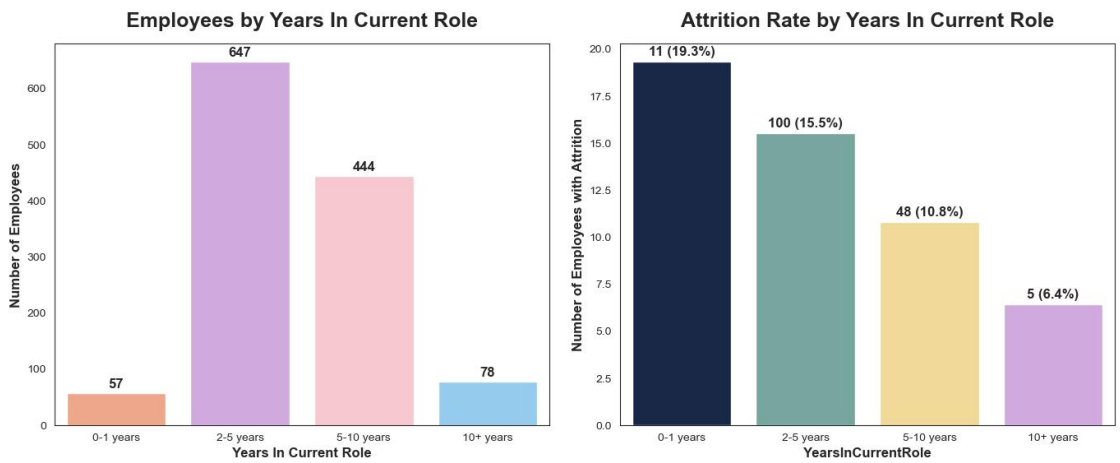
23) Attrition vs Years in Current Role

```
plt.figure(figsize=(16, 6))

# Employee distribution by YearsInCurrentRole
plt.subplot(1, 2, 1)
sns.histplot(df['YearsInCurrentRole'], bins=20, kde=True, color='skyblue')
plt.title('Employee Distribution by Years In Current Role', fontsize=14, fontweight='b
old')
plt.xlabel('YearsInCurrentRole')
plt.ylabel('Number of Employees')

# Attrition distribution by YearsInCurrentRole

plt.subplot(1,2,2)
sns.histplot(x="YearsInCurrentRole",hue="Attrition",data=df,kde=True,palette=['sal
mon', 'skyblue'])
plt.title("Employee Attrition Distribution by Years In Current Role",fontweight="black
",size=14,pad=10)

plt.tight_layout()
plt.show()
```




```
bin_edges = [0, 1, 5, 10, 20]
bin_labels = ['0-1 years', '2-5 years', '5-10 years', "10+ years"]
yicr_groups = pd.cut(df['YearsInCurrentRole'], bins=bin_edges, labels=bin_labels)
yicr_groups_attrition = pd.cut(df[df["Attrition"] == "Yes"]["YearsInCurrentRole"],
                        bins=bin_edges, labels=bin_labels)


#  Employee Distribution by YearsInCurrentRole ---
plt.figure(figsize=(14,6))
plt.subplot(1,2,1)
```

```
value_1 = yicr_groups.value_counts().sort_index()
sns.barplot(x=value_1.index, y=value_1.values, palette=["#FFA07A", "#D4A1E7", "#F
FC0CB", "#87CEFA"])
plt.title("Employees by Years In Current Role", fontweight="bold", pad=15, size=18)
plt.xlabel("Years In Current Role", fontsize=12, fontweight="bold")
plt.ylabel("Number of Employees", fontsize=12, fontweight="bold")
for index, value in enumerate(value_1.values):
    plt.text(index, value + 5, value, ha="center", va="bottom", fontweight="bold", size
=12)

# Attrition Rate by YearsInCurrentRole ---
plt.subplot(1,2,2)
yirc_groups_attrition = pd.cut(df[df["Attrition"] == "Yes"]["YearsInCurrentRole"], bin
s=bin_edges, labels=bin_labels)
value_2 = yirc_groups_attrition.value_counts().sort_index()
attrition_rate = np.round((value_2 / value_1) * 100,1).values

sns.barplot(x=value_2.index,y=attrition_rate, palette=["#11264e","#6faea4","#FEE08
B","#D4A1E7"])
plt.title("Attrition Rate by Years In Current Role", fontweight="bold", pad=15, size=1
8)
plt.xlabel("YearsInCurrentRole", fontsize=12, fontweight="bold")
plt.ylabel("Number of Employees with Attrition", fontsize=12, fontweight="bold")
for i, rate in enumerate(attrition_rate):
    plt.text(i, rate + 0.2, f"{value_2.values[i]} ({rate}%)",
            ha="center", va="bottom", fontweight="bold", size=12)

plt.tight_layout(pad=2.0)
plt.show()
```



Findings & Insights:

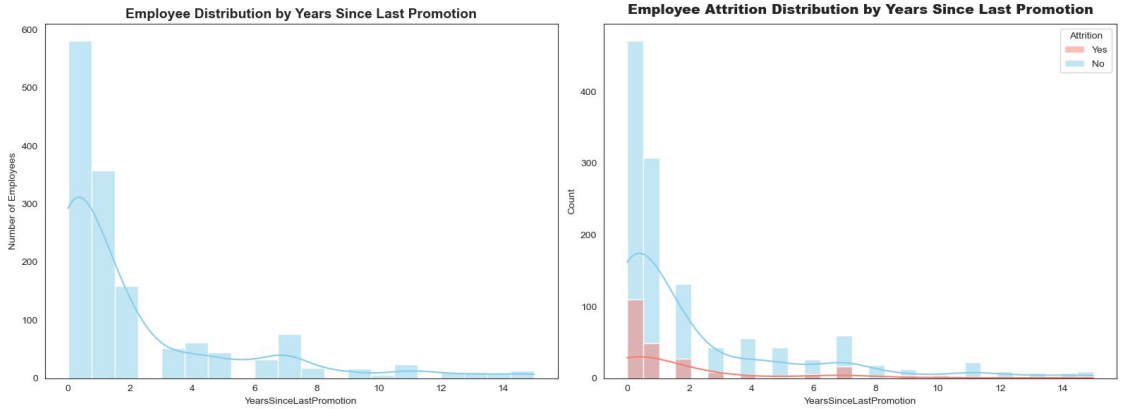
Most employees have worked for 2-5 years for the same role in the company without any role change. Very few employees has worked for 1 or less than 1 year or more than 10 years in the same role. Employees who have worked for 0 to 1 years in

the same role are having very high attrition rate followed by employees who worked for 2-5 years in the same role. Employees who have worked for 10+ years in the same role are having low attrition rate.

```
total_yicrcounts = yicr_groups.value_counts().sort_index()
total_yicrcounts
0-1 years      57
2-5 years     647
5-10 years    444
10+ years      78
Name: YearsInCurrentRole, dtype: int64
attrition_yicrcounts = yicr_groups_attrition.value_counts().sort_index()
attrition_yicrcounts
0-1 years      11
2-5 years     100
5-10 years     48
10+ years       5
Name: YearsInCurrentRole, dtype: int64
attrition_yicrra = (attrition_yicrcounts / total_yicrcounts * 100).round(2)
attrition_yicrra
0-1 years     19.30
2-5 years     15.46
5-10 years    10.81
10+ years      6.41
Name: YearsInCurrentRole, dtype: float64
```

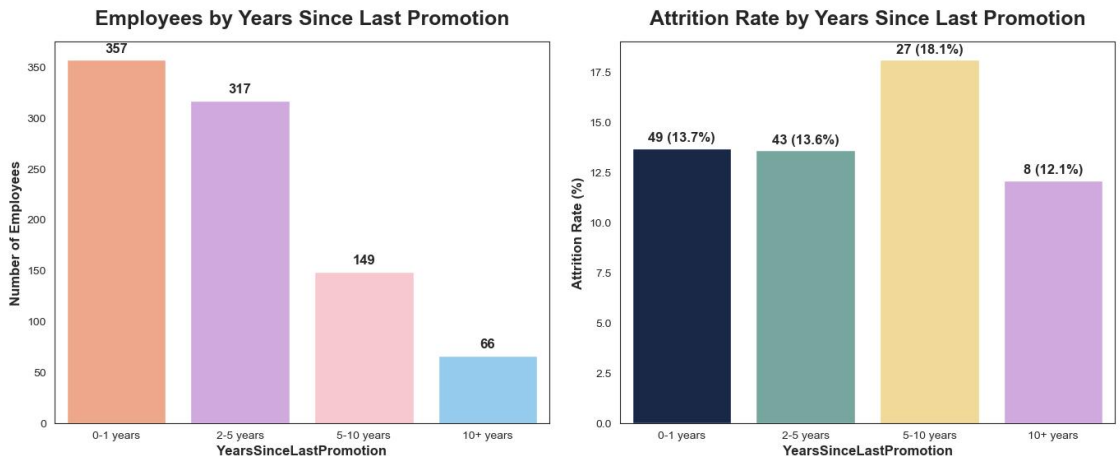
24) Attrition vs Years Since Last Promotion

```
plt.figure(figsize=(16, 6))

#  Employee distribution by YearsSinceLastPromotion
plt.subplot(1, 2, 1)
sns.histplot(df['YearsSinceLastPromotion'], bins=20, kde=True, color='skyblue')
plt.title('Employee Distribution by Years Since Last Promotion', fontsize=14, fontweig
ht='bold')
plt.xlabel('YearsSinceLastPromotion')
plt.ylabel('Number of Employees')

#  Attrition distribution by YearsSinceLastPromotion

plt.subplot(1,2,2)
sns.histplot(x="YearsSinceLastPromotion",hue="Attrition",data=df,kde=True,palette
=['salmon', 'skyblue'])
plt.title("Employee Attrition Distribution by Years Since Last Promotion",fontweight=
"black",size=14,pad=10)

plt.tight_layout()
plt.show()
```




```
bin_edges = [0, 1, 5, 10, 20]
bin_labels = ['0-1 years', '2-5 years', '5-10 years', "10+ years"]
ys_groups = pd.cut(df['YearsSinceLastPromotion'], bins=bin_edges, labels=bin_labels)
ys_groups_attrition = pd.cut(df[df["Attrition"] == "Yes"]["YearsSinceLastPromotion"],
bins=bin_edges, labels=bin_labels)


# Employee Distribution by YearsSinceLastPromotion ---
plt.figure(figsize=(14,6))
plt.subplot(1,2,1)
value_1 = ys_groups.value_counts().sort_index()
sns.barplot(x=value_1.index, y=value_1.values, palette=["#FFA07A", "#D4A1E7", "#F
FC0CB", "#87CEFA"])
plt.title("Employees by Years Since Last Promotion", fontweight="bold", pad=15, size
=18)
plt.xlabel("YearsSinceLastPromotion", fontsize=12, fontweight="bold")
plt.ylabel("Number of Employees", fontsize=12, fontweight="bold")
for index, value in enumerate(value_1.values):
    plt.text(index, value + 5, value, ha="center", va="bottom", fontweight="bold", size
=12)


# Attrition Rate by YearsSinceLastPromotion ---
plt.subplot(1,2,2)
value_2 = ys_groups_attrition.value_counts().sort_index()
attrition_rate = np.round((value_2 / value_1) * 100,1).values

sns.barplot(x=value_2.index, y=attrition_rate,
            palette=["#11264e","#6faea4","#FEE08B","#D4A1E7"])

plt.title("Attrition Rate by Years Since Last Promotion", fontweight="bold", pad=15, s
ize=18)
plt.xlabel("YearsSinceLastPromotion", fontsize=12, fontweight="bold")
plt.ylabel("Attrition Rate (%)", fontsize=12, fontweight="bold")
```

```
for i, rate in enumerate(attrition_rate):
    plt.text(i, rate + 0.2, f"{value_2.values[i]} ({rate}%)",
             ha="center", va="bottom", fontweight="bold", size=12)

plt.tight_layout(pad=2.0)
plt.show()
```

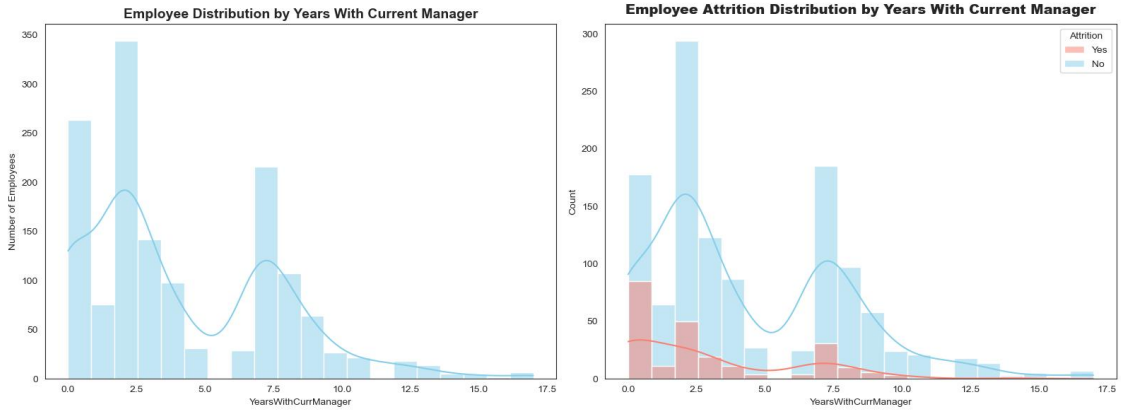


Findings & Insights:

Attrition rates are higher for employees who have not been promoted since 5 to 10 years. Employees who don't get promotion within few years are more tend to leave the company.

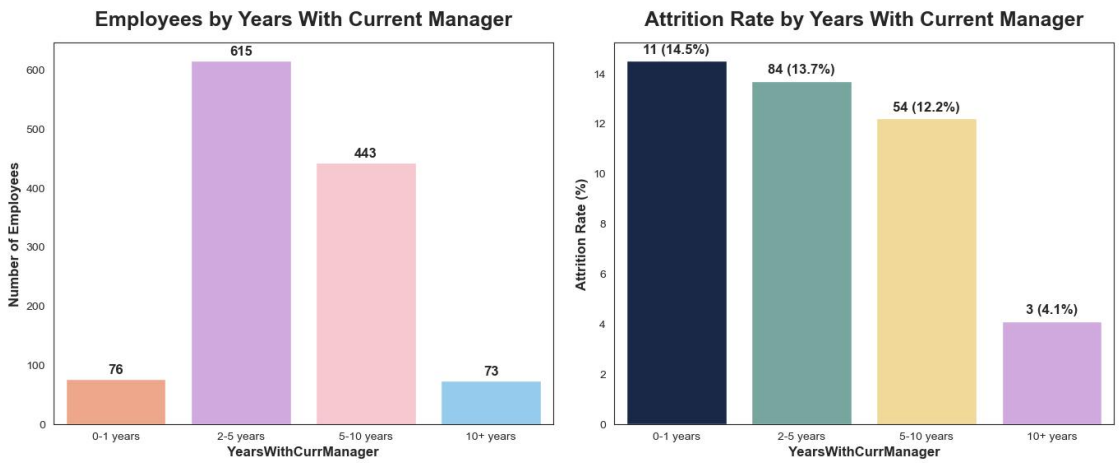
25) Attrition vs Years with Current Manager

```
plt.figure(figsize=(16, 6))

# Employee distribution by YearsWithCurrManager
plt.subplot(1, 2, 1)
sns.histplot(df['YearsWithCurrManager'], bins=20, kde=True, color='skyblue')
plt.title('Employee Distribution by Years With Current Manager', fontsize=14, fontwe
ight='bold')
plt.xlabel('YearsWithCurrManager')
plt.ylabel('Number of Employees')

# Attrition distribution by YearsWithCurrManager

plt.subplot(1,2,2)
sns.histplot(x="YearsWithCurrManager",hue="Attrition",data=df,kde=True,palette=['
salmon', 'skyblue'])
plt.title("Employee Attrition Distribution by Years With Current Manager",fontweight
="black",size=14,pad=10)

plt.tight_layout()
plt.show()
```




```
bin_edges = [0, 1, 5, 10, 20]
bin_labels = ['0-1 years', '2-5 years', '5-10 years', "10+ years"]
ycm_groups = pd.cut(df['YearsWithCurrManager'], bins=bin_edges, labels=bin_labels)
ycm_groups_attrition = pd.cut(df[df["Attrition"] == "Yes"]["YearsWithCurrManager"],

                    bins=bin_edges, labels=bin_labels)


# Employee Distribution by YearsWithCurrManager ---
plt.figure(figsize=(14,6))
plt.subplot(1,2,1)
value_1 = ycm_groups.value_counts().sort_index()
sns.barplot(x=value_1.index, y=value_1.values, palette=["#FFA07A", "#D4A1E7", "#F
FC0CB", "#87CEFA"])
plt.title("Employees by Years With Current Manager", fontweight="bold", pad=15, siz
e=18)
plt.xlabel("YearsWithCurrManager", fontsize=12, fontweight="bold")
plt.ylabel("Number of Employees", fontsize=12, fontweight="bold")
for index, value in enumerate(value_1.values):
   plt.text(index, value + 5, value, ha="center", va="bottom", fontweight="bold", size
=12)


# Attrition Rate by YearsWithCurrManager ---
plt.subplot(1,2,2)
value_2 = ycm_groups_attrition.value_counts().sort_index()
attrition_rate = np.round((value_2 / value_1) * 100,1).values

sns.barplot(x=value_2.index, y=attrition_rate,
        palette=["#11264e","#6faea4","#FEE08B","#D4A1E7"])

plt.title("Attrition Rate by Years With Current Manager", fontweight="bold", pad=15,
 size=18)
plt.xlabel("YearsWithCurrManager", fontsize=12, fontweight="bold")
plt.ylabel("Attrition Rate (%)", fontsize=12, fontweight="bold")
```

```
for i, rate in enumerate(attrition_rate):
    plt.text(i, rate + 0.2, f"{value_2.values[i]} ({rate}%)",
             ha="center", va="bottom", fontweight="bold", size=12)

plt.tight_layout(pad=2.0)
plt.show()
```



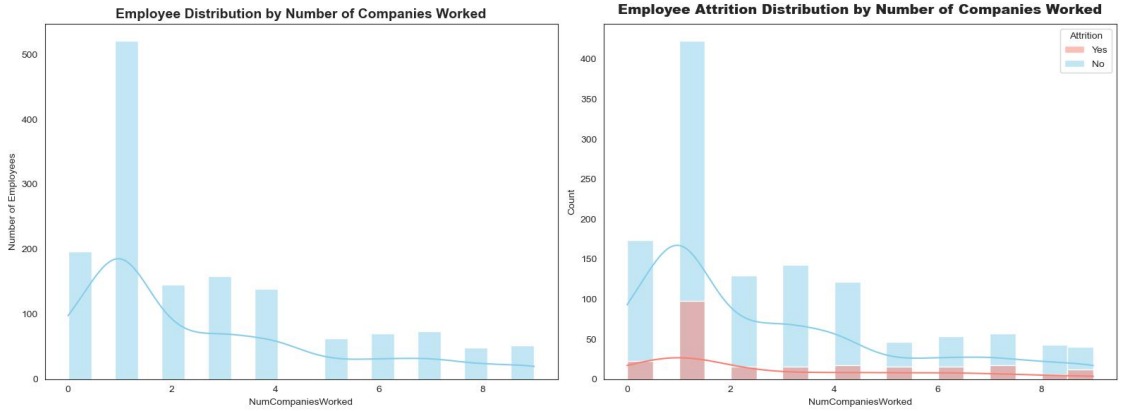
Findings & Insights:

Most of the employees in the company have worked for 2-5 years with the same manager. Employees who have worked only for 1 or less than a year or for 2 to 5 with the same manager are having high attrition rate. That means the relationship with them and their manager was not so great. Employees who worked for more than 10 years with the same manager have low attrition rate. That means that they got along really well with their manager.

26) Attrition vs Number of Companies Worked

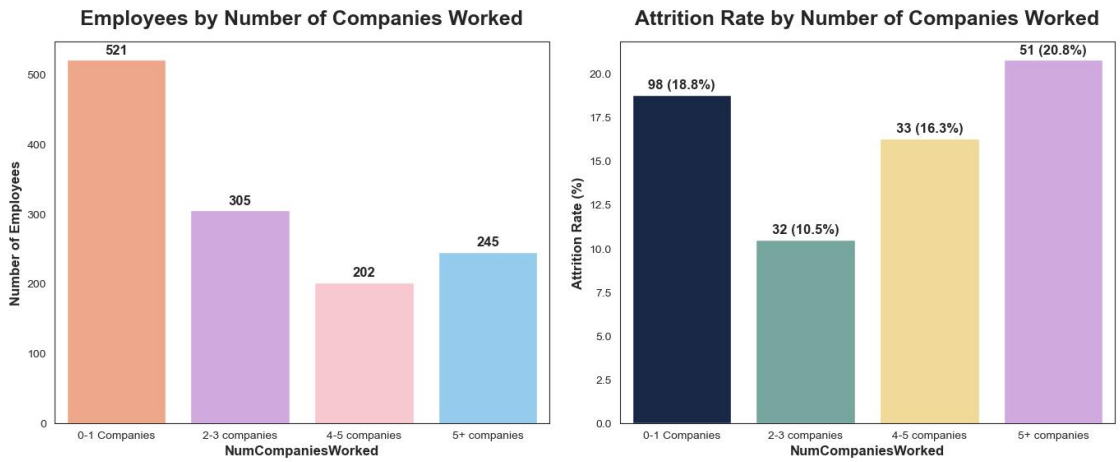
```
plt.figure(figsize=(16, 6))

# Employee distribution by NumCompaniesWorked
plt.subplot(1, 2, 1)
sns.histplot(df['NumCompaniesWorked'], bins=20, kde=True, color='skyblue')
plt.title('Employee Distribution by Number of Companies Worked', fontsize=14, font
weight='bold')
plt.xlabel('NumCompaniesWorked')
plt.ylabel('Number of Employees')

# Attrition distribution by NumCompaniesWorked

plt.subplot(1,2,2)
sns.histplot(x="NumCompaniesWorked",hue="Attrition",data=df,kde=True,palette=
['salmon', 'skyblue'])
plt.title("Employee Attrition Distribution by Number of Companies Worked",fontwei
```

```
ght="black",size=14,pad=10)

plt.tight_layout()
plt.show()
```




```
bin_edges = [0, 1, 3, 5, 10]
bin_labels = ['0-1 Companies', '2-3 companies', '4-5 companies', "5+ companies"]
ncw_groups = pd.cut(df['NumCompaniesWorked'], bins=bin_edges, labels=bin_label
s)
ncw_groups_attrition = pd.cut(df[df["Attrition"] == "Yes"]["NumCompaniesWorked"],

                        bins=bin_edges, labels=bin_labels)


# Employee Distribution by NumCompaniesWorked ---
plt.figure(figsize=(14,6))
plt.subplot(1,2,1)
value_1 = ncw_groups.value_counts().sort_index()
sns.barplot(x=value_1.index, y=value_1.values, palette=["#FFA07A", "#D4A1E7", "#F
FC0CB", "#87CEFA"])
plt.title("Employees by Number of Companies Worked", fontweight="bold", pad=15,
size=18)
plt.xlabel("NumCompaniesWorked", fontsize=12, fontweight="bold")
plt.ylabel("Number of Employees", fontsize=12, fontweight="bold")
for index, value in enumerate(value_1.values):
   plt.text(index, value + 5, value, ha="center", va="bottom", fontweight="bold", size
=12)


# Attrition Rate by NumCompaniesWorked ---
plt.subplot(1,2,2)
value_2 = ncw_groups_attrition.value_counts().sort_index()
attrition_rate = np.round((value_2 / value_1) * 100,1).values

sns.barplot(x=value_2.index, y=attrition_rate,
```

```
        palette=["#11264e","#6faea4","#FEE08B","#D4A1E7"])

plt.title("Attrition Rate by Number of Companies Worked", fontweight="bold", pad=
15, size=18)
plt.xlabel("NumCompaniesWorked", fontsize=12, fontweight="bold")
plt.ylabel("Attrition Rate (%)", fontsize=12, fontweight="bold")


for i, rate in enumerate(attrition_rate):
   plt.text(i, rate + 0.2, f"{value_2.values[i]} ({rate}%)",
         ha="center", va="bottom", fontweight="bold", size=12)

plt.tight_layout(pad=2.0)
plt.show()
```



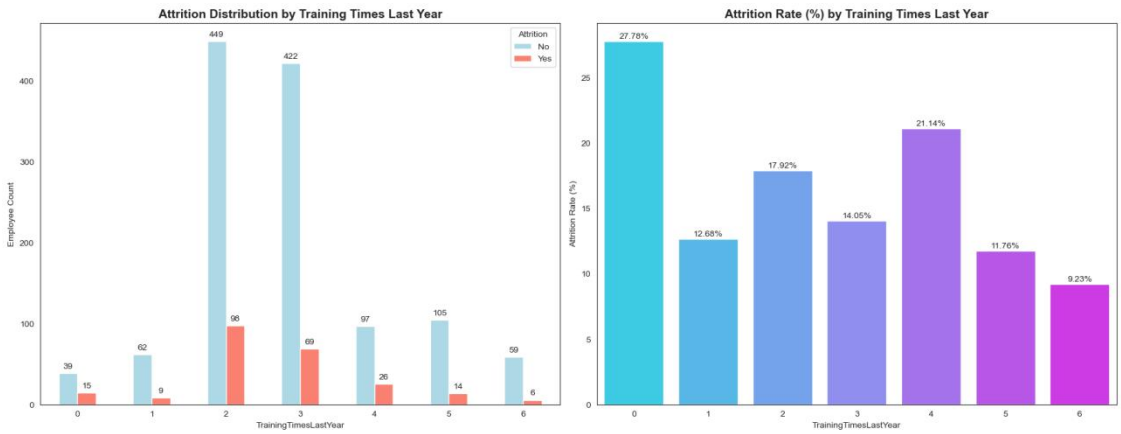
Findings & Insights:

Most of the employees in the company have worked only in 1 company. Employees who worked in more than 4 companies indicate their company switching behaviour. Employees who switches the companies multiple times or worked for more than 4 companies are having higher attrition rate. Many employees who have worked only for one company also tend to leave the company as it is the first company they are working. And if they don't like it there, they tend to leave the company.

27) Attrition vs Training Times Last Year

```
fig, axes = plt.subplots(1, 2, figsize=(18, 7))
plt.subplots_adjust(wspace=0.4)

# Attrition Distribution (Yes/No) by TrainingTimesLastYear
attrition_ttls = df.groupby(['TrainingTimesLastYear', 'Attrition']).size().unstack(fill_val
ue=0)
attrition_ttls.plot(kind='bar', ax=axes[0], color=['lightblue', 'salmon'])
axes[0].set_title('Attrition Distribution by Training Times Last Year', fontsize=14, font
weight='bold')
```

```
axes[0].set_xlabel('TrainingTimesLastYear')
axes[0].set_ylabel('Employee Count')
axes[0].legend(title='Attrition')
axes[0].tick_params(axis='x', rotation=0)
for container in axes[0].containers:
    axes[0].bar_label(container, fmt='%d', label_type='edge', fontsize=10, padding=3)

# Attrition Rate (%) by TrainingTimesLastYear
attrition_rate_ttls = calculate_attrition_rate(df, 'TrainingTimesLastYear')
sns.barplot(
    x='TrainingTimesLastYear',
    y='AttritionRate',
    data=attrition_rate_ttls,
    palette='cool',
    ax=axes[1]
)

axes[1].set_title('Attrition Rate (%) by Training Times Last Year', fontsize=14, fontwei
ght='bold')
axes[1].set_xlabel('TrainingTimesLastYear')
axes[1].set_ylabel('Attrition Rate (%)')


for i, v in enumerate(attrition_rate_ttls['AttritionRate']):
    axes[1].text(i, v + 0.2, f'{v:.2f}%', ha='center', fontsize=10)



plt.tight_layout()
plt.show()
```

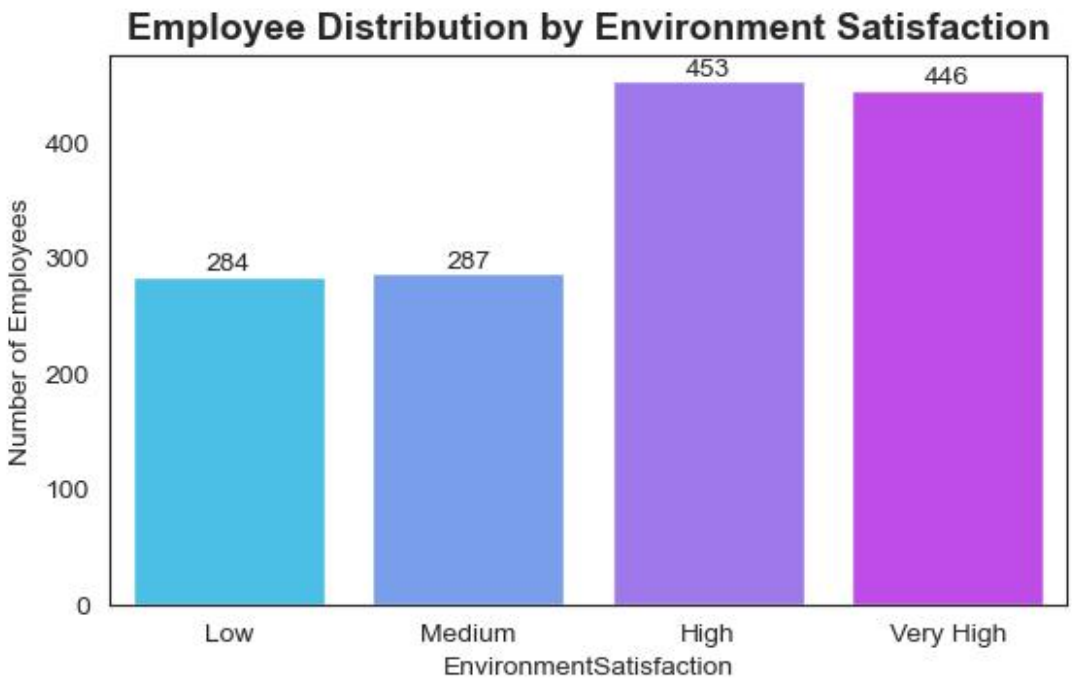


Findings & Insights:

TrainingTimesLastYear means how many times employee took training session last year. Attrition rate is higher for those employees who didn't take the training sessions.

28) Attrition vs Environment Satisfaction

```
df['EnvironmentSatisfaction'].value_counts()
3    453
4    446
2    287
1    284
Name: EnvironmentSatisfaction, dtype: int64

es_labels = {1:"Low",2:"Medium",3:"High",4:"Very High"}


es_counts = df['EnvironmentSatisfaction'].value_counts().sort_index()

sns.set_style("white")

plt.figure(figsize=(6, 4))

sns.barplot(x=[es_labels[i] for i in es_counts.index],
            y=es_counts.values,
            palette='cool')

plt.title('Employee Distribution by Environment Satisfaction', fontsize=14, fontweight
='bold')
plt.xlabel('EnvironmentSatisfaction')
plt.ylabel('Number of Employees')

plt.grid(False)

for i, v in enumerate(es_counts.values):
    plt.text(i, v + 5, str(v), ha='center', fontweight='normal')


plt.tight_layout(pad=2.0)
plt.show()
```

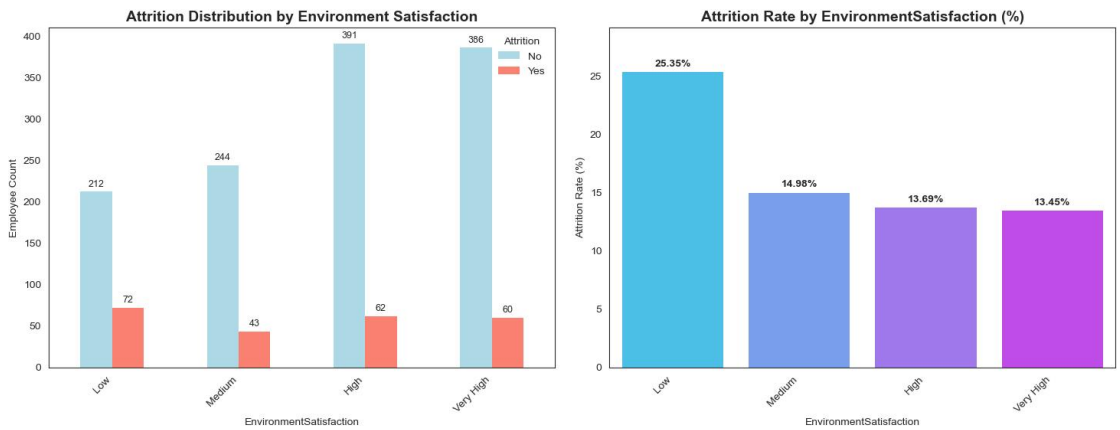


Findings:

Most of the employees have high levels of working environment satisfaction.

```
df[df['Attrition'] == 'Yes']['EnvironmentSatisfaction'].value_counts()
1    72
3    62
4    60
2    43
Name: EnvironmentSatisfaction, dtype: int64
calculate_attrition_rate(df, 'EnvironmentSatisfaction').round(2)
  EnvironmentSatisfaction  AttritionRate
0                       1          25.35
1                       2          14.98
2                       3          13.69
3                       4          13.45

es_labels = {1:"Low",2:"Medium",3:"High",4:"Very High"}



sns.set_style("white")


fig, axes = plt.subplots(1, 2, figsize=(15, 6))
plt.subplots_adjust(wspace=0.4)

#  Attrition Distribution (Yes/No) by EnvironmentSatisfaction
attrition_es = df.groupby(['EnvironmentSatisfaction', 'Attrition']).size().unstack(fill_v
```

```
alue=0)
attrition_es.plot(kind='bar', ax=axes[0], color=['lightblue', 'salmon'], edgecolor='none
')
axes[0].set_title('Attrition Distribution by Environment Satisfaction', fontsize=14, fon
tweight='bold')
axes[0].set_xlabel('EnvironmentSatisfaction')
axes[0].set_ylabel('Employee Count')
axes[0].set_xticklabels([es_labels[i] for i in attrition_es.index], rotation=45, ha='right')
axes[0].legend(title='Attrition', frameon=False)
axes[0].grid(False)
for container in axes[0].containers:
    axes[0].bar_label(container, fmt='%d', label_type='edge', fontsize=9, padding=3)

# Attrition Rate (%) by EnvironmentSatisfaction
attrition_rate_es = calculate_attrition_rate(df, 'EnvironmentSatisfaction')
sns.barplot(x='EnvironmentSatisfaction', y='AttritionRate', data=attrition_rate_es, pa
lette='cool', ax=axes[1], edgecolor='none')
axes[1].set_title('Attrition Rate by EnvironmentSatisfaction (%)', fontsize=14, fontwei
ght='bold')
axes[1].set_xlabel('EnvironmentSatisfaction')
axes[1].set_ylabel('Attrition Rate (%)')
axes[1].set_xticklabels([es_labels[i] for i in attrition_rate_es['EnvironmentSatisfactio
n']], rotation=45, ha='right')
axes[1].set_ylim(0, attrition_rate_es['AttritionRate'].max() * 1.15)
axes[1].grid(False)
for i, v in enumerate(attrition_rate_es['AttritionRate']):
    axes[1].text(i, v + 0.5, f'{v:.2f}%', ha='center', fontweight='bold')

plt.tight_layout(pad=2.0)
plt.show()
```



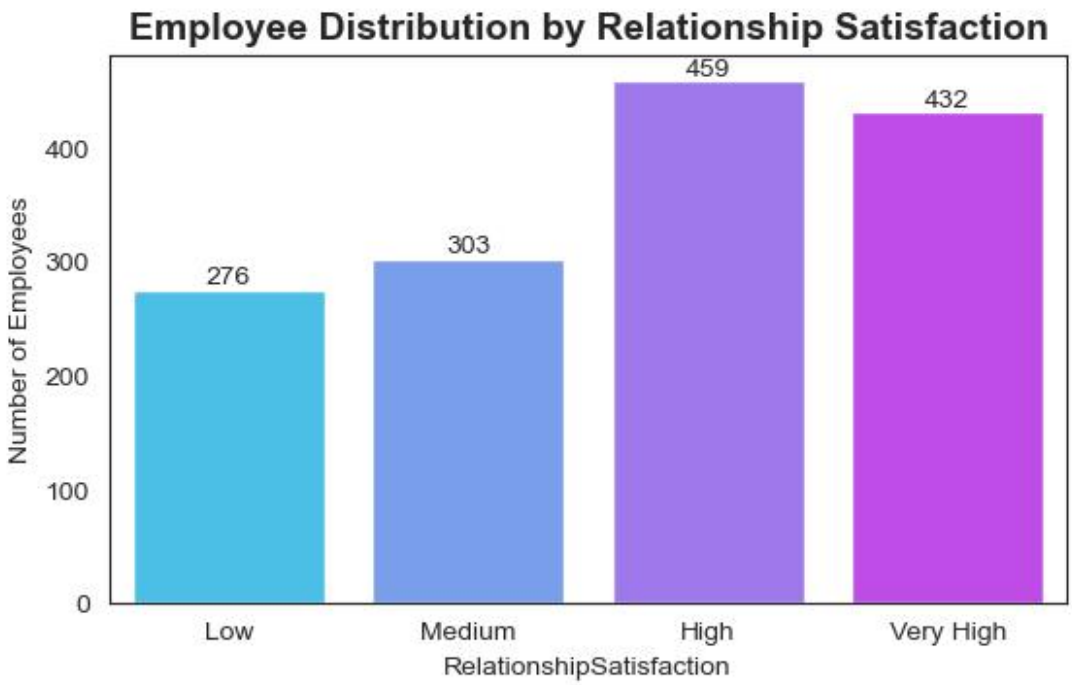
Findings & Insights:

Employees with low levels of working environment satisfaction have higher attrition rates. As the environment satisfaction level increases attrition rate decreases. If the

working environment is stressful, not supportive and toxic, it will be very difficult for the employees to work and grow. So environment satisfaction is important.

29) Attrition vs Relationship Satisfaction

```
df['RelationshipSatisfaction'].value_counts()
3    459
4    432
2    303
1    276
Name: RelationshipSatisfaction, dtype: int64

rs_labels = {1:"Low",2:"Medium",3:"High",4:"Very High"}


rs_counts = df['RelationshipSatisfaction'].value_counts().sort_index()

sns.set_style("white")

plt.figure(figsize=(6, 4))

sns.barplot(x=[rs_labels[i] for i in rs_counts.index],
        y=rs_counts.values,
        palette='cool')

plt.title('Employee Distribution by Relationship Satisfaction', fontsize=14, fontweight
='bold')
plt.xlabel('RelationshipSatisfaction')
plt.ylabel('Number of Employees')

plt.grid(False)

for i, v in enumerate(rs_counts.values):
   plt.text(i, v + 5, str(v), ha='center', fontweight='normal')


plt.tight_layout(pad=2.0)
plt.show()
```



Findings & Insights:

It shows that most of the employees in the company have great relationship satisfaction with their colleagues and management.

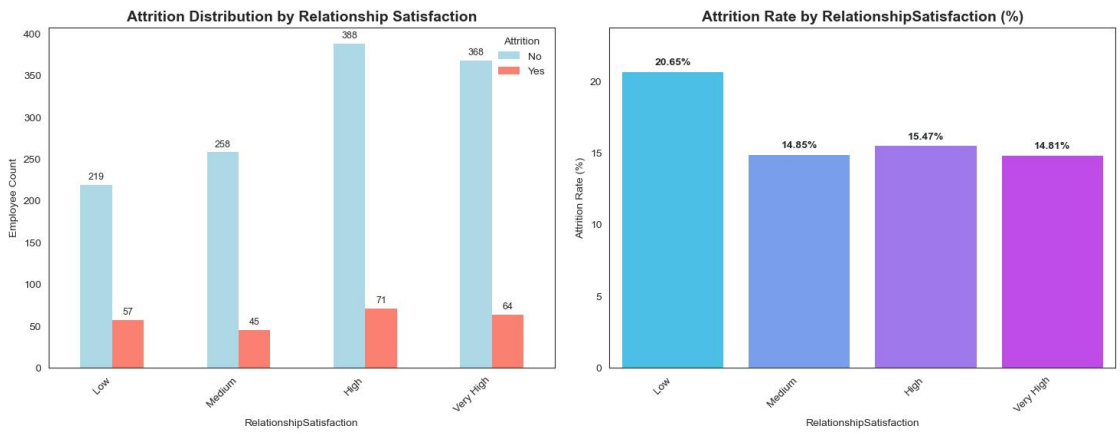
```
df[df['Attrition'] == 'Yes']['RelationshipSatisfaction'].value_counts()
3    71
4    64
1    57
2    45
Name: RelationshipSatisfaction, dtype: int64
```

```
calculate_attrition_rate(df, 'RelationshipSatisfaction').round(2)
```

	RelationshipSatisfaction	AttritionRate
0	1	20.65
1	2	14.85
2	3	15.47
3	4	14.81

```
rs_labels = {1:"Low",2:"Medium",3:"High",4:"Very High"}



sns.set_style("white")


fig, axes = plt.subplots(1, 2, figsize=(15, 6))
plt.subplots_adjust(wspace=0.4)
```

```
# Attrition Distribution (Yes/No) by RelationshipSatisfaction
attrition_rs = df.groupby(['RelationshipSatisfaction', 'Attrition']).size().unstack(fill_val
ue=0)
attrition_rs.plot(kind='bar', ax=axes[0], color=['lightblue', 'salmon'], edgecolor='none
')
axes[0].set_title('Attrition Distribution by Relationship Satisfaction', fontsize=14, font
weight='bold')
axes[0].set_xlabel('RelationshipSatisfaction')
axes[0].set_ylabel('Employee Count')
axes[0].set_xticklabels([rs_labels[i] for i in attrition_rs.index], rotation=45, ha='right')
axes[0].legend(title='Attrition', frameon=False)
axes[0].grid(False)
for container in axes[0].containers:
    axes[0].bar_label(container, fmt='%d', label_type='edge', fontsize=9, padding=3)

# Attrition Rate (%) by RelationshipSatisfaction
attrition_rate_rs = calculate_attrition_rate(df, 'RelationshipSatisfaction')
sns.barplot(x='RelationshipSatisfaction', y='AttritionRate', data=attrition_rate_rs, pal
ette='cool', ax=axes[1], edgecolor='none')
axes[1].set_title('Attrition Rate by RelationshipSatisfaction (%)', fontsize=14, fontwei
ght='bold')
axes[1].set_xlabel('RelationshipSatisfaction')
axes[1].set_ylabel('Attrition Rate (%)')
axes[1].set_xticklabels([rs_labels[i] for i in attrition_rate_rs['RelationshipSatisfaction
']], rotation=45, ha='right')
axes[1].set_ylim(0, attrition_rate_rs['AttritionRate'].max() * 1.15)
axes[1].grid(False)
for i, v in enumerate(attrition_rate_rs['AttritionRate']):
    axes[1].text(i, v + 0.5, f'{v:.2f}%', ha='center', fontweight='bold')

plt.tight_layout(pad=2.0)
plt.show()
```

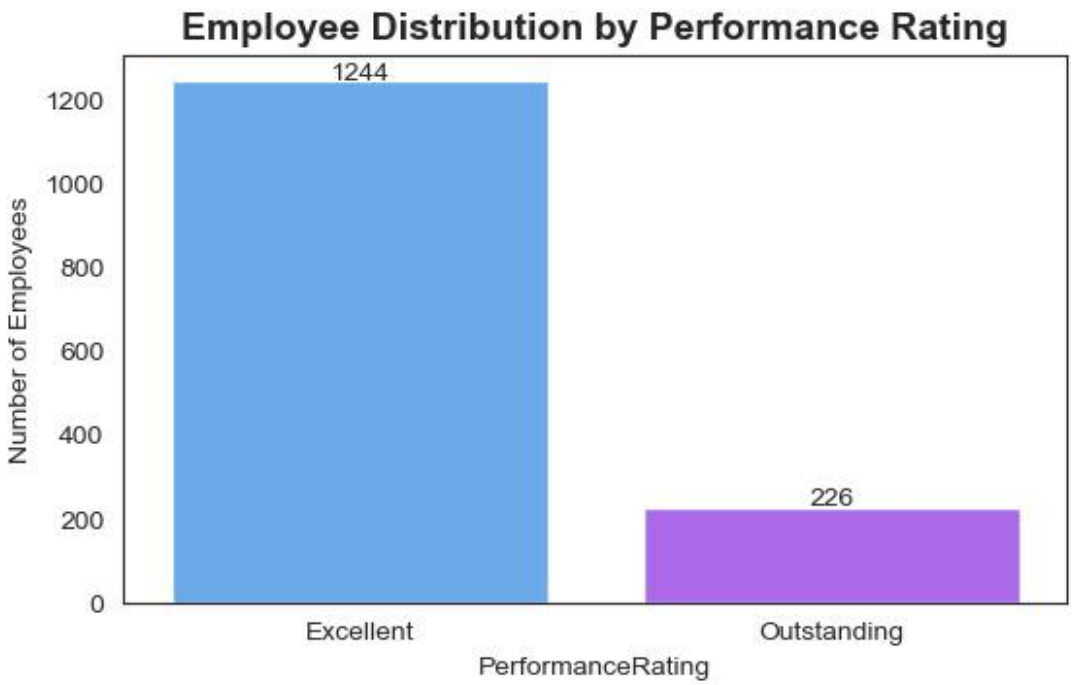


Findings & Insights:

Low relationship satisfaction means not so good relationship with others in the company. Employees with lower relationship satisfaction have higher attrition rates. Employees with great relationship with others are more likely to stay in the company.

30) Attrition vs Performance Rating

```
df['PerformanceRating'].value_counts()
3    1244
4     226
Name: PerformanceRating, dtype: int64

pr_labels = {1:"Low",2:"Good",3:"Excellent",4:"Outstanding"}


pr_counts = df['PerformanceRating'].value_counts().sort_index()

sns.set_style("white")

plt.figure(figsize=(6, 4))

sns.barplot(x=[pr_labels[i] for i in pr_counts.index],
        y=pr_counts.values,
        palette='cool')

plt.title('Employee Distribution by Performance Rating', fontsize=14, fontweight='bol
d')
plt.xlabel('PerformanceRating')
plt.ylabel('Number of Employees')

plt.grid(False)

for i, v in enumerate(pr_counts.values):
   plt.text(i, v + 5, str(v), ha='center', fontweight='normal')


plt.tight_layout(pad=2.0)
plt.show()
```

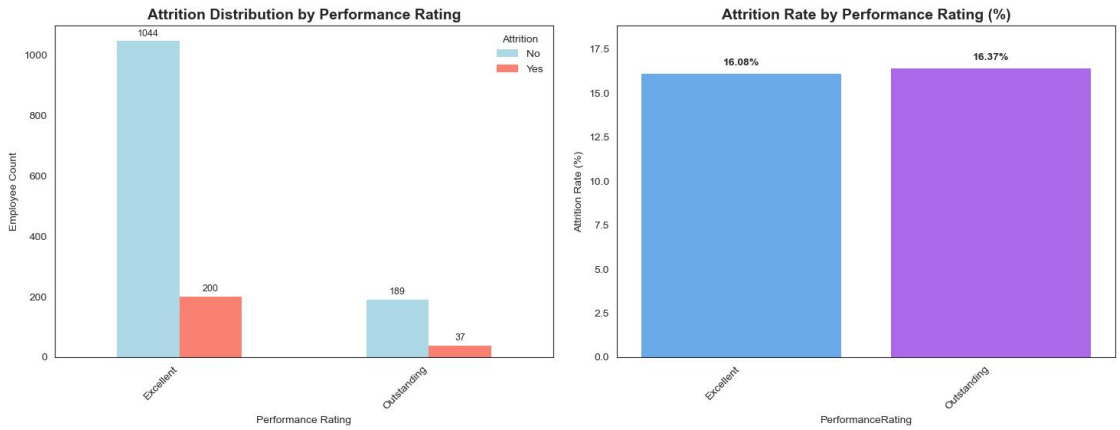


Findings:

Most of the employees in the company have excellent performance rating.

```
df[df['Attrition'] == 'Yes']['PerformanceRating'].value_counts()
3    200
4     37
Name: PerformanceRating, dtype: int64

calculate_attrition_rate(df, 'PerformanceRating').round(2)

   PerformanceRating  AttritionRate
0                  3          16.08
1                  4          16.37

pr_labels = {1:"Low",2:"Good",3:"Excellent",4:"Outstanding"}



sns.set_style("white")


fig, axes = plt.subplots(1, 2, figsize=(15, 6))
plt.subplots_adjust(wspace=0.4)

#  Attrition Distribution (Yes/No) by PerformanceRating
attrition_pr = df.groupby(['PerformanceRating', 'Attrition']).size().unstack(fill_value=
0)
attrition_pr.plot(kind='bar', ax=axes[0], color=['lightblue', 'salmon'], edgecolor='none
```

```
')
axes[0].set_title('Attrition Distribution by Performance Rating', fontsize=14, fontweig
ht='bold')
axes[0].set_xlabel('Performance Rating')
axes[0].set_ylabel('Employee Count')
axes[0].set_xticklabels([pr_labels[i] for i in attrition_pr.index], rotation=45, ha='right')
axes[0].legend(title='Attrition', frameon=False)
axes[0].grid(False)
for container in axes[0].containers:
    axes[0].bar_label(container, fmt='%d', label_type='edge', fontsize=9, padding=3)

#  Attrition Rate (%) by PerformanceRating
attrition_rate_pr = calculate_attrition_rate(df, 'PerformanceRating')
sns.barplot(x='PerformanceRating', y='AttritionRate', data=attrition_rate_pr, palette
='cool', ax=axes[1], edgecolor='none')
axes[1].set_title('Attrition Rate by Performance Rating (%)', fontsize=14, fontweight=
'bold')
axes[1].set_xlabel('PerformanceRating')
axes[1].set_ylabel('Attrition Rate (%)')
axes[1].set_xticklabels([pr_labels[i] for i in attrition_rate_pr['PerformanceRating']], r
otation=45, ha='right')
axes[1].set_ylim(0, attrition_rate_pr['AttritionRate'].max() * 1.15)
axes[1].grid(False)
for i, v in enumerate(attrition_rate_pr['AttritionRate']):
    axes[1].text(i, v + 0.5, f'{v:.2f}%', ha='center', fontweight='bold')

plt.tight_layout(pad=2.0)
plt.show()
```

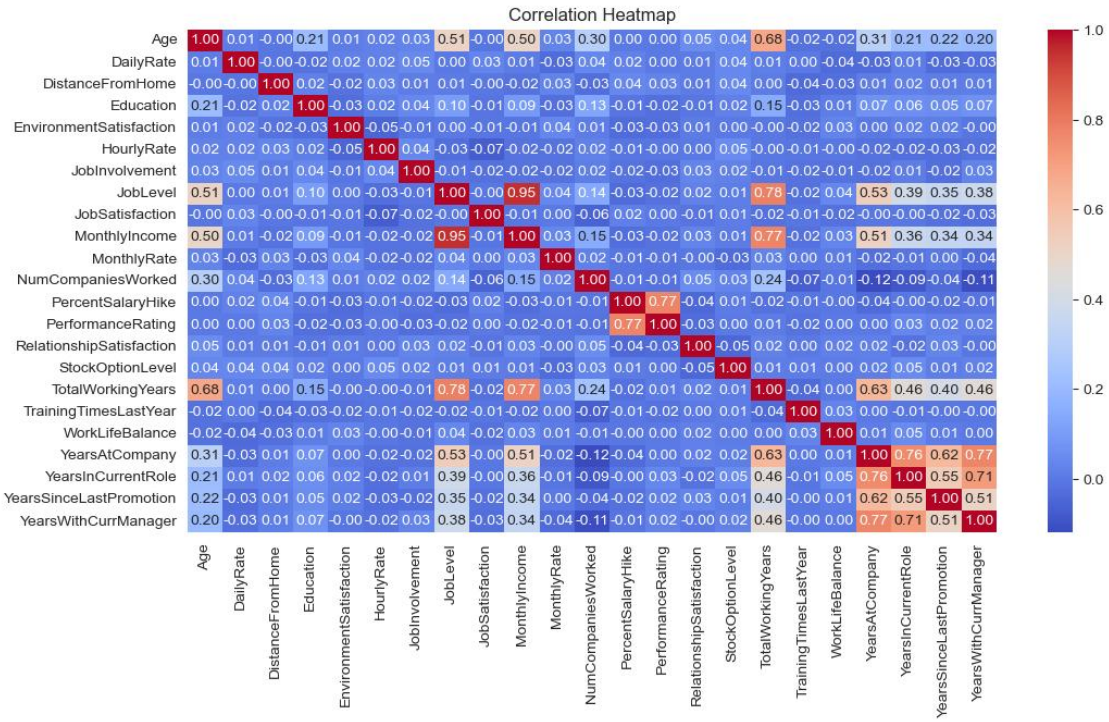


Findings & Insights:

As most of the employees are having excellent performance rating and both the categories of performance ratings such as excellent and outstanding are having almost same attriton rate, we can't generate any meaningful inisghts from this.

Correlation Heatmap

```
df_new=df.copy()
df_new.corr()
plt.figure(figsize=(12,6))
sns.heatmap(df_new.corr(), annot=True, cmap='coolwarm', fmt=".2f")
plt.title("Correlation Heatmap")
plt.show()
```




```
corr_matrix = df_new.corr()

# Convert to list
corr_pairs = corr_matrix.unstack()

# Remove self-correlation
corr_pairs = corr_pairs[corr_pairs.index.get_level_values(0) != corr_pairs.index.get_l
evel_values(1)]

# Drop duplicate pairs
corr_pairs = corr_pairs.drop_duplicates()

# Top 10 correlations (absolute)
top_10_corr = corr_pairs.abs().sort_values(ascending=False).head(10)

print(top_10_corr)
JobLevel          MonthlyIncome            0.950300
                  TotalWorkingYears        0.782208
PercentSalaryHike  PerformanceRating       0.773550
MonthlyIncome      TotalWorkingYears       0.772893
YearsAtCompany     YearsWithCurrManager    0.769212
```

```
YearsInCurrentRole        0.758754
YearsInCurrentRole  YearsWithCurrManager      0.714365
Age                 TotalWorkingYears         0.680381
TotalWorkingYears   YearsAtCompany            0.628133
YearsAtCompany      YearsSinceLastPromotion   0.618409
dtype: float64
```

Findings & Insights:

--Top 10 Correlation--

1) JobLevel & MonthlyIncome 0.95

Salary rises almost directly with job level. Higher positions get much higher pay.

2)JobLevel & TotalWorkingYears 0.78

Employees with more experience usually hold higher job levels.

3)PercentSalaryHike & PerformanceRating 0.77

High performers receive bigger salary hikes.

4)MonthlyIncome & TotalWorkingYears 0.77

More experienced employees earn more.

5)YearsAtCompany & YearsWithCurrManager 0.77

Long-tenure employees usually stay with the same manager.

6)YearsAtCompany & YearsInCurrentRole 0.76

The longer an employee stays, the longer they remain in the same role.

7)YearsInCurrentRole & YearsWithCurrManager 0.71

Employees who remain long in a role usually report to the same manager.

8)Age & TotalWorkingYears 0.68

Older employees naturally have more work experience.

9)TotalWorkingYears & YearsAtCompany 0.63

More experienced people tend to stay longer in the company.

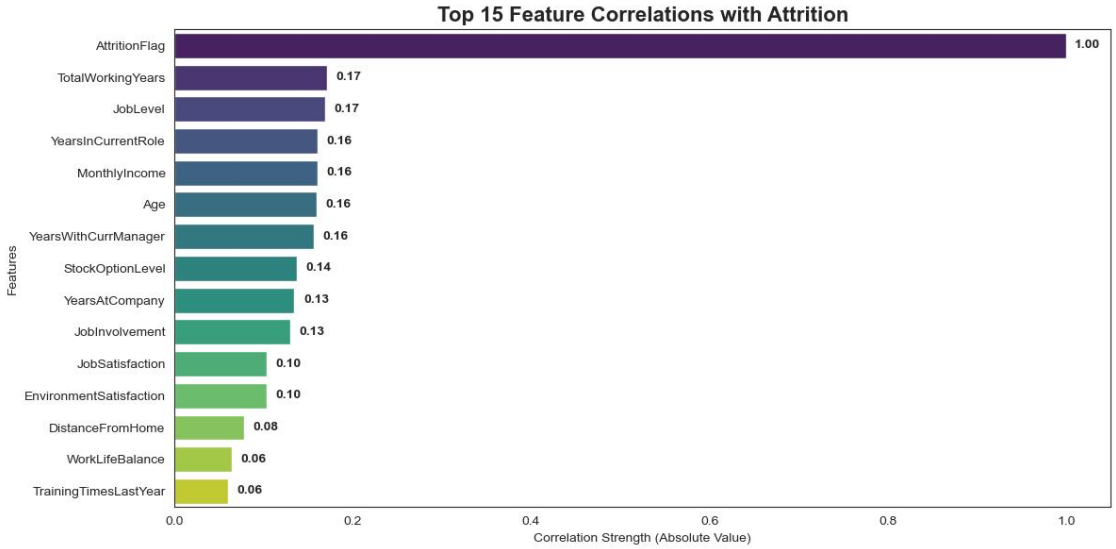
10)YearsAtCompany & YearsSinceLastPromotion 0.62

Employees who stay long often wait many years for promotion.

```
df_new['AttritionFlag'] = df_new['Attrition'].map({'Yes': 1, 'No': 0})
numeric_cols = df_new.select_dtypes(include=['int64', 'float64']).columns
attrition_corr = df_new[numeric_cols].corr()['AttritionFlag'].sort_values(ascending=F
alse)
attrition_corr.abs().sort_values(ascending=False).head(15)
```

```
AttritionFlag            1.000000
TotalWorkingYears        0.171063
JobLevel                 0.169105
YearsInCurrentRole       0.160545
MonthlyIncome            0.159840
Age                      0.159205
YearsWithCurrManager     0.156199
StockOptionLevel         0.137145
YearsAtCompany           0.134392
JobInvolvement           0.130016
JobSatisfaction          0.103481
EnvironmentSatisfaction  0.103369
DistanceFromHome         0.077924
WorkLifeBalance          0.063939
TrainingTimesLastYear    0.059478
Name: AttritionFlag, dtype: float64
```

```
# --- Compute Top Correlations ---
top_corr = attrition_corr.abs().sort_values(ascending=False).head(15)

plt.figure(figsize=(12, 6))
sns.barplot(
    x=top_corr.values,
    y=top_corr.index,
    palette="viridis"
)

plt.title("Top 15 Feature Correlations with Attrition", fontsize=16, fontweight="bold")
plt.xlabel("Correlation Strength (Absolute Value)")
plt.ylabel("Features")


for i, v in enumerate(top_corr.values):
    plt.text(v + 0.01, i, f"{v:.2f}", va='center', fontweight='bold')

plt.tight_layout()
plt.show()
```



Findings & Insights:

#Top 15 Feature Correlations with Attrition

AttritionFlag means

1 = Employee left

0 = Employee stayed

So:

Positive correlation → higher value = more likely to leave

Negative correlation → higher value = more likely to stay

1)TotalWorkingYears 0.17 -Attrition is related to experience level

2)JobLevel 0.17 -Junior employees leave more often

3)YearsInCurrentRole 0.16 -People stuck in the same role are likely to resign

4)MonthlyIncome 0.16 -Low-income employees leave more

5)Age 0.16 -Younger employees leave more

6)YearsWithCurrManager 0.15 -Poor manager stability increases exits

7)StockOptionLevel 0.14 -Fewer stock benefits means higher attrition

8)YearsAtCompany 0.13 -New employees quit faster

9)JobInvolvement 0.13 -Low engagement employees resign more

10)JobSatisfaction 0.10 -Dissatisfied employees leave

11)EnvironmentSatisfaction 0.10 -Poor work environment increases exits

12)DistanceFromHome 0.08 -Long commute pushes attrition

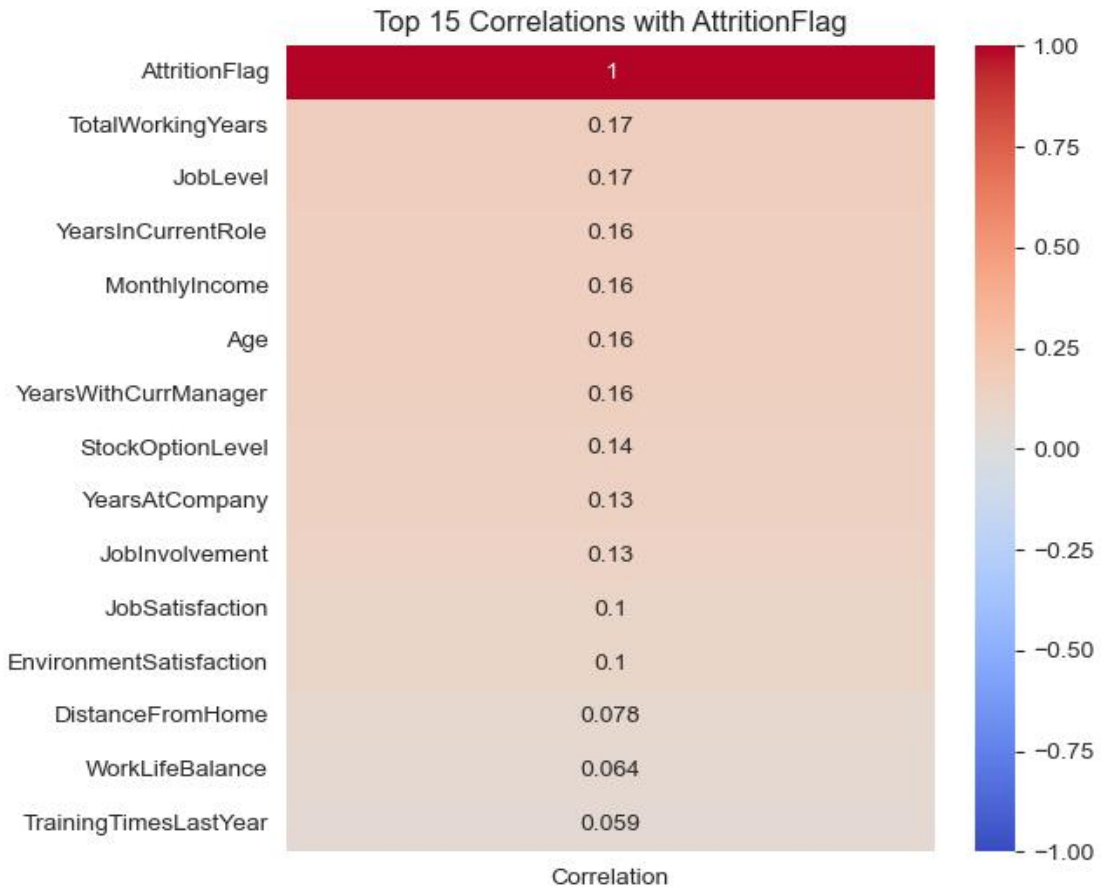
13)WorkLifeBalance 0.06 -Poor balance leads to exits

14)TrainingTimesLastYear 0.06 -Lack of learning causes resignations

Findings & Insights:

Correlation analysis with the Attrition Flag indicates that employees who are younger, in lower job levels, earning lower monthly income, and having fewer total working years are more likely to leave the company. Additionally, lower job involvement, job satisfaction, and longer distance from home also show a positive association with attrition. These results suggest that IBM faces higher turnover risk among early-career employees experiencing slow growth, limited rewards, and lower engagement levels.

```
top_corr_df = top_corr.to_frame(name='Correlation')

plt.figure(figsize=(6, 6))
sns.heatmap(top_corr_df, annot=True, cmap="coolwarm", vmin=-1, vmax=1)
plt.title("Top 15 Correlations with AttritionFlag")
plt.show()
```



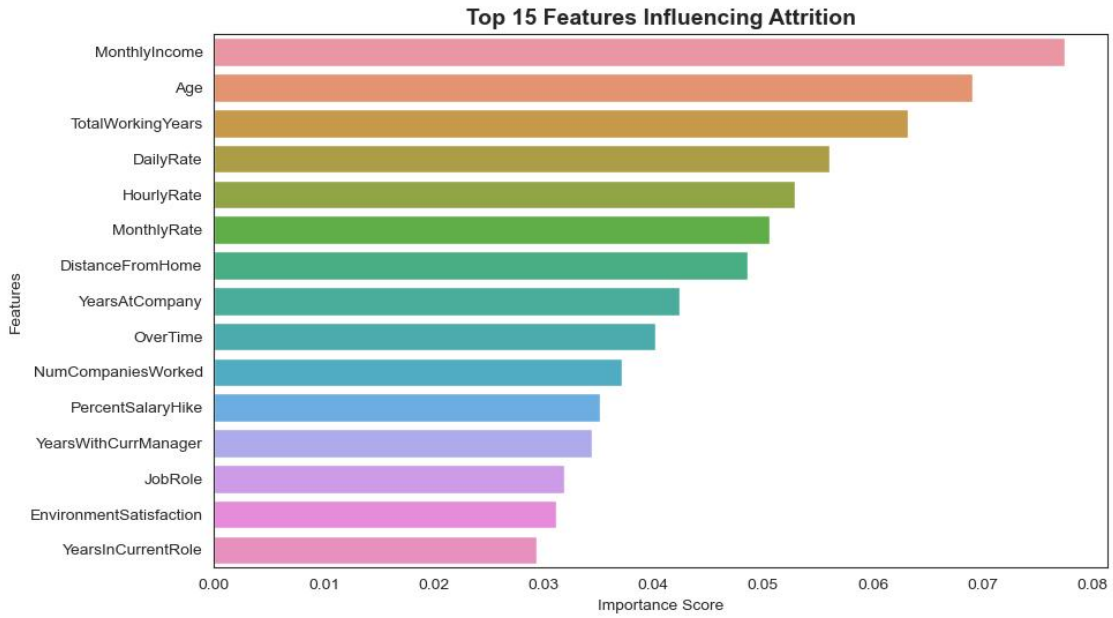
5. **Employee Attrition Prediction Model using Machine Learning**

```
#complete attrition prediction ML pipeline
# 1) Prepare the data
df_new['AttritionFlag'] = df_new['Attrition'].map({'Yes': 1, 'No': 0})
```

```
# 2) Encode categorical variables


le = LabelEncoder()
df_encoded = df_new.copy()

for col in df_encoded.select_dtypes(include='object').columns:
    df_encoded[col] = le.fit_transform(df_encoded[col])
# 3) Split features & target
X = df_encoded.drop(['Attrition', 'AttritionFlag'], axis=1)
y = df_encoded['AttritionFlag']
# 4) Train–Test Split


X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y)
# 5) Scale numeric variables


scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
# 6)  Logistic Regression Model


log_reg = LogisticRegression(max_iter=1000)
log_reg.fit(X_train_scaled, y_train)

y_pred_lr = log_reg.predict(X_test_scaled)
# 7) Random Forest Model



rf = RandomForestClassifier(n_estimators=200, random_state=42)
rf.fit(X_train, y_train)

y_pred_rf = rf.predict(X_test)
# 8) Model Evaluation


print("Logistic Regression Accuracy:", accuracy_score(y_test, y_pred_lr))
print(confusion_matrix(y_test, y_pred_lr))
print(classification_report(y_test, y_pred_lr))

print("\nRandom Forest Accuracy:", accuracy_score(y_test, y_pred_rf))
print(confusion_matrix(y_test, y_pred_rf))
print(classification_report(y_test, y_pred_rf))
```

```
Logistic Regression Accuracy: 0.8741496598639455
[[239   8]
 [ 29  18]]
              precision    recall  f1-score   support

           0       0.89      0.97      0.93       247
           1       0.69      0.38      0.49        47

    accuracy                           0.87       294
   macro avg       0.79      0.68      0.71       294
weighted avg       0.86      0.87      0.86       294


Random Forest Accuracy: 0.8367346938775511
[[240   7]
 [ 41   6]]
              precision    recall  f1-score   support

           0       0.85      0.97      0.91       247
           1       0.46      0.13      0.20        47

    accuracy                           0.84       294
   macro avg       0.66      0.55      0.55       294
weighted avg       0.79      0.84      0.80       294
```

```
# 9) Feature Importance (Random Forest)

#This shows what features drive attrition.


importances = pd.Series(rf.feature_importances_, index=X.columns).sort_values(asc
ending=False)

plt.figure(figsize=(10,6))
sns.barplot(x=importances.values[:15], y=importances.index[:15])
plt.title("Top 15 Features Influencing Attrition", fontsize=14, fontweight="bold")
plt.xlabel("Importance Score")
plt.ylabel("Features")
plt.show()
```



Findings & Insights:

Interpretation of Top Features Influencing Employee Attrition (Random Forest)

The Random Forest model identified the following variables as the most influential in predicting employee attrition, in descending order of importance:

Monthly Income, Age, Total Working Years, Daily Rate, Hourly Rate, Monthly Rate, Distance from Home, Years at Company, Overtime, Number of Companies Worked, Percentage Salary Hike, Years with Current Manager, Job Role, Environment Satisfaction, and Years in Current Role.

These are the main reasons employees leave:

1) Low or unfair compensation – employees feel underpaid compared to their effort or the market.
2) Lack of career growth – no promotions, learning, or future opportunities.
3) Poor work-life balance – long working hours, overtime, or constant stress.
4) Job dissatisfaction – boring roles, unclear responsibilities, or no recognition.
5) Bad management or leadership – lack of support, micromanagement, or poor communication.
6) High job pressure – especially in sales or target-driven roles.
7) Long commute or relocation issues – travel stress affects daily life.
8) Toxic or unsupportive work culture – conflicts, favoritism, or unfair treatment.
9) Better opportunities elsewhere – higher pay, better role, or better company brand.

1. Compensation is the strongest driver

- Monthly Income, Daily Rate, Hourly Rate, Monthly Rate, and Percentage Salary Hike appear repeatedly in the top features.

- This indicates that: Employees with lower pay and smaller salary hikes are significantly more likely to leave.
- Compensation structure is a critical retention factor.

2. Early-career employees are at higher risk

- High importance of Age, Total Working Years, and Number of Companies Worked shows that:
- Younger employees and those with less experience or frequent job changes are more likely to resign.
- This reflects instability and job-hopping behaviour in early careers.

3. Work-life stress increases attrition

- Distance from Home and Overtime are strong predictors, meaning:
- Employees with long commutes and regular overtime experience burnout and are more likely to leave.

4. Career stagnation contributes to exits

- Years at Company, Years with Current Manager, Years in Current Role, and Job Role indicate that employees who remain too long in the same role or under the same manager feel limited growth opportunities.
- This creates dissatisfaction and resignation risk.

5. Work environment still matters

- Environment Satisfaction confirms that poor workplace environment directly contributes to employee turnover.

Feature importance analysis using the Random Forest model reveals that employee attrition at IBM is primarily driven by compensation structure, career stage, workload stress, and growth stagnation. Low income, limited salary hikes, younger age, long commuting distance, frequent overtime, and prolonged time in the same role or under the same manager significantly increase the likelihood of employee resignation. These findings suggest that retention strategies should prioritize competitive compensation, flexible work policies, reduced overtime, and clear career progression pathways, especially for early-career employees.

Insights:

Here are the key reasons for employee attrition based on the analysis:

-Some employees feel underpaid or undervalued despite giving their best effort or performance

-Low salary and insufficient salary hikes

-Limited career growth and delayed promotions

-Long commuting distance and lack of flexible work options

-Frequent overtime and poor work-life balance

-High job pressure, especially in Sales roles

-Low job involvement and poor employee engagement

-Unsatisfactory work environment and management issues

-Frequent business travel causing burnout

-Not so good relationship with colleagues and managers

-Lack of upskilling and development opportunities for early-career employees

In simple words: employees leave when they don’t feel valued, supported, or able to grow.

Recommendations:

-Give salary hikes, rewards and promotion according to their performance.

-Make sure that the company's work culture is supportive and not toxic.

-Revise compensation for entry-level and junior employees to reduce early-career attrition.

-Define clear promotion timelines to prevent career stagnation, especially at mid-level roles.

-Provide upskilling and certification programs for employees with lower education levels.

-Expand hybrid and remote work options to reduce commute-related resignations.

-Implement burnout-prevention strategies in Sales through realistic targets and improved incentives.

-Apply focused retention initiatives for high-attrition roles such as Sales Representatives, Lab Technicians, and HR staff.

-Limit excessive overtime and frequent business travel to improve work-life balance.

-Strengthen employee engagement through recognition programs, regular feedback, and leadership support.

-Deploy balanced machine learning attrition-prediction models to proactively identify employees at risk of leaving.

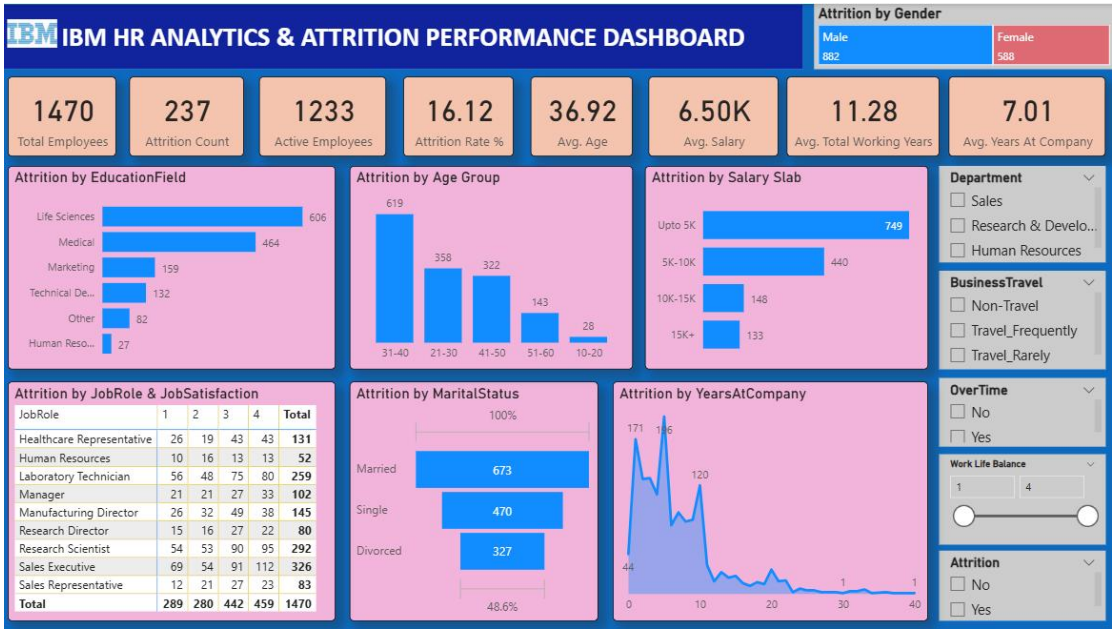
Conclusion:

The analysis reveals that employee attrition at IBM is not random but is strongly influenced by a combination of compensation, career progression, work-life balance, job role pressure, and personal lifestyle factors. Attrition is highest among early-career employees, sales and HR roles, employees with lower education levels, long commuting distances, frequent business travel, low job involvement, and high overtime workloads .

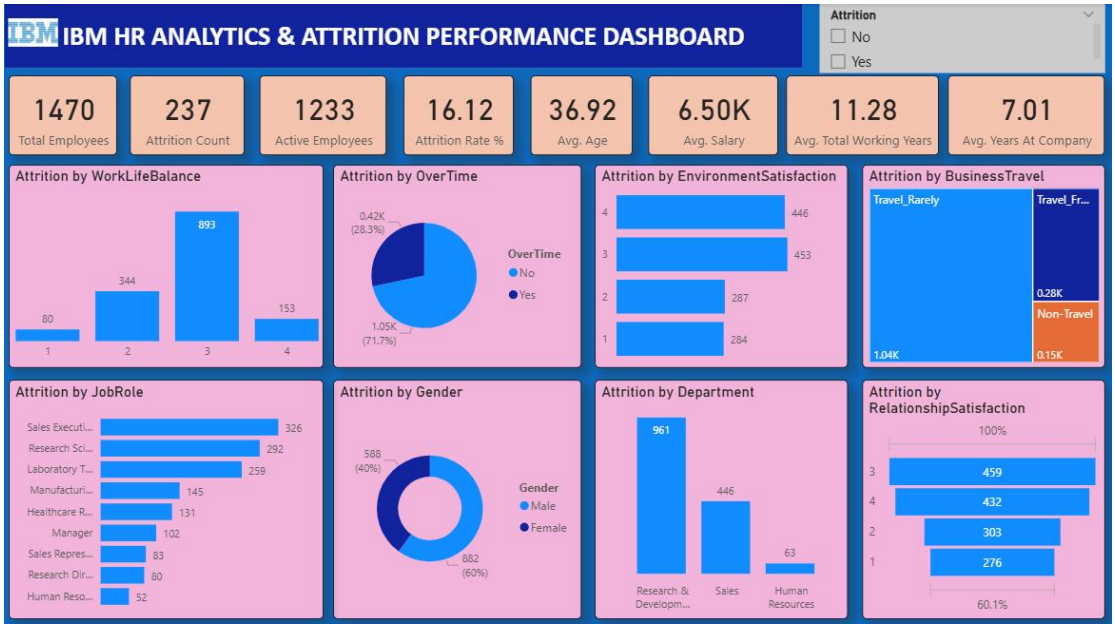
It also shows that employees with higher income, longer tenure, greater job involvement, and better work environments are significantly more likely to stay with the company, while frontline and junior roles face the greatest retention risk .

Overall, the project concludes that IBM can substantially reduce attrition by focusing on improving compensation fairness, offering clear career growth paths, promoting flexible working options, controlling travel and overtime demands, and strengthening employee engagement initiatives, particularly for early-career and high-pressure job roles .

Dashboard:



JobRole	1	2	3	4	Total
Healthcare Representative	26	19	43	43	131
Human Resources	10	16	13	13	52
Laboratory Technician	56	48	75	80	259
Manager	21	21	27	33	102
Manufacturing Director	26	32	49	38	145
Research Director	15	16	27	22	80
Research Scientist	54	53	90	95	292
Sales Executive	69	54	91	112	326
Sales Representative	12	21	27	23	83
Total	**289**	**280**	**442**	**459**	**1470**


IBM HR ANALYTICS & ATTRITION PERFORMANCE DASHBOARD
Attrition
No
Yes
1470
Total Employees
237
Attrition Count
1233
Active Employees
16.12
Attrition Rate %
36.92
Avg. Age
6.50K
Avg. Salary
11.28
Avg. Total Working Years
7.01
Avg. Years At Company
Attrition by WorkLifeBalance
80
344
893
153
1
2
3
4
Attrition by OverTime
0.42K
(28.3%)
1.05K
(71.7%)
OverTime
No
Yes
Attrition by EnvironmentSatisfaction
4
446
3
453
2
287
1
284
Attrition by BusinessTravel
Travel_Rarely
1.04K
Travel_Fr...
0.28K
Non-Travel
0.15K
Attrition by JobRole
Sales Executi...
326
Research Sci...
292
Laboratory T...
259
Manufacturi...
145
Healthcare R...
131
Manager
102
Sales Repres...
83
Research Dir...
80
Human Reso...
52
Attrition by Gender
588
(40%)
882
(60%)
Gender
Male
Female
Attrition by Department
961
446
63
Research &
Developm...
Sales
Human
Resources
Attrition by
RelationshipSatisfaction
100%
3
459
4
432
2
303
1
276
60.1%